Stopa's Selection

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For my love, Nicole

Contents

Life

Chapter 1

What You'll Wish You'd Known

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(I wrote this talk for a high school. I never actually gave it, because the school authorities vetoed the plan to invite me.)

When I said I was speaking at a high school, my friends were curious. What will you say to high school students? So I asked them, what do you wish someone had told you in high school? Their answers were remarkably similar. So I'm going to tell you what we all wish someone had told us.

I'll start by telling you something you don't have to know in high school: what you want to do with your life. People are always asking you this, so you think you're supposed to have an answer. But adults ask this mainly as a conversation starter. They want to know what sort of person you are, and this question is just to get you talking. They ask it the way you might poke a hermit crab in a tide pool, to see what it does.

If I were back in high school and someone asked about my plans, I'd say that my first priority was to learn what the options were. You don't need to be in a rush to choose your life's work. What you need to do is discover what you like. You have to work on stuff you like if you want to be good at what you do.

It might seem that nothing would be easier than deciding what you like, but it turns out to be hard, partly because it's hard to get an accurate picture of most jobs. Being a doctor is not the way it's portrayed on TV. Fortunately you can also watch real doctors, by volunteering in hospitals. [1]

But there are other jobs you can't learn about, because no one is doing them yet. Most of the work I've done in the last ten years didn't exist when I was in high school. The world changes fast, and the rate at which it changes is itself speeding up. In such a world it's not a good idea to have fixed plans.

And yet every May, speakers all over the country fire up the Standard Graduation Speech, the theme of which is: don't give up on your dreams. I know what they mean, but this is a bad way to put it, because it implies you're supposed to be bound by some plan you made early on. The computer world has a name for this: premature optimization. And it is synonymous with disaster. These speakers would do better to say simply, don't give up.

What they really mean is, don't get demoralized. Don't think that you can't do what other people can. And I agree you shouldn't underestimate your potential. People who've done great things tend to seem as if they were a race apart. And most biographies only exaggerate this illusion, partly due to the worshipful attitude biographers inevitably sink into, and partly because, knowing how the story ends, they can't help streamlining the plot till it seems like the subject's life was a matter of destiny, the mere unfolding of some innate genius. In fact I suspect if you had the sixteen year old Shakespeare or Einstein in school with you, they'd seem impressive, but not totally unlike your other friends.

Which is an uncomfortable thought. If they were just like us, then they had to work very hard to do what they did. And that's one reason we like to believe in genius. It gives us an excuse for being lazy. If these guys were able to do what they did only because of some magic Shakespeareness or Einsteinness, then it's not our fault if we can't do something as good.

[1] A doctor friend warns that even this can give an inaccurate picture. "Who knew how much time it would take up, how little autonomy one would have for endless years of training, and how unbelievably annoying it is to carry a beeper?"

I'm not saying there's no such thing as genius. But if you're trying to choose between two theories and one gives you an excuse for being lazy, the other one is probably right.

So far we've cut the Standard Graduation Speech down from "don't give up on your dreams" to "what someone else can do, you can do." But it needs to be cut still further. There is *some* variation in natural ability. Most people overestimate its role, but it does exist. If I were talking to a guy four feet tall whose ambition was to play in the NBA, I'd feel pretty stupid saying, you can do anything if you really try. [2]

We need to cut the Standard Graduation Speech down to, "what someone else with your abilities can do, you can do; and don't underestimate your abilities." But as so often happens, the closer you get to the truth, the messier your sentence gets. We've taken a nice, neat (but wrong) slogan, and churned it up like a mud puddle. It doesn't make a very good speech anymore. But worse still, it doesn't tell you what to do anymore. Someone with your abilities? What are your abilities?

Upwind

I think the solution is to work in the other direction. Instead of working back from a goal, work forward from promising situations. This is what most successful people actually do anyway.

In the graduation-speech approach, you decide where you want to be in twenty years, and then ask: what should I do now to get there? I propose instead that you don't commit to anything in the future, but just look at the options available now, and choose those that will give you the most promising range of options afterward.

It's not so important what you work on, so long as you're not wasting your time. Work on things that interest you and increase your options, and worry later about which you'll take.

[2] His best bet would probably be to become dictator and intimidate the NBA into letting him play. So far the closest anyone has come is Secretary of Labor.

Suppose you're a college freshman deciding whether to major in math or economics. Well, math will give you more options: you can go into almost any field from math. If you major in math it will be easy to get into grad school in economics, but if you major in economics it will be hard to get into grad school in math.

Flying a glider is a good metaphor here. Because a glider doesn't have an engine, you can't fly into the wind without losing a lot of altitude. If you let yourself get far downwind of good places to land, your options narrow uncomfortably. As a rule you want to stay upwind. So I propose that as a replacement for "don't give up on your dreams." Stay upwind.

How do you do that, though? Even if math is upwind of economics, how are you supposed to know that as a high school student?

Well, you don't, and that's what you need to find out. Look for smart people and hard problems. Smart people tend to clump together, and if you can find such a clump, it's probably worthwhile to join it. But it's not straightforward to find these, because there is a lot of faking going on.

To a newly arrived undergraduate, all university departments look much the same. The professors all seem forbiddingly intellectual and publish papers unintelligible to outsiders. But while in some fields the papers are unintelligible because they're full of hard ideas, in others they're deliberately written in an obscure way to seem as if they're saying something important. This may seem a scandalous proposition, but it has been experimentally verified, in the famous *Social Text* affair. Suspecting that the papers published by literary theorists were often just intellectual-sounding nonsense, a physicist deliberately wrote a paper full of intellectual-sounding nonsense, and submitted it to a literary theory journal, which published it.

The best protection is always to be working on hard problems. Writing novels is hard. Reading novels isn't. Hard means worry: if you're not worrying that something you're making will come out badly, or that you won't be able to understand something you're studying, then it isn't hard enough. There has to be suspense.

Well, this seems a grim view of the world, you may think. What I'm telling you is that you should worry? Yes, but it's not as bad as it sounds. It's exhilarating to overcome worries. You don't see faces much happier than people winning gold medals. And you know why they're so happy? Relief.

I'm not saying this is the only way to be happy. Just that some kinds of worry are not as bad as they sound.

Ambition

In practice, "stay upwind" reduces to "work on hard problems." And you can start today. I wish I'd grasped that in high school.

Most people like to be good at what they do. In the so-called real world this need is a powerful force. But high school students rarely benefit from it, because they're given a fake thing to do. When I was in high school, I let myself believe that my job was to be a high school student. And so I let my need to be good at what I did be satisfied by merely doing well in school.

If you'd asked me in high school what the difference was between high school kids and adults, I'd have said it was that adults had to earn a living. Wrong. It's that adults take responsibility for themselves. Making a living is only a small part of it. Far more important is to take intellectual responsibility for oneself.

If I had to go through high school again, I'd treat it like a day job. I don't mean that I'd slack in school. Working at something as a day job doesn't mean doing it badly. It means not being defined by it. I mean I wouldn't think of myself as a high school student, just as a musician with a day job as a waiter doesn't think of himself as a waiter. [3] And

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[3] A day job is one you take to pay the bills so you can do what you really want, like play in a band, or invent relativity.

Treating high school as a day job might actually make it easier for some students when I wasn't working at my day job I'd start trying to do real work.

When I ask people what they regret most about high school, they nearly all say the same thing: that they wasted so much time. If you're wondering what you're doing now that you'll regret most later, that's probably it. [4]

Some people say this is inevitable — that high school students aren't capable of getting anything done yet. But I don't think this is true. And the proof is that you're bored. You probably weren't bored when you were eight. When you're eight it's called "playing" instead of "hanging out," but it's the same thing. And when I was eight, I was rarely bored. Give me a back yard and a few other kids and I could play all day.

The reason this got stale in middle school and high school, I now realize, is that I was ready for something else. Childhood was getting old.

I'm not saying you shouldn't hang out with your friends — that you should all become humorless little robots who do nothing but work. Hanging out with friends is like chocolate cake. You enjoy it more if you eat it occasionally than if you eat nothing but chocolate cake for every meal. No matter how much you like chocolate cake, you'll be pretty queasy after the third meal of it. And that's what the malaise

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to get good grades. If you treat your classes as a game, you won't be demoralized if they seem pointless.

However bad your classes, you need to get good grades in them to get into a decent college. And that *is* worth doing, because universities are where a lot of the clumps of smart people are these days.

[4]The second biggest regret was caring so much about unimportant things. And especially about what other people thought of them.

I think what they really mean, in the latter case, is caring what random people thought of them. Adults care just as much what other people think, but they get to be more selective about the other people.

I have about thirty friends whose opinions I care about, and the opinion of the rest of the world barely affects me. The problem in high school is that your peers are chosen for you by accidents of age and geography, rather than by you based on respect for their judgement.

one feels in high school is: mental queasiness. [5]

You may be thinking, we have to do more than get good grades. We have to have *extracurricular activities.* But you know perfectly well how bogus most of these are. Collecting donations for a charity is an admirable thing to do, but it's not *hard.* It's not getting something done. What I mean by getting something done is learning how to write well, or how to program computers, or what life was really like in preindustrial societies, or how to draw the human face from life. This sort of thing rarely translates into a line item on a college application.

Corruption

It's dangerous to design your life around getting into college, because the people you have to impress to get into college are not a very discerning audience. At most colleges, it's not the professors who decide whether you get in, but admissions officers, and they are nowhere near as smart. They're the NCOs of the intellectual world. They can't tell how smart you are. The mere existence of prep schools is proof of that.

Few parents would pay so much for their kids to go to a school that didn't improve their admissions prospects. Prep schools openly say this is one of their aims. But what that means, if you stop to think about it, is that they can hack the admissions process: that they can take the very same kid and make him seem a more appealing candidate than he would if he went to the local public school. [6]

[5]The key to wasting time is distraction. Without distractions it's too obvious to your brain that you're not doing anything with it, and you start to feel uncomfortable. If you want to measure how dependent you've become on distractions, try this experiment: set aside a chunk of time on a weekend and sit alone and think. You can have a notebook to write your thoughts down in, but nothing else: no friends, TV, music, phone, IM, email, Web, games, books, newspapers, or magazines. Within an hour most people will feel a strong craving for distraction.

[6]I don't mean to imply that the only function of prep schools is to trick admissions officers. They also generally provide a better education. But try this thought experiment: suppose prep schools supplied the same superior education but had a tiny (.001) negative effect on college admissions. How many parents would still send

Right now most of you feel your job in life is to be a promising college applicant. But that means you're designing your life to satisfy a process so mindless that there's a whole industry devoted to subverting it. No wonder you become cynical. The malaise you feel is the same that a producer of reality TV shows or a tobacco industry executive feels. And you don't even get paid a lot.

So what do you do? What you should not do is rebel. That's what I did, and it was a mistake. I didn't realize exactly what was happening to us, but I smelled a major rat. And so I just gave up. Obviously the world sucked, so why bother?

When I discovered that one of our teachers was herself using Cliff's Notes, it seemed par for the course. Surely it meant nothing to get a good grade in such a class.

In retrospect this was stupid. It was like someone getting fouled in a soccer game and saying, hey, you fouled me, that's against the rules, and walking off the field in indignation. Fouls happen. The thing to do when you get fouled is not to lose your cool. Just keep playing.

By putting you in this situation, society has fouled you. Yes, as you suspect, a lot of the stuff you learn in your classes is crap. And yes, as you suspect, the college admissions process is largely a charade. But like many fouls, this one was unintentional. [7] So just keep playing.

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their kids to them?

It might also be argued that kids who went to prep schools, because they've learned more, *are* better college candidates. But this seems empirically false. What you learn in even the best high school is rounding error compared to what you learn in college. Public school kids arrive at college with a slight disadvantage, but they start to pull ahead in the sophomore year.

(I'm not saying public school kids are smarter than preppies, just that they are within any given college. That follows necessarily if you agree prep schools improve kids' admissions prospects.)

[7]Why does society foul you? Indifference, mainly. There are simply no outside forces pushing high school to be good. The air traffic control system works because planes would crash otherwise. Businesses have to deliver because otherwise competitors would take their customers. But no planes crash if your school sucks, and it has no competitors. High school isn't evil; it's random; but random is pretty bad.

Rebellion is almost as stupid as obedience. In either case you let yourself be defined by what they tell you to do. The best plan, I think, is to step onto an orthogonal vector. Don't just do what they tell you, and don't just refuse to. Instead treat school as a day job. As day jobs go, it's pretty sweet. You're done at 3 o'clock, and you can even work on your own stuff while you're there.

Curiosity

And what's your real job supposed to be? Unless you're Mozart, your first task is to figure that out. What are the great things to work on? Where are the imaginative people? And most importantly, what are you interested in? The word "aptitude" is misleading, because it implies something innate. The most powerful sort of aptitude is a consuming interest in some question, and such interests are often acquired tastes.

A distorted version of this idea has filtered into popular culture under the name "passion." I recently saw an ad for waiters saying they wanted people with a "passion for service." The real thing is not something one could have for waiting on tables. And passion is a bad word for it. A better name would be curiosity.

Kids are curious, but the curiosity I mean has a different shape from kid curiosity. Kid curiosity is broad and shallow; they ask why at random about everything. In most adults this curiosity dries up entirely. It has to: you can't get anything done if you're always asking why about everything. But in ambitious adults, instead of drying up, curiosity becomes narrow and deep. The mud flat morphs into a well.

Curiosity turns work into play. For Einstein, relativity wasn't a book full of hard stuff he had to learn for an exam. It was a mystery he was trying to solve. So it probably felt like less work to him to invent it than it would seem to someone now to learn it in a class.

One of the most dangerous illusions you get from school is the idea that doing great things requires a lot of discipline. Most subjects are taught in such a boring way that it's only by discipline that you can flog yourself through them. So I was surprised when, early in college, I read a quote by Wittgenstein saying that he had no self-discipline and had never been able to deny himself anything, not even a cup of coffee.

Now I know a number of people who do great work, and it's the same with all of them. They have little discipline. They're all terrible procrastinators and find it almost impossible to make themselves do anything they're not interested in. One still hasn't sent out his half of the thank-you notes from his wedding, four years ago. Another has 26,000 emails in her inbox.

I'm not saying you can get away with zero self-discipline. You probably need about the amount you need to go running. I'm often reluctant to go running, but once I do, I enjoy it. And if I don't run for several days, I feel ill. It's the same with people who do great things. They know they'll feel bad if they don't work, and they have enough discipline to get themselves to their desks to start working. But once they get started, interest takes over, and discipline is no longer necessary.

Do you think Shakespeare was gritting his teeth and diligently trying to write Great Literature? Of course not. He was having fun. That's why he's so good.

If you want to do good work, what you need is a great curiosity about a promising question. The critical moment for Einstein was when he looked at Maxwell's equations and said, what the hell is going on here?

It can take years to zero in on a productive question, because it can take years to figure out what a subject is really about. To take an extreme example, consider math. Most people think they hate math, but the boring stuff you do in school under the name "mathematics" is not at all like what mathematicians do.

The great mathematician G. H. Hardy said he didn't like math in high school either. He only took it up because he was better at it than the other students. Only later did he realize math was interesting — only later did he start to ask questions instead of merely answering them correctly.

When a friend of mine used to grumble because he had to write a paper for school, his mother would tell him: find a way to make it interesting. That's what you need to do: find a question that makes the world interesting. People who do great things look at the same world everyone else does, but notice some odd detail that's compellingly mysterious.

And not only in intellectual matters. Henry Ford's great question was, why do cars have to be a luxury item? What would happen if you treated them as a commodity? Franz Beckenbauer's was, in effect, why does everyone have to stay in his position? Why can't defenders score goals too?

Now

If it takes years to articulate great questions, what do you do now, at sixteen? Work toward finding one. Great questions don't appear suddenly. They gradually congeal in your head. And what makes them congeal is experience. So the way to find great questions is not to search for them — not to wander about thinking, what great discovery shall I make? You can't answer that; if you could, you'd have made it.

The way to get a big idea to appear in your head is not to hunt for big ideas, but to put in a lot of time on work that interests you, and in the process keep your mind open enough that a big idea can take roost. Einstein, Ford, and Beckenbauer all used this recipe. They all knew their work like a piano player knows the keys. So when something seemed amiss to them, they had the confidence to notice it.

Put in time how and on what? Just pick a project that seems interesting: to master some chunk of material, or to make something, or to answer some question. Choose a project that will take less than a month, and make it something you have the means to finish. Do something hard enough to stretch you, but only just, especially at first.

If you're deciding between two projects, choose whichever seems most fun. If one blows up in your face, start another. Repeat till, like an internal combustion engine, the process becomes self-sustaining, and each project generates the next one. (This could take years.)

It may be just as well not to do a project "for school," if that will restrict you or make it seem like work. Involve your friends if you want, but not too many, and only if they're not flakes. Friends offer moral support (few startups are started by one person), but secrecy also has its advantages. There's something pleasing about a secret project. And you can take more risks, because no one will know if you fail.

Don't worry if a project doesn't seem to be on the path to some goal you're supposed to have. Paths can bend a lot more than you think. So let the path grow out the project. The most important thing is to be excited about it, because it's by doing that you learn.

Don't disregard unseemly motivations. One of the most powerful is the desire to be better than other people at something. Hardy said that's what got him started, and I think the only unusual thing about him is that he admitted it. Another powerful motivator is the desire to do, or know, things you're not supposed to. Closely related is the desire to do something audacious. Sixteen year olds aren't supposed to write novels. So if you try, anything you achieve is on the plus side of the ledger; if you fail utterly, you're doing no worse than expectations. [8]

Beware of bad models. Especially when they excuse laziness. When I was in high school I used to write "existentialist" short stories like ones I'd seen by famous writers. My stories didn't have a lot of plot, but they were very deep. And they were less work to write than entertaining ones would have been. I should have known that was a danger sign. And in fact I found my stories pretty boring; what excited me was the

[8] And then of course there is money. It's not a big factor in high school, because you can't do much that anyone wants. But a lot of great things were created mainly to make money. Samuel Johnson said "no man but a blockhead ever wrote except for money." (Many hope he was exaggerating.)

idea of writing serious, intellectual stuff like the famous writers.

Now I have enough experience to realize that those famous writers actually sucked. Plenty of famous people do; in the short term, the quality of one's work is only a small component of fame. I should have been less worried about doing something that seemed cool, and just done something I liked. That's the actual road to coolness anyway.

A key ingredient in many projects, almost a project on its own, is to find good books. Most books are bad. Nearly all textbooks are bad. [9] So don't assume a subject is to be learned from whatever book on it happens to be closest. You have to search actively for the tiny number of good books.

The important thing is to get out there and do stuff. Instead of waiting to be taught, go out and learn.

Your life doesn't have to be shaped by admissions officers. It could be shaped by your own curiosity. It is for all ambitious adults. And you don't have to wait to start. In fact, you don't have to wait to be an adult. There's no switch inside you that magically flips when you turn a certain age or graduate from some institution. You start being an adult when you decide to take responsibility for your life. You can do that at any age. [10]

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[9] Even college textbooks are bad. When you get to college, you'll find that (with a few stellar exceptions) the textbooks are not written by the leading scholars in the field they describe. Writing college textbooks is unpleasant work, done mostly by people who need the money. It's unpleasant because the publishers exert so much control, and there are few things worse than close supervision by someone who doesn't understand what you're doing. This phenomenon is apparently even worse in the production of high school textbooks.

[10] Your teachers are always telling you to behave like adults. I wonder if they'd like it if you did. You may be loud and disorganized, but you're very docile compared to adults. If you actually started acting like adults, it would be just as if a bunch of adults had been transposed into your bodies. Imagine the reaction of an FBI agent or taxi driver or reporter to being told they had to ask permission to go the bathroom, and only one person could go at a time. To say nothing of the things you're taught. If a bunch of actual adults suddenly found themselves trapped in high school, the first thing they'd do is form a union and renegotiate all the rules

This may sound like bullshit. I'm just a minor, you may think, I have no money, I have to live at home, I have to do what adults tell me all day long. Well, most adults labor under restrictions just as cumbersome, and they manage to get things done. If you think it's restrictive being a kid, imagine having kids.

The only real difference between adults and high school kids is that adults realize they need to get things done, and high school kids don't. That realization hits most people around 23. But I'm letting you in on the secret early. So get to work. Maybe you can be the first generation whose greatest regret from high school isn't how much time you wasted.

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with the administration.

Chapter 2

Life is Short

January 2016

Life is short, as everyone knows. When I was a kid I used to wonder about this. Is life actually short, or are we really complaining about its finiteness? Would we be just as likely to feel life was short if we lived 10 times as long?

Since there didn't seem any way to answer this question, I stopped wondering about it. Then I had kids. That gave me a way to answer the question, and the answer is that life actually is short.

Having kids showed me how to convert a continuous quantity, time, into discrete quantities. You only get 52 weekends with your 2 year old. If Christmas-as-magic lasts from say ages 3 to 10, you only get to watch your child experience it 8 times. And while it's impossible to say what is a lot or a little of a continuous quantity like time, 8 is not a lot of something. If you had a handful of 8 peanuts, or a shelf of 8 books to choose from, the quantity would definitely seem limited, no matter what your lifespan was.

Ok, so life actually is short. Does it make any difference to know that?

It has for me. It means arguments of the form "Life is too short for x" have great force. It's not just a figure of speech to say that life is too short for something. It's not just a synonym for annoying. If you find yourself thinking that life is too short for something, you should try to eliminate it if you can.

When I ask myself what I've found life is too short for, the word that pops into my head is "bullshit." I realize that answer is somewhat tautological. It's almost the definition of bullshit that it's the stuff that life is too short for. And yet bullshit does have a distinctive character. There's something fake about it. It's the junk food of experience. [1]

If you ask yourself what you spend your time on that's bullshit, you probably already know the answer. Unnecessary meetings, pointless disputes, bureaucracy, posturing, dealing with other people's mistakes, traffic jams, addictive but unrewarding pastimes.

There are two ways this kind of thing gets into your life: it's either forced on you, or it tricks you. To some extent you have to put up with the bullshit forced on you by circumstances. You need to make money, and making money consists mostly of errands. Indeed, the law of supply and demand ensures that: the more rewarding some kind of work is, the cheaper people will do it. It may be that less bullshit is forced on you than you think, though. There has always been a stream of people who opt out of the default grind and go live somewhere where opportunities are fewer in the conventional sense, but life feels more authentic. This could become more common.

You can do it on a smaller scale without moving. The amount of time you have to spend on bullshit varies between employers. Most large organizations (and many small ones) are steeped in it. But if you consciously prioritize bullshit avoidance over other factors like money and prestige, you can probably find employers that will waste less of your time.

If you're a freelancer or a small company, you can do this at the level of individual customers. If you fire or avoid toxic customers, you can decrease the amount of bullshit in your life by more than you decrease your income.

But while some amount of bullshit is inevitably forced on you, the bullshit that sneaks into your life by tricking you is no one's fault but

[1] At first I didn't like it that the word that came to mind was one that had other meanings. But then I realized the other meanings are fairly closely related. Bullshit in the sense of things you waste your time on is a lot like intellectual bullshit.

your own. And yet the bullshit you choose may be harder to eliminate than the bullshit that's forced on you. Things that lure you into wasting your time have to be really good at tricking you. An example that will be familiar to a lot of people is arguing online. When someone contradicts you, they're in a sense attacking you. Sometimes pretty overtly. Your instinct when attacked is to defend yourself. But like a lot of instincts, this one wasn't designed for the world we now live in. Counterintuitive as it feels, it's better most of the time not to defend yourself. Otherwise these people are literally taking your life. [2]

Arguing online is only incidentally addictive. There are more dangerous things than that. As I've written before, one byproduct of technical progress is that things we like tend to become more addictive. Which means we will increasingly have to make a conscious effort to avoid addictions — to stand outside ourselves and ask "is this how I want to be spending my time?"

As well as avoiding bullshit, one should actively seek out things that matter. But different things matter to different people, and most have to learn what matters to them. A few are lucky and realize early on that they love math or taking care of animals or writing, and then figure out a way to spend a lot of time doing it. But most people start out with a life that's a mix of things that matter and things that don't, and only gradually learn to distinguish between them.

For the young especially, much of this confusion is induced by the artificial situations they find themselves in. In middle school and high school, what the other kids think of you seems the most important thing in the world. But when you ask adults what they got wrong at that age, nearly all say they cared too much what other kids thought of them.

One heuristic for distinguishing stuff that matters is to ask yourself whether you'll care about it in the future. Fake stuff that matters usu-

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[2] I chose this example deliberately as a note to self. I get attacked a lot online. People tell the craziest lies about me. And I have so far done a pretty mediocre job of suppressing the natural human inclination to say "Hey, that's not true!"

ally has a sharp peak of seeming to matter. That's how it tricks you. The area under the curve is small, but its shape jabs into your consciousness like a pin.

The things that matter aren't necessarily the ones people would call "important." Having coffee with a friend matters. You won't feel later like that was a waste of time.

One great thing about having small children is that they make you spend time on things that matter: them. They grab your sleeve as you're staring at your phone and say "will you play with me?" And odds are that is in fact the bullshit-minimizing option.

If life is short, we should expect its shortness to take us by surprise. And that is just what tends to happen. You take things for granted, and then they're gone. You think you can always write that book, or climb that mountain, or whatever, and then you realize the window has closed. The saddest windows close when other people die. Their lives are short too. After my mother died, I wished I'd spent more time with her. I lived as if she'd always be there. And in her typical quiet way she encouraged that illusion. But an illusion it was. I think a lot of people make the same mistake I did.

The usual way to avoid being taken by surprise by something is to be consciously aware of it. Back when life was more precarious, people used to be aware of death to a degree that would now seem a bit morbid. I'm not sure why, but it doesn't seem the right answer to be constantly reminding oneself of the grim reaper hovering at everyone's shoulder. Perhaps a better solution is to look at the problem from the other end. Cultivate a habit of impatience about the things you most want to do. Don't wait before climbing that mountain or writing that book or visiting your mother. You don't need to be constantly reminding yourself why you shouldn't wait. Just don't wait.

I can think of two more things one does when one doesn't have much of something: try to get more of it, and savor what one has. Both make sense here.

How you live affects how long you live. Most people could do better. Me among them.

But you can probably get even more effect by paying closer attention to the time you have. It's easy to let the days rush by. The "flow" that imaginative people love so much has a darker cousin that prevents you from pausing to savor life amid the daily slurry of errands and alarms. One of the most striking things I've read was not in a book, but the title of one: James Salter's *Burning the Days*.

It is possible to slow time somewhat. I've gotten better at it. Kids help. When you have small children, there are a lot of moments so perfect that you can't help noticing.

It does help too to feel that you've squeezed everything out of some experience. The reason I'm sad about my mother is not just that I miss her but that I think of all the things we could have done that we didn't. My oldest son will be 7 soon. And while I miss the 3 year old version of him, I at least don't have any regrets over what might have been. We had the best time a daddy and a 3 year old ever had.

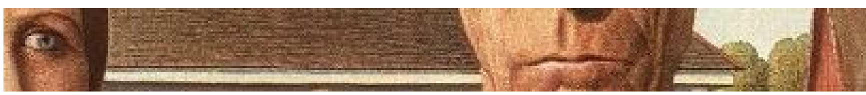
Relentlessly prune bullshit, don't wait to do things that matter, and savor the time you have. That's what you do when life is short.

Thanks to Jessica Livingston and Geoff Ralston for reading drafts of this.

Philosophy

Chapter 3

What You Can't Say



January 2004

Have you ever seen an old photo of yourself and been embarrassed at the way you looked? *Did we actually dress like that?* We did. And we had no idea how silly we looked. It's the nature of fashion to be invisible, in the same way the movement of the earth is invisible to all of us riding on it.

What scares me is that there are moral fashions too. They're just as arbitrary, and just as invisible to most people. But they're much more dangerous. Fashion is mistaken for good design; moral fashion is mistaken for good. Dressing oddly gets you laughed at. Violating moral fashions can get you fired, ostracized, imprisoned, or even killed.

If you could travel back in a time machine, one thing would be true no matter where you went: you'd have to watch what you said. Opinions we consider harmless could have gotten you in big trouble. I've already said at least one thing that would have gotten me in big trouble in most of Europe in the seventeenth century, and did get Galileo in big trouble when he said it — that the earth moves. [1]

It seems to be a constant throughout history: In every period, people

[1]The Inquisition probably never intended to carry out their threat of torture. But that was because Galileo made it clear that he would do whatever they asked. If he had refused, it's hard to imagine they would simply have backed down. Not long before they had burnt the philosopher Giordano Bruno when he proved intransigent.

believed things that were just ridiculous, and believed them so strongly that you would have gotten in terrible trouble for saying otherwise.

Is our time any different? To anyone who has read any amount of history, the answer is almost certainly no. It would be a remarkable coincidence if ours were the first era to get everything just right.

It's tantalizing to think we believe things that people in the future will find ridiculous. What *would* someone coming back to visit us in a time machine have to be careful not to say? That's what I want to study here. But I want to do more than just shock everyone with the heresy du jour. I want to find general recipes for discovering what you can't say, in any era.

The Conformist Test

Let's start with a test: Do you have any opinions that you would be reluctant to express in front of a group of your peers?

If the answer is no, you might want to stop and think about that. If everything you believe is something you're supposed to believe, could that possibly be a coincidence? Odds are it isn't. Odds are you just think what you're told.

The other alternative would be that you independently considered every question and came up with the exact same answers that are now considered acceptable. That seems unlikely, because you'd also have to make the same mistakes. Mapmakers deliberately put slight mistakes in their maps so they can tell when someone copies them. If another map has the same mistake, that's very convincing evidence.

Like every other era in history, our moral map almost certainly contains a few mistakes. And anyone who makes the same mistakes probably didn't do it by accident. It would be like someone claiming they had independently decided in 1972 that bell-bottom jeans were a good idea.

If you believe everything you're supposed to now, how can you be sure you wouldn't also have believed everything you were supposed to if you had grown up among the plantation owners of the pre-Civil War South, or in Germany in the 1930s — or among the Mongols in 1200, for that matter? Odds are you would have.

Back in the era of terms like "well-adjusted," the idea seemed to be that there was something wrong with you if you thought things you didn't dare say out loud. This seems backward. Almost certainly, there is something wrong with you if you *don't* think things you don't dare say out loud.

Trouble

What can't we say? One way to find these ideas is simply to look at things people do say, and get in trouble for. [2]

Of course, we're not just looking for things we can't say. We're looking for things we can't say that are true, or at least have enough chance of being true that the question should remain open. But many of the things people get in trouble for saying probably do make it over this second, lower threshold. No one gets in trouble for saying that 2 + 2 is 5, or that people in Pittsburgh are ten feet tall. Such obviously false statements might be treated as jokes, or at worst as evidence of insanity, but they are not likely to make anyone mad. The statements that make people mad are the ones they worry might be believed. I suspect the statements that make people maddest are those they worry might be true.

If Galileo had said that people in Padua were ten feet tall, he would have been regarded as a harmless eccentric. Saying the earth orbited the sun was another matter. The church knew this would set people thinking.

Certainly, as we look back on the past, this rule of thumb works well.

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[2] Many organizations obligingly publish lists of what you can't say within them. Unfortunately these are usually both incomplete, because there are things so shocking they don't even anticipate anyone saying them, and at the same time so general that they couldn't possibly be enforced literally. It's a rare university speech code that would not, taken literally, forbid Shakespeare.

A lot of the statements people got in trouble for seem harmless now. So it's likely that visitors from the future would agree with at least some of the statements that get people in trouble today. Do we have no Galileos? Not likely.

To find them, keep track of opinions that get people in trouble, and start asking, could this be true? Ok, it may be heretical (or whatever modern equivalent), but might it also be true?

Heresy

This won't get us all the answers, though. What if no one happens to have gotten in trouble for a particular idea yet? What if some idea would be so radioactively controversial that no one would dare express it in public? How can we find these too?

Another approach is to follow that word, heresy. In every period of history, there seem to have been labels that got applied to statements to shoot them down before anyone had a chance to ask if they were true or not. "Blasphemy", "sacrilege", and "heresy" were such labels for a good part of western history, as in more recent times "indecent", "improper", and "unamerican" have been. By now these labels have lost their sting. They always do. By now they're mostly used ironically. But in their time, they had real force.

The word "defeatist", for example, has no particular political connotations now. But in Germany in 1917 it was a weapon, used by Ludendorff in a purge of those who favored a negotiated peace. At the start of World War II it was used extensively by Churchill and his supporters to silence their opponents. In 1940, any argument against Churchill's aggressive policy was "defeatist". Was it right or wrong? Ideally, no one got far enough to ask that.

We have such labels today, of course, quite a lot of them, from the all-purpose "inappropriate" to the dreaded "divisive." In any period, it should be easy to figure out what such labels are, simply by looking at what people call ideas they disagree with besides untrue. When a politician says his opponent is mistaken, that's a straightforward criticism, but when he attacks a statement as "divisive" or "racially insensitive" instead of arguing that it's false, we should start paying attention.

So another way to figure out which of our taboos future generations will laugh at is to start with the labels. Take a label — "sexist", for example — and try to think of some ideas that would be called that. Then for each ask, might this be true?

Just start listing ideas at random? Yes, because they won't really be random. The ideas that come to mind first will be the most plausible ones. They'll be things you've already noticed but didn't let yourself think.

In 1989 some clever researchers tracked the eye movements of radiologists as they scanned chest images for signs of lung cancer. [3] They found that even when the radiologists missed a cancerous lesion, their eyes had usually paused at the site of it. Part of their brain knew there was something there; it just didn't percolate all the way up into conscious knowledge. I think many interesting heretical thoughts are already mostly formed in our minds. If we turn off our self-censorship temporarily, those will be the first to emerge.

Time and Space

If we could look into the future it would be obvious which of our taboos they'd laugh at. We can't do that, but we can do something almost as good: we can look into the past. Another way to figure out what we're getting wrong is to look at what used to be acceptable and is now unthinkable.

Changes between the past and the present sometimes do represent progress. In a field like physics, if we disagree with past generations it's because we're right and they're wrong. But this becomes rapidly less true as you move away from the certainty of the hard sciences. By the time you get to social questions, many changes are just fashion.

[3] Kundel HL, Nodine CF, Krupinski EA, "Searching for lung nodules: Visual dwell indicates locations of false-positive and false-negative decisions." *Investigative Radiology,* 24 (1989), 472-478.

The age of consent fluctuates like hemlines.

We may imagine that we are a great deal smarter and more virtuous than past generations, but the more history you read, the less likely this seems. People in past times were much like us. Not heroes, not barbarians. Whatever their ideas were, they were ideas reasonable people could believe.

So here is another source of interesting heresies. Diff present ideas against those of various past cultures, and see what you get. [4] Some will be shocking by present standards. Ok, fine; but which might also be true?

You don't have to look into the past to find big differences. In our own time, different societies have wildly varying ideas of what's ok and what isn't. So you can try diffing other cultures' ideas against ours as well. (The best way to do that is to visit them.) Any idea that's considered harmless in a significant percentage of times and places, and yet is taboo in ours, is a candidate for something we're mistaken about.

For example, at the high water mark of political correctness in the early 1990s, Harvard distributed to its faculty and staff a brochure saying, among other things, that it was inappropriate to compliment a colleague or student's clothes. No more "nice shirt." I think this principle is rare among the world's cultures, past or present. There are probably more where it's considered especially polite to compliment someone's clothing than where it's considered improper. Odds are this is, in a mild form, an example of one of the taboos a visitor from the future would have to be careful to avoid if he happened to set his time machine for Cambridge, Massachusetts, 1992. [5]

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[4]The verb "diff" is computer jargon, but it's the only word with exactly the sense I want. It comes from the Unix diff utility, which yields a list of all the differences between two files. More generally it means an unselective and microscopically thorough comparison between two versions of something.

[5]It may seem from this that I am some kind of moral relativist. Far from it. I think that "judgemental" is one of the labels that gets used in our time to prevent

Prigs

Of course, if they have time machines in the future they'll probably have a separate reference manual just for Cambridge. This has always been a fussy place, a town of i dotters and t crossers, where you're liable to get both your grammar and your ideas corrected in the same conversation. And that suggests another way to find taboos. Look for prigs, and see what's inside their heads.

Kids' heads are repositories of all our taboos. It seems fitting to us that kids' ideas should be bright and clean. The picture we give them of the world is not merely simplified, to suit their developing minds, but sanitized as well, to suit our ideas of what kids ought to think. [6]

You can see this on a small scale in the matter of dirty words. A lot of my friends are starting to have children now, and they're all trying not to use words like "fuck" and "shit" within baby's hearing, lest baby start using these words too. But these words are part of the language, and adults use them all the time. So parents are giving their kids an inaccurate idea of the language by not using them. Why do they do this? Because they don't think it's fitting that kids should use the whole language. We like children to seem innocent. [7]

Most adults, likewise, deliberately give kids a misleading view of the world. One of the most obvious examples is Santa Claus. We think it's cute for little kids to believe in Santa Claus. I myself think it's cute for little kids to believe in Santa Claus. But one wonders, do we tell

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discussion of ideas, and that our attempts to be "non-judgemental" will seem to future eras one of the most comical things about us.

[6]This makes the world confusing to kids, since what they see disagrees with what they're told. I could never understand *why,* for example, Portuguese "explorers" had started to work their way along the coast of Africa. In fact, they were after slaves.

Bovill, Edward, *The Golden Trade of the Moors,* Oxford, 1963.

[7]The kids soon learn these words from their friends, but they know they're not supposed to use them. So for a while you have a state of affairs like something from a musical comedy, where the parents use these words among their peers, but never in front of the children, and the children use the words among their peers, but never in front of their parents.

them this stuff for their sake, or for ours?

I'm not arguing for or against this idea here. It is probably inevitable that parents should want to dress up their kids' minds in cute little baby outfits. I'll probably do it myself. The important thing for our purposes is that, as a result, a well brought-up teenage kid's brain is a more or less complete collection of all our taboos — and in mint condition, because they're untainted by experience. Whatever we think that will later turn out to be ridiculous, it's almost certainly inside that head.

How do we get at these ideas? By the following thought experiment. Imagine a kind of latter-day Conrad character who has worked for a time as a mercenary in Africa, for a time as a doctor in Nepal, for a time as the manager of a nightclub in Miami. The specifics don't matter — just someone who has seen a lot. Now imagine comparing what's inside this guy's head with what's inside the head of a well-behaved sixteen year old girl from the suburbs. What does he think that would shock her? He knows the world; she knows, or at least embodies, present taboos. Subtract one from the other, and the result is what we can't say.

Mechanism

I can think of one more way to figure out what we can't say: to look at how taboos are created. How do moral fashions arise, and why are they adopted? If we can understand this mechanism, we may be able to see it at work in our own time.

Moral fashions don't seem to be created the way ordinary fashions are. Ordinary fashions seem to arise by accident when everyone imitates the whim of some influential person. The fashion for broad-toed shoes in late fifteenth century Europe began because Charles VIII of France had six toes on one foot. The fashion for the name Gary began when the actor Frank Cooper adopted the name of a tough mill town in Indiana. Moral fashions more often seem to be created deliberately. When there's something we can't say, it's often because some group doesn't want us to.

The prohibition will be strongest when the group is nervous. The irony of Galileo's situation was that he got in trouble for repeating Copernicus's ideas. Copernicus himself didn't. In fact, Copernicus was a canon of a cathedral, and dedicated his book to the pope. But by Galileo's time the church was in the throes of the Counter-Reformation and was much more worried about unorthodox ideas.

To launch a taboo, a group has to be poised halfway between weakness and power. A confident group doesn't need taboos to protect it. It's not considered improper to make disparaging remarks about Americans, or the English. And yet a group has to be powerful enough to enforce a taboo. Coprophiles, as of this writing, don't seem to be numerous or energetic enough to have had their interests promoted to a lifestyle.

I suspect the biggest source of moral taboos will turn out to be power struggles in which one side only barely has the upper hand. That's where you'll find a group powerful enough to enforce taboos, but weak enough to need them.

Most struggles, whatever they're really about, will be cast as struggles between competing ideas. The English Reformation was at bottom a struggle for wealth and power, but it ended up being cast as a struggle to preserve the souls of Englishmen from the corrupting influence of Rome. It's easier to get people to fight for an idea. And whichever side wins, their ideas will also be considered to have triumphed, as if God wanted to signal his agreement by selecting that side as the victor.

We often like to think of World War II as a triumph of freedom over totalitarianism. We conveniently forget that the Soviet Union was also one of the winners.

I'm not saying that struggles are never about ideas, just that they will always be made to seem to be about ideas, whether they are or not. And just as there is nothing so unfashionable as the last, discarded fashion, there is nothing so wrong as the principles of the most recently defeated opponent. Representational art is only now recovering from the approval of both Hitler and Stalin. [8]

Although moral fashions tend to arise from different sources than fashions in clothing, the mechanism of their adoption seems much the same. The early adopters will be driven by ambition: self-consciously cool people who want to distinguish themselves from the common herd. As the fashion becomes established they'll be joined by a second, much larger group, driven by fear. [9] This second group adopt the fashion not because they want to stand out but because they are afraid of standing out.

So if you want to figure out what we can't say, look at the machinery of fashion and try to predict what it would make unsayable. What groups are powerful but nervous, and what ideas would they like to suppress? What ideas were tarnished by association when they ended up on the losing side of a recent struggle? If a self-consciously cool person wanted to differentiate himself from preceding fashions (e.g. from his parents), which of their ideas would he tend to reject? What are conventional-minded people afraid of saying?

This technique won't find us all the things we can't say. I can think of some that aren't the result of any recent struggle. Many of our taboos are rooted deep in the past. But this approach, combined with the preceding four, will turn up a good number of unthinkable ideas.

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[8] A few years ago I worked for a startup whose logo was a solid red circle with a white V in the middle. I really liked this logo. After we'd been using it for a while, I remember thinking, you know, this is a really powerful symbol, a red circle. Red is arguably the most basic color, and the circle the most basic shape. Together they had such visual punch. Why didn't more American companies have a red circle as their logo? Ahh, yes…

[9] The fear is far the stronger of the two forces. Sometimes when I hear someone use the word "gyp" I tell them, with a serious expression, that one can't use that word anymore because it's considered disparaging to Romani (aka Gypsies). In fact dictionaries disagree about its etymology. But the reaction to this joke is nearly always one of slightly terrified compliance. There is something about fashion, in clothing or ideas, that takes away people's confidence: when they learn something new, they feel it was something they should have known already.

Why

Some would ask, why would one want to do this? Why deliberately go poking around among nasty, disreputable ideas? Why look under rocks?

I do it, first of all, for the same reason I did look under rocks as a kid: plain curiosity. And I'm especially curious about anything that's forbidden. Let me see and decide for myself.

Second, I do it because I don't like the idea of being mistaken. If, like other eras, we believe things that will later seem ridiculous, I want to know what they are so that I, at least, can avoid believing them.

Third, I do it because it's good for the brain. To do good work you need a brain that can go anywhere. And you especially need a brain that's in the habit of going where it's not supposed to.

Great work tends to grow out of ideas that others have overlooked, and no idea is so overlooked as one that's unthinkable. Natural selection, for example. It's so simple. Why didn't anyone think of it before? Well, that is all too obvious. Darwin himself was careful to tiptoe around the implications of his theory. He wanted to spend his time thinking about biology, not arguing with people who accused him of being an atheist.

In the sciences, especially, it's a great advantage to be able to question assumptions. The m.o. of scientists, or at least of the good ones, is precisely that: look for places where conventional wisdom is broken, and then try to pry apart the cracks and see what's underneath. That's where new theories come from.

A good scientist, in other words, does not merely ignore conventional wisdom, but makes a special effort to break it. Scientists go looking for trouble. This should be the m.o. of any scholar, but scientists seem much more willing to look under rocks. [10]

[10] I don't mean to suggest that scientists' opinions are inevitably right, just that their willingness to consider unconventional ideas gives them a head start. In other

Why? It could be that the scientists are simply smarter; most physicists could, if necessary, make it through a PhD program in French literature, but few professors of French literature could make it through a PhD program in physics. Or it could be because it's clearer in the sciences whether theories are true or false, and this makes scientists bolder. (Or it could be that, because it's clearer in the sciences whether theories are true or false, you have to be smart to get jobs as a scientist, rather than just a good politician.)

Whatever the reason, there seems a clear correlation between intelligence and willingness to consider shocking ideas. This isn't just because smart people actively work to find holes in conventional thinking. I think conventions also have less hold over them to start with. You can see that in the way they dress.

It's not only in the sciences that heresy pays off. In any competitive field, you can win big by seeing things that others daren't. And in every field there are probably heresies few dare utter. Within the US car industry there is a lot of hand-wringing now about declining market share. Yet the cause is so obvious that any observant outsider could explain it in a second: they make bad cars. And they have for so long that by now the US car brands are antibrands — something you'd buy a car despite, not because of. Cadillac stopped being the Cadillac of cars in about 1970. And yet I suspect no one dares say this. [11]

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respects they are sometimes at a disadvantage. Like other scholars, many scientists have never directly earned a living — never, that is, been paid in return for services rendered. Most scholars live in an anomalous microworld in which money is something doled out by committees instead of a representation for work, and it seems natural to them that national economies should be run along the same lines. As a result, many otherwise intelligent people were socialists in the middle of the twentieth century.

[11] Presumably, within the industry, such thoughts would be considered "negative". Another label, much like "defeatist". Never mind that, one should ask, are they true or not? Indeed, the measure of a healthy organization is probably the degree to which negative thoughts are allowed. In places where great work is being done, the attitude always seems to be critical and sarcastic, not "positive" and "supportive". The people I know who do great work think that they suck, but that everyone else

Otherwise these companies would have tried to fix the problem.

Training yourself to think unthinkable thoughts has advantages beyond the thoughts themselves. It's like stretching. When you stretch before running, you put your body into positions much more extreme than any it will assume during the run. If you can think things so outside the box that they'd make people's hair stand on end, you'll have no trouble with the small trips outside the box that people call innovative.

Pensieri Stretti

When you find something you can't say, what do you do with it? My advice is, don't say it. Or at least, pick your battles.

Suppose in the future there is a movement to ban the color yellow. Proposals to paint anything yellow are denounced as "yellowist", as is anyone suspected of liking the color. People who like orange are tolerated but viewed with suspicion. Suppose you realize there is nothing wrong with yellow. If you go around saying this, you'll be denounced as a yellowist too, and you'll find yourself having a lot of arguments with anti-yellowists. If your aim in life is to rehabilitate the color yellow, that may be what you want. But if you're mostly interested in other questions, being labelled as a yellowist will just be a distraction. Argue with idiots, and you become an idiot.

The most important thing is to be able to think what you want, not to say what you want. And if you feel you have to say everything you think, it may inhibit you from thinking improper thoughts. I think it's better to follow the opposite policy. Draw a sharp line between your thoughts and your speech. Inside your head, anything is allowed. Within my head I make a point of encouraging the most outrageous thoughts I can imagine. But, as in a secret society, nothing that happens within the building should be told to outsiders. The first rule of Fight Club is, you do not talk about Fight Club.

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sucks even more.

When Milton was going to visit Italy in the 1630s, Sir Henry Wootton, who had been ambassador to Venice, told him his motto should be *"i pensieri stretti & il viso sciolto."* Closed thoughts and an open face. Smile at everyone, and don't tell them what you're thinking. This was wise advice. Milton was an argumentative fellow, and the Inquisition was a bit restive at that time. But I think the difference between Milton's situation and ours is only a matter of degree. Every era has its heresies, and if you don't get imprisoned for them you will at least get in enough trouble that it becomes a complete distraction.

I admit it seems cowardly to keep quiet. When I read about the harassment to which the Scientologists subject their critics [12], or that pro-Israel groups are "compiling dossiers" on those who speak out against Israeli human rights abuses [13], or about people being sued for violating the DMCA [14], part of me wants to say, "All right, you bastards, bring it on." The problem is, there are so many things you can't say. If you said them all you'd have no time left for your real work. You'd have to turn into Noam Chomsky. [15]

The trouble with keeping your thoughts secret, though, is that you lose the advantages of discussion. Talking about an idea leads to more ideas. So the optimal plan, if you can manage it, is to have a few trusted friends you can speak openly to. This is not just a way to develop ideas; it's also a good rule of thumb for choosing friends. The people you can say heretical things to without getting jumped on are also the most interesting to know.

Viso Sciolto?

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[12] Behar, Richard, "The Thriving Cult of Greed and Power," *Time*, 6 May 1991.

[13] Healy, Patrick, "Summers hits 'anti-Semitic' actions," *Boston Globe*, 20 September 2002.

[14] "Tinkerers' champion," *The Economist*, 20 June 2002.

[15] By this I mean you'd have to become a professional controversialist, not that Noam Chomsky's opinions = what you can't say. If you actually said the things you can't say, you'd shock conservatives and liberals about equally—just as, if you went back to Victorian England in a time machine, your ideas would shock Whigs and Tories about equally.

I don't think we need the *viso sciolto* so much as the *pensieri stretti.* Perhaps the best policy is to make it plain that you don't agree with whatever zealotry is current in your time, but not to be too specific about what you disagree with. Zealots will try to draw you out, but you don't have to answer them. If they try to force you to treat a question on their terms by asking "are you with us or against us?" you can always just answer "neither".

Better still, answer "I haven't decided." That's what Larry Summers did when a group tried to put him in this position. Explaining himself later, he said "I don't do litmus tests." [16] A lot of the questions people get hot about are actually quite complicated. There is no prize for getting the answer quickly.

If the anti-yellowists seem to be getting out of hand and you want to fight back, there are ways to do it without getting yourself accused of being a yellowist. Like skirmishers in an ancient army, you want to avoid directly engaging the main body of the enemy's troops. Better to harass them with arrows from a distance.

One way to do this is to ratchet the debate up one level of abstraction. If you argue against censorship in general, you can avoid being accused of whatever heresy is contained in the book or film that someone is trying to censor. You can attack labels with meta-labels: labels that refer to the use of labels to prevent discussion. The spread of the term "political correctness" meant the beginning of the end of political correctness, because it enabled one to attack the phenomenon as a whole without being accused of any of the specific heresies it sought to suppress.

Another way to counterattack is with metaphor. Arthur Miller undermined the House Un-American Activities Committee by writing a play, "The Crucible," about the Salem witch trials. He never referred directly to the committee and so gave them no way to reply. What could HUAC do, defend the Salem witch trials? And yet Miller's

[16] Traub, James, "Harvard Radical," *New York Times Magazine,* 24 August 2003.

metaphor stuck so well that to this day the activities of the committee are often described as a "witch-hunt."

Best of all, probably, is humor. Zealots, whatever their cause, invariably lack a sense of humor. They can't reply in kind to jokes. They're as unhappy on the territory of humor as a mounted knight on a skating rink. Victorian prudishness, for example, seems to have been defeated mainly by treating it as a joke. Likewise its reincarnation as political correctness. "I am glad that I managed to write 'The Crucible,'" Arthur Miller wrote, "but looking back I have often wished I'd had the temperament to do an absurd comedy, which is what the situation deserved." [17]

ABQ

A Dutch friend says I should use Holland as an example of a tolerant society. It's true they have a long tradition of comparative open-mindedness. For centuries the low countries were the place to go to say things you couldn't say anywhere else, and this helped to make the region a center of scholarship and industry (which have been closely tied for longer than most people realize). Descartes, though claimed by the French, did much of his thinking in Holland.

And yet, I wonder. The Dutch seem to live their lives up to their necks in rules and regulations. There's so much you can't do there; is there really nothing you can't say?

Certainly the fact that they value open-mindedness is no guarantee. Who thinks they're not open-minded? Our hypothetical prim miss from the suburbs thinks she's open-minded. Hasn't she been taught to be? Ask anyone, and they'll say the same thing: they're pretty open-minded, though they draw the line at things that are really wrong. (Some tribes may avoid "wrong" as judgemental, and may instead use a more neutral sounding euphemism like "negative" or "destructive".)

When people are bad at math, they know it, because they get the

[17] Miller, Arthur, *The Crucible in History and Other Essays,* Methuen, 2000.

wrong answers on tests. But when people are bad at open-mindedness they don't know it. In fact they tend to think the opposite. Remember, it's the nature of fashion to be invisible. It wouldn't work otherwise. Fashion doesn't seem like fashion to someone in the grip of it. It just seems like the right thing to do. It's only by looking from a distance that we see oscillations in people's idea of the right thing to do, and can identify them as fashions.

Time gives us such distance for free. Indeed, the arrival of new fashions makes old fashions easy to see, because they seem so ridiculous by contrast. From one end of a pendulum's swing, the other end seems especially far away.

To see fashion in your own time, though, requires a conscious effort. Without time to give you distance, you have to create distance yourself. Instead of being part of the mob, stand as far away from it as you can and watch what it's doing. And pay especially close attention whenever an idea is being suppressed. Web filters for children and employees often ban sites containing pornography, violence, and hate speech. What counts as pornography and violence? And what, exactly, is "hate speech?" This sounds like a phrase out of *1984*.

Labels like that are probably the biggest external clue. If a statement is false, that's the worst thing you can say about it. You don't need to say that it's heretical. And if it isn't false, it shouldn't be suppressed. So when you see statements being attacked as x-ist or y-ic (substitute your current values of x and y), whether in 1630 or 2030, that's a sure sign that something is wrong. When you hear such labels being used, ask why.

Especially if you hear yourself using them. It's not just the mob you need to learn to watch from a distance. You need to be able to watch your own thoughts from a distance. That's not a radical idea, by the way; it's the main difference between children and adults. When a child gets angry because he's tired, he doesn't know what's happening. An adult can distance himself enough from the situation to say "never mind, I'm just tired." I don't see why one couldn't, by a similar process, learn to recognize and discount the effects of moral fashions.

You have to take that extra step if you want to think clearly. But it's harder, because now you're working against social customs instead of with them. Everyone encourages you to grow up to the point where you can discount your own bad moods. Few encourage you to continue to the point where you can discount society's bad moods.

How can you see the wave, when you're the water? Always be questioning. That's the only defence. What can't you say? And why?

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Economics

Chapter 4

How to Make Wealth

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If you wanted to get rich, how would you do it? I think your best bet would be to start or join a startup. That's been a reliable way to get rich for hundreds of years. The word "startup" dates from the 1960s, but what happens in one is very similar to the venture-backed trading voyages of the Middle Ages.

Startups usually involve technology, so much so that the phrase "high-tech startup" is almost redundant. A startup is a small company that takes on a hard technical problem.

Lots of people get rich knowing nothing more than that. You don't have to know physics to be a good pitcher. But I think it could give you an edge to understand the underlying principles. Why do startups have to be small? Will a startup inevitably stop being a startup as it grows larger? And why do they so often work on developing new technology? Why are there so many startups selling new drugs or computer software, and none selling corn oil or laundry detergent?

The Proposition

Economically, you can think of a startup as a way to compress your whole working life into a few years. Instead of working at a low intensity for forty years, you work as hard as you possibly can for four. This pays especially well in technology, where you earn a premium for working fast.

Here is a brief sketch of the economic proposition. If you're a good hacker in your mid twenties, you can get a job paying about $80,000 per year. So on average such a hacker must be able to do at least $80,000 worth of work per year for the company just to break even. You could probably work twice as many hours as a corporate employee, and if you focus you can probably get three times as much done in an hour. [1] You should get another multiple of two, at least, by eliminating the drag of the pointy-haired middle manager who would be your boss in a big company. Then there is one more multiple: how much smarter are you than your job description expects you to be? Suppose another multiple of three. Combine all these multipliers, and I'm claiming you could be 36 times more productive than you're expected to be in a random corporate job. [2] If a fairly good hacker is worth $80,000 a year at a big company, then a smart hacker working very hard without any corporate bullshit to slow him down should be able to do work worth about $3 million a year.

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[1] One valuable thing you tend to get only in startups is *uninterruptability*. Different kinds of work have different time quanta. Someone proofreading a manuscript could probably be interrupted every fifteen minutes with little loss of productivity. But the time quantum for hacking is very long: it might take an hour just to load a problem into your head. So the cost of having someone from personnel call you about a form you forgot to fill out can be huge.

This is why hackers give you such a baleful stare as they turn from their screen to answer your question. Inside their heads a giant house of cards is tottering.

The mere possibility of being interrupted deters hackers from starting hard projects. This is why they tend to work late at night, and why it's next to impossible to write great software in a cubicle (except late at night).

One great advantage of startups is that they don't yet have any of the people who interrupt you. There is no personnel department, and thus no form nor anyone to call you about it.

[2] Faced with the idea that people working for startups might be 20 or 30 times as productive as those working for large companies, executives at large companies will naturally wonder, how could I get the people working for me to do that? The answer is simple: pay them to.

Internally most companies are run like Communist states. If you believe in free markets, why not turn your company into one?

Hypothesis: A company will be maximally profitable when each employee is paid in proportion to the wealth they generate.

Like all back-of-the-envelope calculations, this one has a lot of wiggle room. I wouldn't try to defend the actual numbers. But I stand by the structure of the calculation. I'm not claiming the multiplier is precisely 36, but it is certainly more than 10, and probably rarely as high as 100.

If $3 million a year seems high, remember that we're talking about the limit case: the case where you not only have zero leisure time but indeed work so hard that you endanger your health.

Startups are not magic. They don't change the laws of wealth creation. They just represent a point at the far end of the curve. There is a conservation law at work here: if you want to make a million dollars, you have to endure a million dollars' worth of pain. For example, one way to make a million dollars would be to work for the Post Office your whole life, and save every penny of your salary. Imagine the stress of working for the Post Office for fifty years. In a startup you compress all this stress into three or four years. You do tend to get a certain bulk discount if you buy the economy-size pain, but you can't evade the fundamental conservation law. If starting a startup were easy, everyone would do it.

Millions, not Billions

If $3 million a year seems high to some people, it will seem low to others. Three *million?* How do I get to be a billionaire, like Bill Gates?

So let's get Bill Gates out of the way right now. It's not a good idea to use famous rich people as examples, because the press only write about the very richest, and these tend to be outliers. Bill Gates is a smart, determined, and hardworking man, but you need more than that to make as much money as he has. You also need to be very lucky.

There is a large random factor in the success of any company. So the guys you end up reading about in the papers are the ones who are very smart, totally dedicated, *and* win the lottery. Certainly Bill is smart and dedicated, but Microsoft also happens to have been the beneficiary of one of the most spectacular blunders in the history of business: the licensing deal for DOS. No doubt Bill did everything he could to steer IBM into making that blunder, and he has done an excellent job of exploiting it, but if there had been one person with a brain on IBM's side, Microsoft's future would have been very different. Microsoft at that stage had little leverage over IBM. They were effectively a component supplier. If IBM had required an exclusive license, as they should have, Microsoft would still have signed the deal. It would still have meant a lot of money for them, and IBM could easily have gotten an operating system elsewhere.

Instead IBM ended up using all its power in the market to give Microsoft control of the PC standard. From that point, all Microsoft had to do was execute. They never had to bet the company on a bold decision. All they had to do was play hardball with licensees and copy more innovative products reasonably promptly.

If IBM hadn't made this mistake, Microsoft would still have been a successful company, but it could not have grown so big so fast. Bill Gates would be rich, but he'd be somewhere near the bottom of the Forbes 400 with the other guys his age.

There are a lot of ways to get rich, and this essay is about only one of them. This essay is about how to make money by creating wealth and getting paid for it. There are plenty of other ways to get money, including chance, speculation, marriage, inheritance, theft, extortion, fraud, monopoly, graft, lobbying, counterfeiting, and prospecting. Most of the greatest fortunes have probably involved several of these.

The advantage of creating wealth, as a way to get rich, is not just that it's more legitimate (many of the other methods are now illegal) but that it's more *straightforward*. You just have to do something people want.

Money Is Not Wealth

If you want to create wealth, it will help to understand what it is.

Wealth is not the same thing as money. [3] Wealth is as old as human history. Far older, in fact; ants have wealth. Money is a comparatively recent invention.

Wealth is the fundamental thing. Wealth is stuff we want: food, clothes, houses, cars, gadgets, travel to interesting places, and so on. You can have wealth without having money. If you had a magic machine that could on command make you a car or cook you dinner or do your laundry, or do anything else you wanted, you wouldn't need money. Whereas if you were in the middle of Antarctica, where there is nothing to buy, it wouldn't matter how much money you had.

Wealth is what you want, not money. But if wealth is the important thing, why does everyone talk about making money? It is a kind of shorthand: money is a way of moving wealth, and in practice they are usually interchangeable. But they are not the same thing, and unless you plan to get rich by counterfeiting, talking about *making money* can make it harder to understand how to make money.

Money is a side effect of specialization. In a specialized society, most of the things you need, you can't make for yourself. If you want a potato or a pencil or a place to live, you have to get it from someone else.

How do you get the person who grows the potatoes to give you some? By giving him something he wants in return. But you can't get very far by trading things directly with the people who need them. If you make violins, and none of the local farmers wants one, how will you eat?

The solution societies find, as they get more specialized, is to make the trade into a two-step process. Instead of trading violins directly

[3] Until recently even governments sometimes didn't grasp the distinction between money and wealth. Adam Smith (*Wealth of Nations*, v:i) mentions several that tried to preserve their "wealth" by forbidding the export of gold or silver. But having more of the medium of exchange would not make a country richer; if you have more money chasing the same amount of material wealth, the only result is higher prices.

for potatoes, you trade violins for, say, silver, which you can then trade again for anything else you need. The intermediate stuff– the *medium of exchange*-- can be anything that's rare and portable. Historically metals have been the most common, but recently we've been using a medium of exchange, called the *dollar*, that doesn't physically exist. It works as a medium of exchange, however, because its rarity is guaranteed by the U.S. Government.

The advantage of a medium of exchange is that it makes trade work. The disadvantage is that it tends to obscure what trade really means. People think that what a business does is make money. But money is just the intermediate stage– just a shorthand– for whatever people want. What most businesses really do is make wealth. They do something people want. [4]

The Pie Fallacy

A surprising number of people retain from childhood the idea that there is a fixed amount of wealth in the world. There is, in any normal family, a fixed amount of *money* at any moment. But that's not the same thing.

When wealth is talked about in this context, it is often described as a pie. "You can't make the pie larger," say politicians. When you're

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[4] There are many senses of the word "wealth," not all of them material. I'm not trying to make a deep philosophical point here about which is the true kind. I'm writing about one specific, rather technical sense of the word "wealth." What people will give you money for. This is an interesting sort of wealth to study, because it is the kind that prevents you from starving. And what people will give you money for depends on them, not you.

When you're starting a business, it's easy to slide into thinking that customers want what you do. During the Internet Bubble I talked to a woman who, because she liked the outdoors, was starting an "outdoor portal." You know what kind of business you should start if you like the outdoors? One to recover data from crashed hard disks.

What's the connection? None at all. Which is precisely my point. If you want to create wealth (in the narrow technical sense of not starving) then you should be especially skeptical about any plan that centers on things you like doing. That is where your idea of what's valuable is least likely to coincide with other people's.

talking about the amount of money in one family's bank account, or the amount available to a government from one year's tax revenue, this is true. If one person gets more, someone else has to get less.

I can remember believing, as a child, that if a few rich people had all the money, it left less for everyone else. Many people seem to continue to believe something like this well into adulthood. This fallacy is usually there in the background when you hear someone talking about how x percent of the population have y percent of the wealth. If you plan to start a startup, then whether you realize it or not, you're planning to disprove the Pie Fallacy.

What leads people astray here is the abstraction of money. Money is not wealth. It's just something we use to move wealth around. So although there may be, in certain specific moments (like your family, this month) a fixed amount of money available to trade with other people for things you want, there is not a fixed amount of wealth in the world. *You can make more wealth.* Wealth has been getting created and destroyed (but on balance, created) for all of human history.

Suppose you own a beat-up old car. Instead of sitting on your butt next summer, you could spend the time restoring your car to pristine condition. In doing so you create wealth. The world is– and you specifically are– one pristine old car the richer. And not just in some metaphorical way. If you sell your car, you'll get more for it.

In restoring your old car you have made yourself richer. You haven't made anyone else poorer. So there is obviously not a fixed pie. And in fact, when you look at it this way, you wonder why anyone would think there was. [5]

Kids know, without knowing they know, that they can create wealth.

[5] In the average car restoration you probably do make everyone else microscopically poorer, by doing a small amount of damage to the environment. While environmental costs should be taken into account, they don't make wealth a zero-sum game. For example, if you repair a machine that's broken because a part has come unscrewed, you create wealth with no environmental cost.

[5b] This essay was written before Firefox.

If you need to give someone a present and don't have any money, you make one. But kids are so bad at making things that they consider home-made presents to be a distinct, inferior, sort of thing to store-bought ones– a mere expression of the proverbial thought that counts. And indeed, the lumpy ashtrays we made for our parents did not have much of a resale market.

Craftsmen

The people most likely to grasp that wealth can be created are the ones who are good at making things, the craftsmen. Their hand-made objects become store-bought ones. But with the rise of industrialization there are fewer and fewer craftsmen. One of the biggest remaining groups is computer programmers.

A programmer can sit down in front of a computer and *create wealth*. A good piece of software is, in itself, a valuable thing. There is no manufacturing to confuse the issue. Those characters you type are a complete, finished product. If someone sat down and wrote a web browser that didn't suck (a fine idea, by the way), the world would be that much richer. [5b]

Everyone in a company works together to create wealth, in the sense of making more things people want. Many of the employees (e.g. the people in the mailroom or the personnel department) work at one remove from the actual making of stuff. Not the programmers. They literally think the product, one line at a time. And so it's clearer to programmers that wealth is something that's made, rather than being distributed, like slices of a pie, by some imaginary Daddy.

It's also obvious to programmers that there are huge variations in the rate at which wealth is created. At Viaweb we had one programmer who was a sort of monster of productivity. I remember watching what he did one long day and estimating that he had added several hundred thousand dollars to the market value of the company. A great programmer, on a roll, could create a million dollars worth of wealth in a couple weeks. A mediocre programmer over the same period will generate zero or even negative wealth (e.g. by introducing bugs).

This is why so many of the best programmers are libertarians. In our world, you sink or swim, and there are no excuses. When those far removed from the creation of wealth– undergraduates, reporters, politicians– hear that the richest 5% of the people have half the total wealth, they tend to think *injustice!* An experienced programmer would be more likely to think *is that all?* The top 5% of programmers probably write 99% of the good software.

Wealth can be created without being sold. Scientists, till recently at least, effectively donated the wealth they created. We are all richer for knowing about penicillin, because we're less likely to die from infections. Wealth is whatever people want, and not dying is certainly something we want. Hackers often donate their work by writing open source software that anyone can use for free. I am much the richer for the operating system FreeBSD, which I'm running on the computer I'm using now, and so is Yahoo, which runs it on all their servers.

What a Job Is

In industrialized countries, people belong to one institution or another at least until their twenties. After all those years you get used to the idea of belonging to a group of people who all get up in the morning, go to some set of buildings, and do things that they do not, ordinarily, enjoy doing. Belonging to such a group becomes part of your identity: name, age, role, institution. If you have to introduce yourself, or someone else describes you, it will be as something like, John Smith, age 10, a student at such and such elementary school, or John Smith, age 20, a student at such and such college.

When John Smith finishes school he is expected to get a job. And what getting a job seems to mean is joining another institution. Superficially it's a lot like college. You pick the companies you want to work for and apply to join them. If one likes you, you become a member of this new group. You get up in the morning and go to a new set of buildings, and do things that you do not, ordinarily, enjoy doing. There are a few differences: life is not as much fun, and you get paid, instead of paying, as you did in college. But the similarities feel greater than the differences. John Smith is now John Smith, 22, a software developer at such and such corporation.

In fact John Smith's life has changed more than he realizes. Socially, a company looks much like college, but the deeper you go into the underlying reality, the more different it gets.

What a company does, and has to do if it wants to continue to exist, is earn money. And the way most companies make money is by creating wealth. Companies can be so specialized that this similarity is concealed, but it is not only manufacturing companies that create wealth. A big component of wealth is location. Remember that magic machine that could make you cars and cook you dinner and so on? It would not be so useful if it delivered your dinner to a random location in central Asia. If wealth means what people want, companies that move things also create wealth. Ditto for many other kinds of companies that don't make anything physical. Nearly all companies exist to do something people want.

And that's what you do, as well, when you go to work for a company. But here there is another layer that tends to obscure the underlying reality. In a company, the work you do is averaged together with a lot of other people's. You may not even be aware you're doing something people want. Your contribution may be indirect. But the company as a whole must be giving people something they want, or they won't make any money. And if they are paying you x dollars a year, then on average you must be contributing at least x dollars a year worth of work, or the company will be spending more than it makes, and will go out of business.

Someone graduating from college thinks, and is told, that he needs to get a job, as if the important thing were becoming a member of an institution. A more direct way to put it would be: you need to start doing something people want. You don't need to join a company to do that. All a company is is a group of people working together to do something people want. It's doing something people want that matters, not joining the group. [6]

For most people the best plan probably is to go to work for some existing company. But it is a good idea to understand what's happening when you do this. A job means doing something people want, averaged together with everyone else in that company.

Working Harder

That averaging gets to be a problem. I think the single biggest problem afflicting large companies is the difficulty of assigning a value to each person's work. For the most part they punt. In a big company you get paid a fairly predictable salary for working fairly hard. You're expected not to be obviously incompetent or lazy, but you're not expected to devote your whole life to your work.

It turns out, though, that there are economies of scale in how much of your life you devote to your work. In the right kind of business, someone who really devoted himself to work could generate ten or even a hundred times as much wealth as an average employee. A programmer, for example, instead of chugging along maintaining and updating an existing piece of software, could write a whole new piece of software, and with it create a new source of revenue.

Companies are not set up to reward people who want to do this. You can't go to your boss and say, I'd like to start working ten times as hard, so will you please pay me ten times as much? For one thing, the official fiction is that you are already working as hard as you can. But a more serious problem is that the company has no way of measuring the value of your work.

[6] Many people feel confused and depressed in their early twenties. Life seemed so much more fun in college. Well, of course it was. Don't be fooled by the surface similarities. You've gone from guest to servant. It's possible to have fun in this new world. Among other things, you now get to go behind the doors that say "authorized personnel only." But the change is a shock at first, and all the worse if you're not consciously aware of it.

Salesmen are an exception. It's easy to measure how much revenue they generate, and they're usually paid a percentage of it. If a salesman wants to work harder, he can just start doing it, and he will automatically get paid proportionally more.

There is one other job besides sales where big companies can hire first-rate people: in the top management jobs. And for the same reason: their performance can be measured. The top managers are held responsible for the performance of the entire company. Because an ordinary employee's performance can't usually be measured, he is not expected to do more than put in a solid effort. Whereas top management, like salespeople, have to actually come up with the numbers. The CEO of a company that tanks cannot plead that he put in a solid effort. If the company does badly, he's done badly.

A company that could pay all its employees so straightforwardly would be enormously successful. Many employees would work harder if they could get paid for it. More importantly, such a company would attract people who wanted to work especially hard. It would crush its competitors.

Unfortunately, companies can't pay everyone like salesmen. Salesmen work alone. Most employees' work is tangled together. Suppose a company makes some kind of consumer gadget. The engineers build a reliable gadget with all kinds of new features; the industrial designers design a beautiful case for it; and then the marketing people convince everyone that it's something they've got to have. How do you know how much of the gadget's sales are due to each group's efforts? Or, for that matter, how much is due to the creators of past gadgets that gave the company a reputation for quality? There's no way to untangle all their contributions. Even if you could read the minds of the consumers, you'd find these factors were all blurred together.

If you want to go faster, it's a problem to have your work tangled together with a large number of other people's. In a large group, your performance is not separately measurable– and the rest of the group slows you down.

Measurement and Leverage

To get rich you need to get yourself in a situation with two things, measurement and leverage. You need to be in a position where your performance can be measured, or there is no way to get paid more by doing more. And you have to have leverage, in the sense that the decisions you make have a big effect.

Measurement alone is not enough. An example of a job with measurement but not leverage is doing piecework in a sweatshop. Your performance is measured and you get paid accordingly, but you have no scope for decisions. The only decision you get to make is how fast you work, and that can probably only increase your earnings by a factor of two or three.

An example of a job with both measurement and leverage would be lead actor in a movie. Your performance can be measured in the gross of the movie. And you have leverage in the sense that your performance can make or break it.

CEOs also have both measurement and leverage. They're measured, in that the performance of the company is their performance. And they have leverage in that their decisions set the whole company moving in one direction or another.

I think everyone who gets rich by their own efforts will be found to be in a situation with measurement and leverage. Everyone I can think of does: CEOs, movie stars, hedge fund managers, professional athletes. A good hint to the presence of leverage is the possibility of failure. Upside must be balanced by downside, so if there is big potential for gain there must also be a terrifying possibility of loss. CEOs, stars, fund managers, and athletes all live with the sword hanging over their heads; the moment they start to suck, they're out. If you're in a job that feels safe, you are not going to get rich, because if there is no danger there is almost certainly no leverage.

But you don't have to become a CEO or a movie star to be in a situation with measurement and leverage. All you need to do is be part of a small group working on a hard problem.

Smallness = Measurement

If you can't measure the value of the work done by individual employees, you can get close. You can measure the value of the work done by small groups.

One level at which you can accurately measure the revenue generated by employees is at the level of the whole company. When the company is small, you are thereby fairly close to measuring the contributions of individual employees. A viable startup might only have ten employees, which puts you within a factor of ten of measuring individual effort.

Starting or joining a startup is thus as close as most people can get to saying to one's boss, I want to work ten times as hard, so please pay me ten times as much. There are two differences: you're not saying it to your boss, but directly to the customers (for whom your boss is only a proxy after all), and you're not doing it individually, but along with a small group of other ambitious people.

It will, ordinarily, be a group. Except in a few unusual kinds of work, like acting or writing books, you can't be a company of one person. And the people you work with had better be good, because it's their work that yours is going to be averaged with.

A big company is like a giant galley driven by a thousand rowers. Two things keep the speed of the galley down. One is that individual rowers don't see any result from working harder. The other is that, in a group of a thousand people, the average rower is likely to be pretty average.

If you took ten people at random out of the big galley and put them in a boat by themselves, they could probably go faster. They would have both carrot and stick to motivate them. An energetic rower would be encouraged by the thought that he could have a visible effect on the speed of the boat. And if someone was lazy, the others would be more likely to notice and complain.

But the real advantage of the ten-man boat shows when you take the ten *best* rowers out of the big galley and put them in a boat together. They will have all the extra motivation that comes from being in a small group. But more importantly, by selecting that small a group you can get the best rowers. Each one will be in the top 1%. It's a much better deal for them to average their work together with a small group of their peers than to average it with everyone.

That's the real point of startups. Ideally, you are getting together with a group of other people who also want to work a lot harder, and get paid a lot more, than they would in a big company. And because startups tend to get founded by self-selecting groups of ambitious people who already know one another (at least by reputation), the level of measurement is more precise than you get from smallness alone. A startup is not merely ten people, but ten people like you.

Steve Jobs once said that the success or failure of a startup depends on the first ten employees. I agree. If anything, it's more like the first five. Being small is not, in itself, what makes startups kick butt, but rather that small groups can be select. You don't want small in the sense of a village, but small in the sense of an all-star team.

The larger a group, the closer its average member will be to the average for the population as a whole. So all other things being equal, a very able person in a big company is probably getting a bad deal, because his performance is dragged down by the overall lower performance of the others. Of course, all other things often are not equal: the able person may not care about money, or may prefer the stability of a large company. But a very able person who does care about money will ordinarily do better to go off and work with a small group of peers.

Technology = Leverage

Startups offer anyone a way to be in a situation with measurement and leverage. They allow measurement because they're small, and they offer leverage because they make money by inventing new technology.

What is technology? It's *technique*. It's the way we all do things. And when you discover a new way to do things, its value is multiplied by all the people who use it. It is the proverbial fishing rod, rather than the fish. That's the difference between a startup and a restaurant or a barber shop. You fry eggs or cut hair one customer at a time. Whereas if you solve a technical problem that a lot of people care about, you help everyone who uses your solution. That's leverage.

If you look at history, it seems that most people who got rich by creating wealth did it by developing new technology. You just can't fry eggs or cut hair fast enough. What made the Florentines rich in 1200 was the discovery of new techniques for making the high-tech product of the time, fine woven cloth. What made the Dutch rich in 1600 was the discovery of shipbuilding and navigation techniques that enabled them to dominate the seas of the Far East.

Fortunately there is a natural fit between smallness and solving hard problems. The leading edge of technology moves fast. Technology that's valuable today could be worthless in a couple years. Small companies are more at home in this world, because they don't have layers of bureaucracy to slow them down. Also, technical advances tend to come from unorthodox approaches, and small companies are less constrained by convention.

Big companies can develop technology. They just can't do it quickly. Their size makes them slow and prevents them from rewarding employees for the extraordinary effort required. So in practice big companies only get to develop technology in fields where large capital requirements prevent startups from competing with them, like microprocessors, power plants, or passenger aircraft. And even in those fields they depend heavily on startups for components and ideas.

It's obvious that biotech or software startups exist to solve hard technical problems, but I think it will also be found to be true in businesses that don't seem to be about technology. McDonald's, for example, grew big by designing a system, the McDonald's franchise, that could then be reproduced at will all over the face of the earth. A McDonald's franchise is controlled by rules so precise that it is practically a piece of software. Write once, run everywhere. Ditto for Wal-Mart. Sam Walton got rich not by being a retailer, but by designing a new kind of store.

Use difficulty as a guide not just in selecting the overall aim of your company, but also at decision points along the way. At Viaweb one of our rules of thumb was *run upstairs.* Suppose you are a little, nimble guy being chased by a big, fat, bully. You open a door and find yourself in a staircase. Do you go up or down? I say up. The bully can probably run downstairs as fast as you can. Going upstairs his bulk will be more of a disadvantage. Running upstairs is hard for you but even harder for him.

What this meant in practice was that we deliberately sought hard problems. If there were two features we could add to our software, both equally valuable in proportion to their difficulty, we'd always take the harder one. Not just because it was more valuable, but *because it was harder.* We delighted in forcing bigger, slower competitors to follow us over difficult ground. Like guerillas, startups prefer the difficult terrain of the mountains, where the troops of the central government can't follow. I can remember times when we were just exhausted after wrestling all day with some horrible technical problem. And I'd be delighted, because something that was hard for us would be impossible for our competitors.

This is not just a good way to run a startup. It's what a startup is. Venture capitalists know about this and have a phrase for it: *barriers to entry.* If you go to a VC with a new idea and ask him to invest in it, one of the first things he'll ask is, how hard would this be for someone else to develop? That is, how much difficult ground have you put between yourself and potential pursuers? [7] And you had better have a convincing explanation of why your technology would be hard to duplicate. Otherwise as soon as some big company becomes aware

[7] When VCs asked us how long it would take another startup to duplicate our software, we used to reply that they probably wouldn't be able to at all. I think this made us seem naive, or liars.

of it, they'll make their own, and with their brand name, capital, and distribution clout, they'll take away your market overnight. You'd be like guerillas caught in the open field by regular army forces.

One way to put up barriers to entry is through patents. But patents may not provide much protection. Competitors commonly find ways to work around a patent. And if they can't, they may simply violate it and invite you to sue them. A big company is not afraid to be sued; it's an everyday thing for them. They'll make sure that suing them is expensive and takes a long time. Ever heard of Philo Farnsworth? He invented television. The reason you've never heard of him is that his company was not the one to make money from it. [8] The company that did was RCA, and Farnsworth's reward for his efforts was a decade of patent litigation.

Here, as so often, the best defense is a good offense. If you can develop technology that's simply too hard for competitors to duplicate, you don't need to rely on other defenses. Start by picking a hard problem, and then at every decision point, take the harder choice. [9]

The Catch(es)

If it were simply a matter of working harder than an ordinary employee and getting paid proportionately, it would obviously be a good deal to start a startup. Up to a point it would be more fun. I don't think many people like the slow pace of big companies, the inter-

[8] Few technologies have one clear inventor. So as a rule, if you know the "inventor" of something (the telephone, the assembly line, the airplane, the light bulb, the transistor) it is because their company made money from it, and the company's PR people worked hard to spread the story. If you don't know who invented something (the automobile, the television, the computer, the jet engine, the laser), it's because other companies made all the money.

[9] This is a good plan for life in general. If you have two choices, choose the harder. If you're trying to decide whether to go out running or sit home and watch TV, go running. Probably the reason this trick works so well is that when you have two choices and one is harder, the only reason you're even considering the other is laziness. You know in the back of your mind what's the right thing to do, and this trick merely forces you to acknowledge it.

minable meetings, the water-cooler conversations, the clueless middle managers, and so on.

Unfortunately there are a couple catches. One is that you can't choose the point on the curve that you want to inhabit. You can't decide, for example, that you'd like to work just two or three times as hard, and get paid that much more. When you're running a startup, your competitors decide how hard you work. And they pretty much all make the same decision: as hard as you possibly can.

The other catch is that the payoff is only on average proportionate to your productivity. There is, as I said before, a large random multiplier in the success of any company. So in practice the deal is not that you're 30 times as productive and get paid 30 times as much. It is that you're 30 times as productive, and get paid between zero and a thousand times as much. If the mean is 30x, the median is probably zero. Most startups tank, and not just the dogfood portals we all heard about during the Internet Bubble. It's common for a startup to be developing a genuinely good product, take slightly too long to do it, run out of money, and have to shut down.

A startup is like a mosquito. A bear can absorb a hit and a crab is armored against one, but a mosquito is designed for one thing: to score. No energy is wasted on defense. The defense of mosquitos, as a species, is that there are a lot of them, but this is little consolation to the individual mosquito.

Startups, like mosquitos, tend to be an all-or-nothing proposition. And you don't generally know which of the two you're going to get till the last minute. Viaweb came close to tanking several times. Our trajectory was like a sine wave. Fortunately we got bought at the top of the cycle, but it was damned close. While we were visiting Yahoo in California to talk about selling the company to them, we had to borrow a conference room to reassure an investor who was about to back out of a new round of funding that we needed to stay alive.

The all-or-nothing aspect of startups was not something we wanted.

Viaweb's hackers were all extremely risk-averse. If there had been some way just to work super hard and get paid for it, without having a lottery mixed in, we would have been delighted. We would have much preferred a 100% chance of $1 million to a 20% chance of $10 million, even though theoretically the second is worth twice as much. Unfortunately, there is not currently any space in the business world where you can get the first deal.

The closest you can get is by selling your startup in the early stages, giving up upside (and risk) for a smaller but guaranteed payoff. We had a chance to do this, and stupidly, as we then thought, let it slip by. After that we became comically eager to sell. For the next year or so, if anyone expressed the slightest curiosity about Viaweb we would try to sell them the company. But there were no takers, so we had to keep going.

It would have been a bargain to buy us at an early stage, but companies doing acquisitions are not looking for bargains. A company big enough to acquire startups will be big enough to be fairly conservative, and within the company the people in charge of acquisitions will be among the more conservative, because they are likely to be business school types who joined the company late. They would rather overpay for a safe choice. So it is easier to sell an established startup, even at a large premium, than an early-stage one.

Get Users

I think it's a good idea to get bought, if you can. Running a business is different from growing one. It is just as well to let a big company take over once you reach cruising altitude. It's also financially wiser, because selling allows you to diversify. What would you think of a financial advisor who put all his client's assets into one volatile stock?

How do you get bought? Mostly by doing the same things you'd do if you didn't intend to sell the company. Being profitable, for example. But getting bought is also an art in its own right, and one that we spent a lot of time trying to master.

Potential buyers will always delay if they can. The hard part about getting bought is getting them to act. For most people, the most powerful motivator is not the hope of gain, but the fear of loss. For potential acquirers, the most powerful motivator is the prospect that one of their competitors will buy you. This, as we found, causes CEOs to take redeyes. The second biggest is the worry that, if they don't buy you now, you'll continue to grow rapidly and will cost more to acquire later, or even become a competitor.

In both cases, what it all comes down to is users. You'd think that a company about to buy you would do a lot of research and decide for themselves how valuable your technology was. Not at all. What they go by is the number of users you have.

In effect, acquirers assume the customers know who has the best technology. And this is not as stupid as it sounds. Users are the only real proof that you've created wealth. Wealth is what people want, and if people aren't using your software, maybe it's not just because you're bad at marketing. Maybe it's because you haven't made what they want.

Venture capitalists have a list of danger signs to watch out for. Near the top is the company run by techno-weenies who are obsessed with solving interesting technical problems, instead of making users happy. In a startup, you're not just trying to solve problems. You're trying to solve problems *that users care about.*

So I think you should make users the test, just as acquirers do. Treat a startup as an optimization problem in which performance is measured by number of users. As anyone who has tried to optimize software knows, the key is measurement. When you try to guess where your program is slow, and what would make it faster, you almost always guess wrong.

Number of users may not be the perfect test, but it will be very close. It's what acquirers care about. It's what revenues depend on. It's what makes competitors unhappy. It's what impresses reporters, and potential new users. Certainly it's a better test than your a priori notions of what problems are important to solve, no matter how technically adept you are.

Among other things, treating a startup as an optimization problem will help you avoid another pitfall that VCs worry about, and rightly–taking a long time to develop a product. Now we can recognize this as something hackers already know to avoid: premature optimization. Get a version 1.0 out there as soon as you can. Until you have some users to measure, you're optimizing based on guesses.

The ball you need to keep your eye on here is the underlying principle that wealth is what people want. If you plan to get rich by creating wealth, you have to know what people want. So few businesses really pay attention to making customers happy. How often do you walk into a store, or call a company on the phone, with a feeling of dread in the back of your mind? When you hear "your call is important to us, please stay on the line," do you think, oh good, now everything will be all right?

A restaurant can afford to serve the occasional burnt dinner. But in technology, you cook one thing and that's what everyone eats. So any difference between what people want and what you deliver is multiplied. You please or annoy customers wholesale. The closer you can get to what they want, the more wealth you generate.

Wealth and Power

Making wealth is not the only way to get rich. For most of human history it has not even been the most common. Until a few centuries ago, the main sources of wealth were mines, slaves and serfs, land, and cattle, and the only ways to acquire these rapidly were by inheritance, marriage, conquest, or confiscation. Naturally wealth had a bad reputation.

Two things changed. The first was the rule of law. For most of the world's history, if you did somehow accumulate a fortune, the ruler or his henchmen would find a way to steal it. But in medieval Europe something new happened. A new class of merchants and manufacturers began to collect in towns. [10] Together they were able to withstand the local feudal lord. So for the first time in our history, the bullies stopped stealing the nerds' lunch money. This was naturally a great incentive, and possibly indeed the main cause of the second big change, industrialization.

A great deal has been written about the causes of the Industrial Revolution. But surely a necessary, if not sufficient, condition was that people who made fortunes be able to enjoy them in peace. [11] One piece of evidence is what happened to countries that tried to return to the old model, like the Soviet Union, and to a lesser extent Britain under the labor governments of the 1960s and early 1970s. Take away the incentive of wealth, and technical innovation grinds to a halt.

Remember what a startup is, economically: a way of saying, I want to work faster. Instead of accumulating money slowly by being paid a regular wage for fifty years, I want to get it over with as soon as possible. So governments that forbid you to accumulate wealth are in effect decreeing that you work slowly. They're willing to let you earn $3 million over fifty years, but they're not willing to let you work so hard that you can do it in two. They are like the corporate boss that you can't go to and say, I want to work ten times as hard, so please pay me ten times a much. Except this is not a boss you can escape by starting your own company.

The problem with working slowly is not just that technical innovation

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[10] It is probably no accident that the middle class first appeared in northern Italy and the low countries, where there were no strong central governments. These two regions were the richest of their time and became the twin centers from which Renaissance civilization radiated. If they no longer play that role, it is because other places, like the United States, have been truer to the principles they discovered.

[11] It may indeed be a sufficient condition. But if so, why didn't the Industrial Revolution happen earlier? Two possible (and not incompatible) answers: (a) It did. The Industrial Revolution was one in a series. (b) Because in medieval towns, monopolies and guild regulations initially slowed the development of new means of production.

happens slowly. It's that it tends not to happen at all. It's only when you're deliberately looking for hard problems, as a way to use speed to the greatest advantage, that you take on this kind of project. Developing new technology is a pain in the ass. It is, as Edison said, one percent inspiration and ninety-nine percent perspiration. Without the incentive of wealth, no one wants to do it. Engineers will work on sexy projects like fighter planes and moon rockets for ordinary salaries, but more mundane technologies like light bulbs or semiconductors have to be developed by entrepreneurs.

Startups are not just something that happened in Silicon Valley in the last couple decades. Since it became possible to get rich by creating wealth, everyone who has done it has used essentially the same recipe: measurement and leverage, where measurement comes from working with a small group, and leverage from developing new techniques. The recipe was the same in Florence in 1200 as it is in Santa Clara today.

Understanding this may help to answer an important question: why Europe grew so powerful. Was it something about the geography of Europe? Was it that Europeans are somehow racially superior? Was it their religion? The answer (or at least the proximate cause) may be that the Europeans rode on the crest of a powerful new idea: allowing those who made a lot of money to keep it.

Once you're allowed to do that, people who want to get rich can do it by generating wealth instead of stealing it. The resulting technological growth translates not only into wealth but into military power. The theory that led to the stealth plane was developed by a Soviet mathematician. But because the Soviet Union didn't have a computer industry, it remained for them a theory; they didn't have hardware capable of executing the calculations fast enough to design an actual airplane.

In that respect the Cold War teaches the same lesson as World War II and, for that matter, most wars in recent history. Don't let a ruling class of warriors and politicians squash the entrepreneurs. The same recipe that makes individuals rich makes countries powerful. Let the nerds keep their lunch money, and you rule the world.

Chapter 5

Mind the Gap

May 2004

When people care enough about something to do it well, those who do it best tend to be far better than everyone else. There's a huge gap between Leonardo and second-rate contemporaries like Borgognone. You see the same gap between Raymond Chandler and the average writer of detective novels. A top-ranked professional chess player could play ten thousand games against an ordinary club player without losing once.

Like chess or painting or writing novels, making money is a very specialized skill. But for some reason we treat this skill differently. No one complains when a few people surpass all the rest at playing chess or writing novels, but when a few people make more money than the rest, we get editorials saying this is wrong.

Why? The pattern of variation seems no different than for any other skill. What causes people to react so strongly when the skill is making money?

I think there are three reasons we treat making money as different: the misleading model of wealth we learn as children; the disreputable way in which, till recently, most fortunes were accumulated; and the worry that great variations in income are somehow bad for society. As far as I can tell, the first is mistaken, the second outdated, and the third empirically false. Could it be that, in a modern democracy, variation in income is actually a sign of health?

The Daddy Model of Wealth

When I was five I thought electricity was created by electric sockets. I didn't realize there were power plants out there generating it. Likewise, it doesn't occur to most kids that wealth is something that has to be generated. It seems to be something that flows from parents.

Because of the circumstances in which they encounter it, children tend to misunderstand wealth. They confuse it with money. They think that there is a fixed amount of it. And they think of it as something that's distributed by authorities (and so should be distributed equally), rather than something that has to be created (and might be created unequally).

In fact, wealth is not money. Money is just a convenient way of trading one form of wealth for another. Wealth is the underlying stuff—the goods and services we buy. When you travel to a rich or poor country, you don't have to look at people's bank accounts to tell which kind you're in. You can *see* wealth—in buildings and streets, in the clothes and the health of the people.

Where does wealth come from? People make it. This was easier to grasp when most people lived on farms, and made many of the things they wanted with their own hands. Then you could see in the house, the herds, and the granary the wealth that each family created. It was obvious then too that the wealth of the world was not a fixed quantity that had to be shared out, like slices of a pie. If you wanted more wealth, you could make it.

This is just as true today, though few of us create wealth directly for ourselves (except for a few vestigial domestic tasks). Mostly we create wealth for other people in exchange for money, which we then trade for the forms of wealth we want. [1]

[1] Part of the reason this subject is so contentious is that some of those most vocal on the subject of wealth—university students, heirs, professors, politicians, and journalists—have the least experience creating it. (This phenomenon will be familiar to anyone who has overheard conversations about sports in a bar.)

Students are mostly still on the parental dole, and have not stopped to think about where that money comes from. Heirs will be on the parental dole for life. Professors

Because kids are unable to create wealth, whatever they have has to be given to them. And when wealth is something you're given, then of course it seems that it should be distributed equally. [2] As in most families it is. The kids see to that. "Unfair," they cry, when one sibling gets more than another.

In the real world, you can't keep living off your parents. If you want something, you either have to make it, or do something of equivalent value for someone else, in order to get them to give you enough money to buy it. In the real world, wealth is (except for a few specialists like thieves and speculators) something you have to create, not something that's distributed by Daddy. And since the ability and desire to create it vary from person to person, it's not made equally.

You get paid by doing or making something people want, and those who make more money are often simply better at doing what people want. Top actors make a lot more money than B-list actors. The B-list actors might be almost as charismatic, but when people go to the theater and look at the list of movies playing, they want that extra oomph that the big stars have.

Doing what people want is not the only way to get money, of course. You could also rob banks, or solicit bribes, or establish a monopoly.

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and politicians live within socialist eddies of the economy, at one remove from the creation of wealth, and are paid a flat rate regardless of how hard they work. And journalists as part of their professional code segregate themselves from the revenue-collecting half of the businesses they work for (the ad sales department). Many of these people never come face to face with the fact that the money they receive represents wealth—wealth that, except in the case of journalists, someone else created earlier. They live in a world in which income *is* doled out by a central authority according to some abstract notion of fairness (or randomly, in the case of heirs), rather than given by other people in return for something they wanted, so it may seem to them unfair that things don't work the same in the rest of the economy.

(Some professors do create a great deal of wealth for society. But the money they're paid isn't a *quid pro quo*. It's more in the nature of an investment.)

[2]When one reads about the origins of the Fabian Society, it sounds like something cooked up by the high-minded Edwardian child-heroes of Edith Nesbit's *The Wouldbegoods*.

Such tricks account for some variation in wealth, and indeed for some of the biggest individual fortunes, but they are not the root cause of variation in income. The root cause of variation in income, as Occam's Razor implies, is the same as the root cause of variation in every other human skill.

In the United States, the CEO of a large public company makes about 100 times as much as the average person. [3] Basketball players make about 128 times as much, and baseball players 72 times as much. Editorials quote this kind of statistic with horror. But I have no trouble imagining that one person could be 100 times as productive as another. In ancient Rome the price of *slaves* varied by a factor of 50 depending on their skills. [4] And that's without considering motivation, or the extra leverage in productivity that you can get from modern technology.

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[3] According to a study by the Corporate Library, the median total compensation, including salary, bonus, stock grants, and the exercise of stock options, of S&P 500 CEOs in 2002 was $3.65 million. According to *Sports Illustrated*, the average NBA player's salary during the 2002-03 season was $4.54 million, and the average major league baseball player's salary at the start of the 2003 season was $2.56 million. According to the Bureau of Labor Statistics, the mean annual wage in the US in 2002 was $35,560.

[4] In the early empire the price of an ordinary adult slave seems to have been about 2,000 sestertii (e.g. Horace, *Sat.* ii.7.43). A servant girl cost 600 (Martial vi.66), while Columella (iii.3.8) says that a skilled vine-dresser was worth 8,000. A doctor, P. Decimus Eros Merula, paid 50,000 sestertii for his freedom (Dessau, *Inscriptiones* 7812). Seneca (*Ep.* xxvii.7) reports that one Calvisius Sabinus paid 100,000 sestertii apiece for slaves learned in the Greek classics. Pliny (*Hist. Nat.* vii.39) says that the highest price paid for a slave up to his time was 700,000 sestertii, for the linguist (and presumably teacher) Daphnis, but that this had since been exceeded by actors buying their own freedom.

Classical Athens saw a similar variation in prices. An ordinary laborer was worth about 125 to 150 drachmae. Xenophon (*Mem.* ii.5) mentions prices ranging from 50 to 6,000 drachmae (for the manager of a silver mine).

For more on the economics of ancient slavery see:

Jones, A. H. M., "Slavery in the Ancient World," *Economic History Review*, 2:9 (1956), 185-199, reprinted in Finley, M. I. (ed.), *Slavery in Classical Antiquity*, Heffer, 1964.

Editorials about athletes' or CEOs' salaries remind me of early Christian writers, arguing from first principles about whether the Earth was round, when they could just walk outside and check. [5] How much someone's work is worth is not a policy question. It's something the market already determines.

"Are they really worth 100 of us?" editorialists ask. Depends on what you mean by worth. If you mean worth in the sense of what people will pay for their skills, the answer is yes, apparently.

A few CEOs' incomes reflect some kind of wrongdoing. But are there not others whose incomes really do reflect the wealth they generate? Steve Jobs saved a company that was in a terminal decline. And not merely in the way a turnaround specialist does, by cutting costs; he had to decide what Apple's next products should be. Few others could have done it. And regardless of the case with CEOs, it's hard to see how anyone could argue that the salaries of professional basketball players don't reflect supply and demand.

It may seem unlikely in principle that one individual could really generate so much more wealth than another. The key to this mystery is to revisit that question, are they really worth 100 of us? *Would* a basketball team trade one of their players for 100 random people? What would Apple's next product look like if you replaced Steve Jobs with a committee of 100 random people? [6] These things don't scale linearly. Perhaps the CEO or the professional athlete has only ten times (whatever that means) the skill and determination of an ordinary person. But it makes all the difference that it's concentrated in one individual.

When we say that one kind of work is overpaid and another underpaid, what are we really saying? In a free market, prices are determined by what buyers want. People like baseball more than poetry, so baseball players make more than poets. To say that a certain kind of work is underpaid is thus identical with saying that people want the wrong

[5] Eratosthenes (276—195 BC) used shadow lengths in different cities to estimate the Earth's circumference. He was off by only about 2%.

[6] No, and Windows, respectively.

things.

Well, of course people want the wrong things. It seems odd to be surprised by that. And it seems even odder to say that it's *unjust* that certain kinds of work are underpaid. [7] Then you're saying that it's unjust that people want the wrong things. It's lamentable that people prefer reality TV and corndogs to Shakespeare and steamed vegetables, but unjust? That seems like saying that blue is heavy, or that up is circular.

The appearance of the word "unjust" here is the unmistakable spectral signature of the Daddy Model. Why else would this idea occur in this odd context? Whereas if the speaker were still operating on the Daddy Model, and saw wealth as something that flowed from a common source and had to be shared out, rather than something generated by doing what other people wanted, this is exactly what you'd get on noticing that some people made much more than others.

When we talk about "unequal distribution of income," we should also ask, where does that income come from? [8] Who made the wealth it represents? Because to the extent that income varies simply according to how much wealth people create, the distribution may be unequal,

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[7] One of the biggest divergences between the Daddy Model and reality is the valuation of hard work. In the Daddy Model, hard work is in itself deserving. In reality, wealth is measured by what one delivers, not how much effort it costs. If I paint someone's house, the owner shouldn't pay me extra for doing it with a toothbrush.

It will seem to someone still implicitly operating on the Daddy Model that it is unfair when someone works hard and doesn't get paid much. To help clarify the matter, get rid of everyone else and put our worker on a desert island, hunting and gathering fruit. If he's bad at it he'll work very hard and not end up with much food. Is this unfair? Who is being unfair to him?

[8] Part of the reason for the tenacity of the Daddy Model may be the dual meaning of "distribution." When economists talk about "distribution of income," they mean statistical distribution. But when you use the phrase frequently, you can't help associating it with the other sense of the word (as in e.g. "distribution of alms"), and thereby subconsciously seeing wealth as something that flows from some central tap. The word "regressive" as applied to tax rates has a similar effect, at least on me; how can anything *regressive* be good?

but it's hardly unjust.

Stealing It

The second reason we tend to find great disparities of wealth alarming is that for most of human history the usual way to accumulate a fortune was to steal it: in pastoral societies by cattle raiding; in agricultural societies by appropriating others' estates in times of war, and taxing them in times of peace.

In conflicts, those on the winning side would receive the estates confiscated from the losers. In England in the 1060s, when William the Conqueror distributed the estates of the defeated Anglo-Saxon nobles to his followers, the conflict was military. By the 1530s, when Henry VIII distributed the estates of the monasteries to his followers, it was mostly political. [9] But the principle was the same. Indeed, the same principle is at work now in Zimbabwe.

In more organized societies, like China, the ruler and his officials used taxation instead of confiscation. But here too we see the same principle: the way to get rich was not to create wealth, but to serve a ruler powerful enough to appropriate it.

This started to change in Europe with the rise of the middle class. Now we think of the middle class as people who are neither rich nor poor, but originally they were a distinct group. In a feudal society, there are just two classes: a warrior aristocracy, and the serfs who work their estates. The middle class were a new, third group who lived in towns and supported themselves by manufacturing and trade.

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[9]"From the beginning of the reign Thomas Lord Roos was an assiduous courtier of the young Henry VIII and was soon to reap the rewards. In 1525 he was made a Knight of the Garter and given the Earldom of Rutland. In the thirties his support of the breach with Rome, his zeal in crushing the Pilgrimage of Grace, and his readiness to vote the death-penalty in the succession of spectacular treason trials that punctuated Henry's erratic matrimonial progress made him an obvious candidate for grants of monastic property."

Stone, Lawrence, *Family and Fortune: Studies in Aristocratic Finance in the Sixteenth and Seventeenth Centuries*, Oxford University Press, 1973, p. 166.

Starting in the tenth and eleventh centuries, petty nobles and former serfs banded together in towns that gradually became powerful enough to ignore the local feudal lords. [10] Like serfs, the middle class made a living largely by creating wealth. (In port cities like Genoa and Pisa, they also engaged in piracy.) But unlike serfs they had an incentive to create a lot of it. Any wealth a serf created belonged to his master. There was not much point in making more than you could hide. Whereas the independence of the townsmen allowed them to keep whatever wealth they created.

Once it became possible to get rich by creating wealth, society as a whole started to get richer very rapidly. Nearly everything we have was created by the middle class. Indeed, the other two classes have effectively disappeared in industrial societies, and their names been given to either end of the middle class. (In the original sense of the word, Bill Gates is middle class.)

But it was not till the Industrial Revolution that wealth creation definitively replaced corruption as the best way to get rich. In England, at least, corruption only became unfashionable (and in fact only started to be called "corruption") when there started to be other, faster ways to get rich.

Seventeenth-century England was much like the third world today, in that government office was a recognized route to wealth. The great fortunes of that time still derived more from what we would now call corruption than from commerce. [11] By the nineteenth century that

[10]There is archaeological evidence for large settlements earlier, but it's hard to say what was happening in them.

Hodges, Richard and David Whitehouse, *Mohammed, Charlemagne and the Origins of Europe*, Cornell University Press, 1983.

[11]William Cecil and his son Robert were each in turn the most powerful minister of the crown, and both used their position to amass fortunes among the largest of their times. Robert in particular took bribery to the point of treason. "As Secretary of State and the leading advisor to King James on foreign policy, [he] was a special recipient of favour, being offered large bribes by the Dutch not to make peace with Spain, and large bribes by Spain to make peace." (Stone, *op. cit.*, p. 17.)

had changed. There continued to be bribes, as there still are everywhere, but politics had by then been left to men who were driven more by vanity than greed. Technology had made it possible to create wealth faster than you could steal it. The prototypical rich man of the nineteenth century was not a courtier but an industrialist.

With the rise of the middle class, wealth stopped being a zero-sum game. Jobs and Wozniak didn't have to make us poor to make themselves rich. Quite the opposite: they created things that made our lives materially richer. They had to, or we wouldn't have paid for them.

But since for most of the world's history the main route to wealth was to steal it, we tend to be suspicious of rich people. Idealistic undergraduates find their unconsciously preserved child's model of wealth confirmed by eminent writers of the past. It is a case of the mistaken meeting the outdated.

"Behind every great fortune, there is a crime," Balzac wrote. Except he didn't. What he actually said was that a great fortune with no apparent cause was probably due to a crime well enough executed that it had been forgotten. If we were talking about Europe in 1000, or most of the third world today, the standard misquotation would be spot on. But Balzac lived in nineteenth-century France, where the Industrial Revolution was well advanced. He knew you could make a fortune without stealing it. After all, he did himself, as a popular novelist. [12]

Only a few countries (by no coincidence, the richest ones) have reached this stage. In most, corruption still has the upper hand. In most, the fastest way to get wealth is by stealing it. And so when we see increasing differences in income in a rich country, there is a tendency to worry that it's sliding back toward becoming another Venezuela. I think the opposite is happening. I think you're seeing a country a full step ahead of Venezuela.

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[12] Though Balzac made a lot of money from writing, he was notoriously improvident and was troubled by debts all his life.

The Lever of Technology

Will technology increase the gap between rich and poor? It will certainly increase the gap between the productive and the unproductive. That's the whole point of technology. With a tractor an energetic farmer could plow six times as much land in a day as he could with a team of horses. But only if he mastered a new kind of farming.

I've seen the lever of technology grow visibly in my own time. In high school I made money by mowing lawns and scooping ice cream at Baskin-Robbins. This was the only kind of work available at the time. Now high school kids could write software or design web sites. But only some of them will; the rest will still be scooping ice cream.

I remember very vividly when in 1985 improved technology made it possible for me to buy a computer of my own. Within months I was using it to make money as a freelance programmer. A few years before, I couldn't have done this. A few years before, there was no such *thing* as a freelance programmer. But Apple created wealth, in the form of powerful, inexpensive computers, and programmers immediately set to work using it to create more.

As this example suggests, the rate at which technology increases our productive capacity is probably exponential, rather than linear. So we should expect to see ever-increasing variation in individual productivity as time goes on. Will that increase the gap between rich and the poor? Depends which gap you mean.

Technology should increase the gap in income, but it seems to decrease other gaps. A hundred years ago, the rich led a different *kind* of life from ordinary people. They lived in houses full of servants, wore elaborately uncomfortable clothes, and travelled about in carriages drawn by teams of horses which themselves required their own houses and servants. Now, thanks to technology, the rich live more like the average person.

Cars are a good example of why. It's possible to buy expensive, handmade cars that cost hundreds of thousands of dollars. But there is not much point. Companies make more money by building a large number of ordinary cars than a small number of expensive ones. So a company making a mass-produced car can afford to spend a lot more on its design. If you buy a custom-made car, something will always be breaking. The only point of buying one now is to advertise that you can.

Or consider watches. Fifty years ago, by spending a lot of money on a watch you could get better performance. When watches had mechanical movements, expensive watches kept better time. Not any more. Since the invention of the quartz movement, an ordinary Timex is more accurate than a Patek Philippe costing hundreds of thousands of dollars. [13] Indeed, as with expensive cars, if you're determined to spend a lot of money on a watch, you have to put up with some inconvenience to do it: as well as keeping worse time, mechanical watches have to be wound.

The only thing technology can't cheapen is brand. Which is precisely why we hear ever more about it. Brand is the residue left as the substantive differences between rich and poor evaporate. But what label you have on your stuff is a much smaller matter than having it versus not having it. In 1900, if you kept a carriage, no one asked what year or brand it was. If you had one, you were rich. And if you weren't rich, you took the omnibus or walked. Now even the poorest Americans drive cars, and it is only because we're so well trained by advertising that we can even recognize the especially expensive ones. [14]

The same pattern has played out in industry after industry. If there is enough demand for something, technology will make it cheap enough to sell in large volumes, and the mass-produced versions will be, if not

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[13] A Timex will gain or lose about .5 seconds per day. The most accurate mechanical watch, the Patek Philippe 10 Day Tourbillon, is rated at -1.5 to +2 seconds. Its retail price is about $220,000.

[14] If asked to choose which was more expensive, a well-preserved 1989 Lincoln Town Car ten-passenger limousine ($5,000) or a 2004 Mercedes S600 sedan ($122,000), the average Edwardian might well guess wrong.

better, at least more convenient. [15] And there is nothing the rich like more than convenience. The rich people I know drive the same cars, wear the same clothes, have the same kind of furniture, and eat the same foods as my other friends. Their houses are in different neighborhoods, or if in the same neighborhood are different sizes, but within them life is similar. The houses are made using the same construction techniques and contain much the same objects. It's inconvenient to do something expensive and custom.

The rich spend their time more like everyone else too. Bertie Wooster seems long gone. Now, most people who are rich enough not to work do anyway. It's not just social pressure that makes them; idleness is lonely and demoralizing.

Nor do we have the social distinctions there were a hundred years ago. The novels and etiquette manuals of that period read now like descriptions of some strange tribal society. "With respect to the continuance of friendships..." hints *Mrs. Beeton's Book of Household Management* (1880), "it may be found necessary, in some cases, for a mistress to relinquish, on assuming the responsibility of a household, many of

[15] To say anything meaningful about income trends, you have to talk about real income, or income as measured in what it can buy. But the usual way of calculating real income ignores much of the growth in wealth over time, because it depends on a consumer price index created by bolting end to end a series of numbers that are only locally accurate, and that don't include the prices of new inventions until they become so common that their prices stabilize.

So while we might think it was very much better to live in a world with antibiotics or air travel or an electric power grid than without, real income statistics calculated in the usual way will prove to us that we are only slightly richer for having these things.

Another approach would be to ask, if you were going back to the year x in a time machine, how much would you have to spend on trade goods to make your fortune? For example, if you were going back to 1970 it would certainly be less than $500, because the processing power you can get for $500 today would have been worth at least $150 million in 1970. The function goes asymptotic fairly quickly, because for times over a hundred years or so you could get all you needed in present-day trash. In 1800 an empty plastic drink bottle with a screw top would have seemed a miracle of workmanship.

those commenced in the earlier part of her life." A woman who married a rich man was expected to drop friends who didn't. You'd seem a barbarian if you behaved that way today. You'd also have a very boring life. People still tend to segregate themselves somewhat, but much more on the basis of education than wealth. [16]

Materially and socially, technology seems to be decreasing the gap between the rich and the poor, not increasing it. If Lenin walked around the offices of a company like Yahoo or Intel or Cisco, he'd think communism had won. Everyone would be wearing the same clothes, have the same kind of office (or rather, cubicle) with the same furnishings, and address one another by their first names instead of by honorifics. Everything would seem exactly as he'd predicted, until he looked at their bank accounts. Oops.

Is it a problem if technology increases that gap? It doesn't seem to be so far. As it increases the gap in income, it seems to decrease most other gaps.

Alternative to an Axiom

One often hears a policy criticized on the grounds that it would increase the income gap between rich and poor. As if it were an axiom that this would be bad. It might be true that increased variation in income would be bad, but I don't see how we can say it's *axiomatic.*

Indeed, it may even be false, in industrial democracies. In a society

[16] Some will say this amounts to the same thing, because the rich have better opportunities for education. That's a valid point. It is still possible, to a degree, to buy your kids' way into top colleges by sending them to private schools that in effect hack the college admissions process.

According to a 2002 report by the National Center for Education Statistics, about 1.7% of American kids attend private, non-sectarian schools. At Princeton, 36% of the class of 2007 came from such schools. (Interestingly, the number at Harvard is significantly lower, about 28%.) Obviously this is a huge loophole. It does at least seem to be closing, not widening.

Perhaps the designers of admissions processes should take a lesson from the example of computer security, and instead of just assuming that their system can't be hacked, measure the degree to which it is.

of serfs and warlords, certainly, variation in income is a sign of an underlying problem. But serfdom is not the only cause of variation in income. A 747 pilot doesn't make 40 times as much as a checkout clerk because he is a warlord who somehow holds her in thrall. His skills are simply much more valuable.

I'd like to propose an alternative idea: that in a modern society, increasing variation in income is a sign of health. Technology seems to increase the variation in productivity at faster than linear rates. If we don't see corresponding variation in income, there are three possible explanations: (a) that technical innovation has stopped, (b) that the people who would create the most wealth aren't doing it, or (c) that they aren't getting paid for it.

I think we can safely say that (a) and (b) would be bad. If you disagree, try living for a year using only the resources available to the average Frankish nobleman in 800, and report back to us. (I'll be generous and not send you back to the stone age.)

The only option, if you're going to have an increasingly prosperous society without increasing variation in income, seems to be (c), that people will create a lot of wealth without being paid for it. That Jobs and Wozniak, for example, will cheerfully work 20-hour days to produce the Apple computer for a society that allows them, after taxes, to keep just enough of their income to match what they would have made working 9 to 5 at a big company.

Will people create wealth if they can't get paid for it? Only if it's fun. People will write operating systems for free. But they won't install them, or take support calls, or train customers to use them. And at least 90% of the work that even the highest tech companies do is of this second, unedifying kind.

All the unfun kinds of wealth creation slow dramatically in a society that confiscates private fortunes. We can confirm this empirically. Suppose you hear a strange noise that you think may be due to a nearby fan. You turn the fan off, and the noise stops. You turn the fan back on, and the noise starts again. Off, quiet. On, noise. In the absence of other information, it would seem the noise is caused by the fan.

At various times and places in history, whether you could accumulate a fortune by creating wealth has been turned on and off. Northern Italy in 800, off (warlords would steal it). Northern Italy in 1100, on. Central France in 1100, off (still feudal). England in 1800, on. England in 1974, off (98% tax on investment income). United States in 1974, on. We've even had a twin study: West Germany, on; East Germany, off. In every case, the creation of wealth seems to appear and disappear like the noise of a fan as you switch on and off the prospect of keeping it.

There is some momentum involved. It probably takes at least a generation to turn people into East Germans (luckily for England). But if it were merely a fan we were studying, without all the extra baggage that comes from the controversial topic of wealth, no one would have any doubt that the fan was causing the noise.

If you suppress variations in income, whether by stealing private fortunes, as feudal rulers used to do, or by taxing them away, as some modern governments have done, the result always seems to be the same. Society as a whole ends up poorer.

If I had a choice of living in a society where I was materially much better off than I am now, but was among the poorest, or in one where I was the richest, but much worse off than I am now, I'd take the first option. If I had children, it would arguably be immoral not to. It's absolute poverty you want to avoid, not relative poverty. If, as the evidence so far implies, you have to have one or the other in your society, take relative poverty.

You need rich people in your society not so much because in spending their money they create jobs, but because of what they have to do to *get* rich. I'm not talking about the trickle-down effect here. I'm not saying that if you let Henry Ford get rich, he'll hire you as a waiter at his next party. I'm saying that he'll make you a tractor to replace your horse.

Writing

Chapter 6

The Age of the Essay

September 2004

Remember the essays you had to write in high school? Topic sentence, introductory paragraph, supporting paragraphs, conclusion. The conclusion being, say, that Ahab in *Moby Dick* was a Christ-like figure.

Oy. So I'm going to try to give the other side of the story: what an essay really is, and how you write one. Or at least, how I write one.

Mods

The most obvious difference between real essays and the things one has to write in school is that real essays are not exclusively about English literature. Certainly schools should teach students how to write. But due to a series of historical accidents the teaching of writing has gotten mixed together with the study of literature. And so all over the country students are writing not about how a baseball team with a small budget might compete with the Yankees, or the role of color in fashion, or what constitutes a good dessert, but about symbolism in Dickens.

With the result that writing is made to seem boring and pointless. Who cares about symbolism in Dickens? Dickens himself would be more interested in an essay about color or baseball.

How did things get this way? To answer that we have to go back almost a thousand years. Around 1100, Europe at last began to catch its breath after centuries of chaos, and once they had the luxury of curiosity they rediscovered what we call "the classics." The effect was rather as if we were visited by beings from another solar system. These earlier civilizations were so much more sophisticated that for the next several centuries the main work of European scholars, in almost every field, was to assimilate what they knew.

During this period the study of ancient texts acquired great prestige. It seemed the essence of what scholars did. As European scholarship gained momentum it became less and less important; by 1350 someone who wanted to learn about science could find better teachers than Aristotle in his own era. [1] But schools change slower than scholarship. In the 19th century the study of ancient texts was still the backbone of the curriculum.

The time was then ripe for the question: if the study of ancient texts is a valid field for scholarship, why not modern texts? The answer, of course, is that the original raison d'etre of classical scholarship was a kind of intellectual archaeology that does not need to be done in the case of contemporary authors. But for obvious reasons no one wanted to give that answer. The archaeological work being mostly done, it implied that those studying the classics were, if not wasting their time, at least working on problems of minor importance.

And so began the study of modern literature. There was a good deal of resistance at first. The first courses in English literature seem to have been offered by the newer colleges, particularly American ones. Dartmouth, the University of Vermont, Amherst, and University College, London taught English literature in the 1820s. But Harvard didn't have a professor of English literature until 1876, and Oxford not till

[1] I'm thinking of Oresme (c. 1323-82). But it's hard to pick a date, because there was a sudden drop-off in scholarship just as Europeans finished assimilating classical science. The cause may have been the plague of 1347; the trend in scientific progress matches the population curve.

1885. (Oxford had a chair of Chinese before it had one of English.) [2]

What tipped the scales, at least in the US, seems to have been the idea that professors should do research as well as teach. This idea (along with the PhD, the department, and indeed the whole concept of the modern university) was imported from Germany in the late 19th century. Beginning at Johns Hopkins in 1876, the new model spread rapidly.

Writing was one of the casualties. Colleges had long taught English composition. But how do you do research on composition? The professors who taught math could be required to do original math, the professors who taught history could be required to write scholarly articles about history, but what about the professors who taught rhetoric or composition? What should they do research on? The closest thing seemed to be English literature. [3]

And so in the late 19th century the teaching of writing was inherited by English professors. This had two drawbacks: (a) an expert on literature need not himself be a good writer, any more than an art historian has to be a good painter, and (b) the subject of writing now tends to be literature, since that's what the professor is interested in.

High schools imitate universities. The seeds of our miserable high school experiences were sown in 1892, when the National Education

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[2] Parker, William R. "Where Do College English Departments Come From?" *College English* 28 (1966-67), pp. 339-351. Reprinted in Gray, Donald J. (ed). *The Department of English at Indiana University Bloomington 1868-1970.* Indiana University Publications.

Daniels, Robert V. *The University of Vermont: The First Two Hundred Years.* University of Vermont, 1991.

Mueller, Friedrich M. Letter to the *Pall Mall Gazette.* 1886/87. Reprinted in Bacon, Alan (ed). *The Nineteenth-Century History of English Studies.* Ashgate, 1998.

[3] I'm compressing the story a bit. At first literature took a back seat to philology, which (a) seemed more serious and (b) was popular in Germany, where many of the leading scholars of that generation had been trained.

In some cases the writing teachers were transformed *in situ* into English professors. Francis James Child, who had been Boylston Professor of Rhetoric at Harvard since 1851, became in 1876 the university's first professor of English.

Association "formally recommended that literature and composition be unified in the high school course." [4] The 'riting component of the 3 Rs then morphed into English, with the bizarre consequence that high school students now had to write about English literature– to write, without even realizing it, imitations of whatever English professors had been publishing in their journals a few decades before.

It's no wonder if this seems to the student a pointless exercise, because we're now three steps removed from real work: the students are imitating English professors, who are imitating classical scholars, who are merely the inheritors of a tradition growing out of what was, 700 years ago, fascinating and urgently needed work.

No Defense

The other big difference between a real essay and the things they make you write in school is that a real essay doesn't take a position and then defend it. That principle, like the idea that we ought to be writing about literature, turns out to be another intellectual hangover of long forgotten origins.

It's often mistakenly believed that medieval universities were mostly seminaries. In fact they were more law schools. And at least in our tradition lawyers are advocates, trained to take either side of an argument and make as good a case for it as they can. Whether cause or effect, this spirit pervaded early universities. The study of rhetoric, the art of arguing persuasively, was a third of the undergraduate curriculum. [5] And after the lecture the most common form of discussion was the disputation. This is at least nominally preserved in our present-day

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[4] Parker, *op. cit.*, p. 25.

[5] The undergraduate curriculum or *trivium* (whence "trivial") consisted of Latin grammar, rhetoric, and logic. Candidates for masters' degrees went on to study the *quadrivium* of arithmetic, geometry, music, and astronomy. Together these were the seven liberal arts.

The study of rhetoric was inherited directly from Rome, where it was considered the most important subject. It would not be far from the truth to say that education in the classical world meant training landowners' sons to speak well enough to defend their interests in political and legal disputes.

thesis defense: most people treat the words thesis and dissertation as interchangeable, but originally, at least, a thesis was a position one took and the dissertation was the argument by which one defended it.

Defending a position may be a necessary evil in a legal dispute, but it's not the best way to get at the truth, as I think lawyers would be the first to admit. It's not just that you miss subtleties this way. The real problem is that you can't change the question.

And yet this principle is built into the very structure of the things they teach you to write in high school. The topic sentence is your thesis, chosen in advance, the supporting paragraphs the blows you strike in the conflict, and the conclusion– uh, what is the conclusion? I was never sure about that in high school. It seemed as if we were just supposed to restate what we said in the first paragraph, but in different enough words that no one could tell. Why bother? But when you understand the origins of this sort of "essay," you can see where the conclusion comes from. It's the concluding remarks to the jury.

Good writing should be convincing, certainly, but it should be convincing because you got the right answers, not because you did a good job of arguing. When I give a draft of an essay to friends, there are two things I want to know: which parts bore them, and which seem unconvincing. The boring bits can usually be fixed by cutting. But I don't try to fix the unconvincing bits by arguing more cleverly. I need to talk the matter over.

At the very least I must have explained something badly. In that case, in the course of the conversation I'll be forced to come up a with a clearer explanation, which I can just incorporate in the essay. More often than not I have to change what I was saying as well. But the aim is never to be convincing per se. As the reader gets smarter, convincing and true become identical, so if I can convince smart readers I must be near the truth.

The sort of writing that attempts to persuade may be a valid (or at least inevitable) form, but it's historically inaccurate to call it an essay.

An essay is something else.

Trying

To understand what a real essay is, we have to reach back into history again, though this time not so far. To Michel de Montaigne, who in 1580 published a book of what he called "essais." He was doing something quite different from what lawyers do, and the difference is embodied in the name. *Essayer* is the French verb meaning "to try" and an *essai* is an attempt. An essay is something you write to try to figure something out.

Figure out what? You don't know yet. And so you can't begin with a thesis, because you don't have one, and may never have one. An essay doesn't begin with a statement, but with a question. In a real essay, you don't take a position and defend it. You notice a door that's ajar, and you open it and walk in to see what's inside.

If all you want to do is figure things out, why do you need to write anything, though? Why not just sit and think? Well, there precisely is Montaigne's great discovery. Expressing ideas helps to form them. Indeed, helps is far too weak a word. Most of what ends up in my essays I only thought of when I sat down to write them. That's why I write them.

In the things you write in school you are, in theory, merely explaining yourself to the reader. In a real essay you're writing for yourself. You're thinking out loud.

But not quite. Just as inviting people over forces you to clean up your apartment, writing something that other people will read forces you to think well. So it does matter to have an audience. The things I've written just for myself are no good. They tend to peter out. When I run into difficulties, I find I conclude with a few vague questions and then drift off to get a cup of tea.

Many published essays peter out in the same way. Particularly the sort written by the staff writers of newsmagazines. Outside writers tend to supply editorials of the defend-a-position variety, which make a beeline toward a rousing (and foreordained) conclusion. But the staff writers feel obliged to write something "balanced." Since they're writing for a popular magazine, they start with the most radioactively controversial questions, from which– because they're writing for a popular magazine– they then proceed to recoil in terror. Abortion, for or against? This group says one thing. That group says another. One thing is certain: the question is a complex one. (But don't get mad at us. We didn't draw any conclusions.)

The River

Questions aren't enough. An essay has to come up with answers. They don't always, of course. Sometimes you start with a promising question and get nowhere. But those you don't publish. Those are like experiments that get inconclusive results. An essay you publish ought to tell the reader something he didn't already know.

But *what* you tell him doesn't matter, so long as it's interesting. I'm sometimes accused of meandering. In defend-a-position writing that would be a flaw. There you're not concerned with truth. You already know where you're going, and you want to go straight there, blustering through obstacles, and hand-waving your way across swampy ground. But that's not what you're trying to do in an essay. An essay is supposed to be a search for truth. It would be suspicious if it didn't meander.

The Meander (aka Menderes) is a river in Turkey. As you might expect, it winds all over the place. But it doesn't do this out of frivolity. The path it has discovered is the most economical route to the sea. [6]

The river's algorithm is simple. At each step, flow down. For the essayist this translates to: flow interesting. Of all the places to go next, choose the most interesting. One can't have quite as little foresight as a river. I always know generally what I want to write about. But not the specific conclusions I want to reach; from paragraph to paragraph

[6] Trevor Blackwell points out that this isn't strictly true, because the outside edges of curves erode faster.

I let the ideas take their course.

This doesn't always work. Sometimes, like a river, one runs up against a wall. Then I do the same thing the river does: backtrack. At one point in this essay I found that after following a certain thread I ran out of ideas. I had to go back seven paragraphs and start over in another direction.

Fundamentally an essay is a train of thought– but a cleaned-up train of thought, as dialogue is cleaned-up conversation. Real thought, like real conversation, is full of false starts. It would be exhausting to read. You need to cut and fill to emphasize the central thread, like an illustrator inking over a pencil drawing. But don't change so much that you lose the spontaneity of the original.

Err on the side of the river. An essay is not a reference work. It's not something you read looking for a specific answer, and feel cheated if you don't find it. I'd much rather read an essay that went off in an unexpected but interesting direction than one that plodded dutifully along a prescribed course.

Surprise

So what's interesting? For me, interesting means surprise. Interfaces, as Geoffrey James has said, should follow the principle of least astonishment. A button that looks like it will make a machine stop should make it stop, not speed up. Essays should do the opposite. Essays should aim for maximum surprise.

I was afraid of flying for a long time and could only travel vicariously. When friends came back from faraway places, it wasn't just out of politeness that I asked what they saw. I really wanted to know. And I found the best way to get information out of them was to ask what surprised them. How was the place different from what they expected? This is an extremely useful question. You can ask it of the most unobservant people, and it will extract information they didn't even know they were recording.

Surprises are things that you not only didn't know, but that contradict things you thought you knew. And so they're the most valuable sort of fact you can get. They're like a food that's not merely healthy, but counteracts the unhealthy effects of things you've already eaten.

How do you find surprises? Well, therein lies half the work of essay writing. (The other half is expressing yourself well.) The trick is to use yourself as a proxy for the reader. You should only write about things you've thought about a lot. And anything you come across that surprises you, who've thought about the topic a lot, will probably surprise most readers.

For example, in a recent essay I pointed out that because you can only judge computer programmers by working with them, no one knows who the best programmers are overall. I didn't realize this when I began that essay, and even now I find it kind of weird. That's what you're looking for.

So if you want to write essays, you need two ingredients: a few topics you've thought about a lot, and some ability to ferret out the unexpected.

What should you think about? My guess is that it doesn't matter– that anything can be interesting if you get deeply enough into it. One possible exception might be things that have deliberately had all the variation sucked out of them, like working in fast food. In retrospect, was there anything interesting about working at Baskin-Robbins? Well, it was interesting how important color was to the customers. Kids a certain age would point into the case and say that they wanted yellow. Did they want French Vanilla or Lemon? They would just look at you blankly. They wanted yellow. And then there was the mystery of why the perennial favorite Pralines 'n' Cream was so appealing. (I think now it was the salt.) And the difference in the way fathers and mothers bought ice cream for their kids: the fathers like benevolent kings bestowing largesse, the mothers harried, giving in to pressure. So, yes, there does seem to be some material even in fast food.

I didn't notice those things at the time, though. At sixteen I was about as observant as a lump of rock. I can see more now in the fragments of memory I preserve of that age than I could see at the time from having it all happening live, right in front of me.

Observation

So the ability to ferret out the unexpected must not merely be an inborn one. It must be something you can learn. How do you learn it?

To some extent it's like learning history. When you first read history, it's just a whirl of names and dates. Nothing seems to stick. But the more you learn, the more hooks you have for new facts to stick onto—which means you accumulate knowledge at an exponential rate. Once you remember that Normans conquered England in 1066, it will catch your attention when you hear that other Normans conquered southern Italy at about the same time. Which will make you wonder about Normandy, and take note when a third book mentions that Normans were not, like most of what is now called France, tribes that flowed in as the Roman empire collapsed, but Vikings (norman = north man) who arrived four centuries later in 911. Which makes it easier to remember that Dublin was also established by Vikings in the 840s. Etc, etc squared.

Collecting surprises is a similar process. The more anomalies you've seen, the more easily you'll notice new ones. Which means, oddly enough, that as you grow older, life should become more and more surprising. When I was a kid, I used to think adults had it all figured out. I had it backwards. Kids are the ones who have it all figured out. They're just mistaken.

When it comes to surprises, the rich get richer. But (as with wealth) there may be habits of mind that will help the process along. It's good to have a habit of asking questions, especially questions beginning with Why. But not in the random way that three year olds ask why. There are an infinite number of questions. How do you find the fruitful ones?

I find it especially useful to ask why about things that seem wrong. For example, why should there be a connection between humor and misfortune? Why do we find it funny when a character, even one we like, slips on a banana peel? There's a whole essay's worth of surprises there for sure.

If you want to notice things that seem wrong, you'll find a degree of skepticism helpful. I take it as an axiom that we're only achieving 1% of what we could. This helps counteract the rule that gets beaten into our heads as children: that things are the way they are because that is how things have to be. For example, everyone I've talked to while writing this essay felt the same about English classes– that the whole process seemed pointless. But none of us had the balls at the time to hypothesize that it was, in fact, all a mistake. We all thought there was just something we weren't getting.

I have a hunch you want to pay attention not just to things that seem wrong, but things that seem wrong in a humorous way. I'm always pleased when I see someone laugh as they read a draft of an essay. But why should I be? I'm aiming for good ideas. Why should good ideas be funny? The connection may be surprise. Surprises make us laugh, and surprises are what one wants to deliver.

I write down things that surprise me in notebooks. I never actually get around to reading them and using what I've written, but I do tend to reproduce the same thoughts later. So the main value of notebooks may be what writing things down leaves in your head.

People trying to be cool will find themselves at a disadvantage when collecting surprises. To be surprised is to be mistaken. And the essence of cool, as any fourteen year old could tell you, is *nil admirari.* When you're mistaken, don't dwell on it; just act like nothing's wrong and maybe no one will notice.

One of the keys to coolness is to avoid situations where inexperience may make you look foolish. If you want to find surprises you should do the opposite. Study lots of different things, because some of the most interesting surprises are unexpected connections between different fields. For example, jam, bacon, pickles, and cheese, which are among the most pleasing of foods, were all originally intended as methods of preservation. And so were books and paintings.

Whatever you study, include history– but social and economic history, not political history. History seems to me so important that it's misleading to treat it as a mere field of study. Another way to describe it is *all the data we have so far.*

Among other things, studying history gives one confidence that there are good ideas waiting to be discovered right under our noses. Swords evolved during the Bronze Age out of daggers, which (like their flint predecessors) had a hilt separate from the blade. Because swords are longer the hilts kept breaking off. But it took five hundred years before someone thought of casting hilt and blade as one piece.

Disobedience

Above all, make a habit of paying attention to things you're not supposed to, either because they're "inappropriate," or not important, or not what you're supposed to be working on. If you're curious about something, trust your instincts. Follow the threads that attract your attention. If there's something you're really interested in, you'll find they have an uncanny way of leading back to it anyway, just as the conversation of people who are especially proud of something always tends to lead back to it.

For example, I've always been fascinated by comb-overs, especially the extreme sort that make a man look as if he's wearing a beret made of his own hair. Surely this is a lowly sort of thing to be interested in– the sort of superficial quizzing best left to teenage girls. And yet there is something underneath. The key question, I realized, is how does the comber-over not see how odd he looks? And the answer is that he got to look that way *incrementally.* What began as combing his hair a little carefully over a thin patch has gradually, over 20 years, grown into a monstrosity. Gradualness is very powerful. And that power can be used for constructive purposes too: just as you can trick yourself into looking like a freak, you can trick yourself into creating something so grand that you would never have dared to *plan* such a thing. Indeed, this is just how most good software gets created. You start by writing a stripped-down kernel (how hard can it be?) and gradually it grows into a complete operating system. Hence the next leap: could you do the same thing in painting, or in a novel?

See what you can extract from a frivolous question? If there's one piece of advice I would give about writing essays, it would be: don't do as you're told. Don't believe what you're supposed to. Don't write the essay readers expect; one learns nothing from what one expects. And don't write the way they taught you to in school.

The most important sort of disobedience is to write essays at all. Fortunately, this sort of disobedience shows signs of becoming rampant. It used to be that only a tiny number of officially approved writers were allowed to write essays. Magazines published few of them, and judged them less by what they said than who wrote them; a magazine might publish a story by an unknown writer if it was good enough, but if they published an essay on x it had to be by someone who was at least forty and whose job title had x in it. Which is a problem, because there are a lot of things insiders can't say precisely because they're insiders.

The Internet is changing that. Anyone can publish an essay on the Web, and it gets judged, as any writing should, by what it says, not who wrote it. Who are you to write about x? You are whatever you wrote.

Popular magazines made the period between the spread of literacy and the arrival of TV the golden age of the short story. The Web may well make this the golden age of the essay. And that's certainly not something I realized when I started writing this.

Thanks to Ken Anderson, Trevor Blackwell, Sarah Harlin, Jessica Livingston, Jackie McDonough, and Robert Morris for reading drafts of this.

Art

Chapter 7

How Art Can Be Good

December 2006

I grew up believing that taste is just a matter of personal preference. Each person has things they like, but no one's preferences are any better than anyone else's. There is no such thing as *good* taste.

Like a lot of things I grew up believing, this turns out to be false, and I'm going to try to explain why.

One problem with saying there's no such thing as good taste is that it also means there's no such thing as good art. If there were good art, then people who liked it would have better taste than people who didn't. So if you discard taste, you also have to discard the idea of art being good, and artists being good at making it.

It was pulling on that thread that unravelled my childhood faith in relativism. When you're trying to make things, taste becomes a practical matter. You have to decide what to do next. Would it make the painting better if I changed that part? If there's no such thing as better, it doesn't matter what you do. In fact, it doesn't matter if you paint at all. You could just go out and buy a ready-made blank canvas. If there's no such thing as good, that would be just as great an achievement as the ceiling of the Sistine Chapel. Less laborious, certainly, but if you can achieve the same level of performance with less effort, surely that's more impressive, not less.

Yet that doesn't seem quite right, does it?

Audience

I think the key to this puzzle is to remember that art has an audience. Art has a purpose, which is to interest its audience. Good art (like good anything) is art that achieves its purpose particularly well. The meaning of "interest" can vary. Some works of art are meant to shock, and others to please; some are meant to jump out at you, and others to sit quietly in the background. But all art has to work on an audience, and—here's the critical point—members of the audience share things in common.

For example, nearly all humans find human faces engaging. It seems to be wired into us. Babies can recognize faces practically from birth. In fact, faces seem to have co-evolved with our interest in them; the face is the body's billboard. So all other things being equal, a painting with faces in it will interest people more than one without. [1]

One reason it's easy to believe that taste is merely personal preference is that, if it isn't, how do you pick out the people with better taste? There are billions of people, each with their own opinion; on what grounds can you prefer one to another? [2]

But if audiences have a lot in common, you're not in a position of having to choose one out of a random set of individual biases, because the set isn't random. All humans find faces engaging—practically by

[1] This is not to say, of course, that good paintings must have faces in them, just that everyone's visual piano has that key on it. There are situations in which you want to avoid faces, precisely because they attract so much attention. But you can see how universally faces work by their prevalence in advertising.

[2] The other reason it's easy to believe is that it makes people feel good. To a kid, this idea is crack. In every other respect they're constantly being told that they have a lot to learn. But in this they're perfect. Their opinion carries the same weight as any adult's. You should probably question anything you believed as a kid that you'd want to believe this much.

definition: face recognition is in our DNA. And so having a notion of good art, in the sense of art that does its job well, doesn't require you to pick out a few individuals and label their opinions as correct. No matter who you pick, they'll find faces engaging.

Of course, space aliens probably wouldn't find human faces engaging. But there might be other things they shared in common with us. The most likely source of examples is math. I expect space aliens would agree with us most of the time about which of two proofs was better. Erdos thought so. He called a maximally elegant proof one out of God's book, and presumably God's book is universal. [3]

Once you start talking about audiences, you don't have to argue simply that there are or aren't standards of taste. Instead tastes are a series of concentric rings, like ripples in a pond. There are some things that will appeal to you and your friends, others that will appeal to most people your age, others that will appeal to most humans, and perhaps others that would appeal to most sentient beings (whatever that means).

The picture is slightly more complicated than that, because in the middle of the pond there are overlapping sets of ripples. For example, there might be things that appealed particularly to men, or to people from a certain culture.

If good art is art that interests its audience, then when you talk about art being good, you also have to say for what audience. So is it meaningless to talk about art simply being good or bad? No, because one audience is the set of all possible humans. I think that's the audience people are implicitly talking about when they say a work of art is good: they mean it would engage any human. [4]

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[3] It's conceivable that the elegance of proofs is quantifiable, in the sense that there may be some formal measure that turns out to coincide with mathematicians' judgements. Perhaps it would be worth trying to make a formal language for proofs in which those considered more elegant consistently came out shorter (perhaps after being macroexpanded or compiled).

[4] Maybe it would be possible to make art that would appeal to space aliens, but

And that is a meaningful test, because although, like any everyday concept, "human" is fuzzy around the edges, there are a lot of things practically all humans have in common. In addition to our interest in faces, there's something special about primary colors for nearly all of us, because it's an artifact of the way our eyes work. Most humans will also find images of 3D objects engaging, because that also seems to be built into our visual perception. [5] And beneath that there's edge-finding, which makes images with definite shapes more engaging than mere blur.

Humans have a lot more in common than this, of course. My goal is not to compile a complete list, just to show that there's some solid ground here. People's preferences aren't random. So an artist working on a painting and trying to decide whether to change some part of it doesn't have to think "Why bother? I might as well flip a coin." Instead he can ask "What would make the painting more interesting to people?" And the reason you can't equal Michelangelo by going out and buying a blank canvas is that the ceiling of the Sistine Chapel is more interesting to people.

A lot of philosophers have had a hard time believing it was possible for there to be objective standards for art. It seemed obvious that beauty, for example, was something that happened in the head of the observer, not something that was a property of objects. It was thus "subjective" rather than "objective." But in fact if you narrow the definition of beauty to something that works a certain way on humans, and you observe how much humans have in common, it turns out to be a property of objects after all. You don't have to choose between something being a property of the subject or the object if subjects all react similarly. Being good art is thus a property of objects as much

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I'm not going to get into that because (a) it's too hard to answer, and (b) I'm satisfied if I can establish that good art is a meaningful idea for human audiences.

[5] If early abstract paintings seem more interesting than later ones, it may be because the first abstract painters were trained to paint from life, and their hands thus tended to make the kind of gestures you use in representing physical things. In effect they were saying "scaramara" instead of "uebfgbsb."

as, say, being toxic to humans is: it's good art if it consistently affects humans in a certain way.

Error

So could we figure out what the best art is by taking a vote? After all, if appealing to humans is the test, we should be able to just ask them, right?

Well, not quite. For products of nature that might work. I'd be willing to eat the apple the world's population had voted most delicious, and I'd probably be willing to visit the beach they voted most beautiful, but having to look at the painting they voted the best would be a crapshoot.

Man-made stuff is different. For one thing, artists, unlike apple trees, often deliberately try to trick us. Some tricks are quite subtle. For example, any work of art sets expectations by its level of finish. You don't expect photographic accuracy in something that looks like a quick sketch. So one widely used trick, especially among illustrators, is to intentionally make a painting or drawing look like it was done faster than it was. The average person looks at it and thinks: how amazingly skillful. It's like saying something clever in a conversation as if you'd thought of it on the spur of the moment, when in fact you'd worked it out the day before.

Another much less subtle influence is brand. If you go to see the Mona Lisa, you'll probably be disappointed, because it's hidden behind a thick glass wall and surrounded by a frenzied crowd taking pictures of themselves in front of it. At best you can see it the way you see a friend across the room at a crowded party. The Louvre might as well replace it with copy; no one would be able to tell. And yet the Mona Lisa is a small, dark painting. If you found people who'd never seen an image of it and sent them to a museum in which it was hanging among other paintings with a tag labelling it as a portrait by an unknown fifteenth century artist, most would walk by without giving it a second look.

For the average person, brand dominates all other factors in the judgement of art. Seeing a painting they recognize from reproductions is so overwhelming that their response to it as a painting is drowned out.

And then of course there are the tricks people play on themselves. Most adults looking at art worry that if they don't like what they're supposed to, they'll be thought uncultured. This doesn't just affect what they claim to like; they actually make themselves like things they're supposed to.

That's why you can't just take a vote. Though appeal to people is a meaningful test, in practice you can't measure it, just as you can't find north using a compass with a magnet sitting next to it. There are sources of error so powerful that if you take a vote, all you're measuring is the error.

We can, however, approach our goal from another direction, by using ourselves as guinea pigs. You're human. If you want to know what the basic human reaction to a piece of art would be, you can at least approach that by getting rid of the sources of error in your own judgements.

For example, while anyone's reaction to a famous painting will be warped at first by its fame, there are ways to decrease its effects. One is to come back to the painting over and over. After a few days the fame wears off, and you can start to see it as a painting. Another is to stand close. A painting familiar from reproductions looks more familiar from ten feet away; close in you see details that get lost in reproductions, and which you're therefore seeing for the first time.

There are two main kinds of error that get in the way of seeing a work of art: biases you bring from your own circumstances, and tricks played by the artist. Tricks are straightforward to correct for. Merely being aware of them usually prevents them from working. For example, when I was ten I used to be very impressed by airbrushed lettering that looked like shiny metal. But once you study how it's done, you see that it's a pretty cheesy trick—one of the sort that relies on pushing a few visual buttons really hard to temporarily overwhelm the viewer.

It's like trying to convince someone by shouting at them.

The way not to be vulnerable to tricks is to explicitly seek out and catalog them. When you notice a whiff of dishonesty coming from some kind of art, stop and figure out what's going on. When someone is obviously pandering to an audience that's easily fooled, whether it's someone making shiny stuff to impress ten year olds, or someone making conspicuously avant-garde stuff to impress would-be intellectuals, learn how they do it. Once you've seen enough examples of specific types of tricks, you start to become a connoisseur of trickery in general, just as professional magicians are.

What counts as a trick? Roughly, it's something done with contempt for the audience. For example, the guys designing Ferraris in the 1950s were probably designing cars that they themselves admired. Whereas I suspect over at General Motors the marketing people are telling the designers, "Most people who buy SUVs do it to seem manly, not to drive off-road. So don't worry about the suspension; just make that sucker as big and tough-looking as you can." [6]

I think with some effort you can make yourself nearly immune to tricks. It's harder to escape the influence of your own circumstances, but you can at least move in that direction. The way to do it is to travel widely, in both time and space. If you go and see all the different kinds of things people like in other cultures, and learn about all the different things people have liked in the past, you'll probably find it changes what you like. I doubt you could ever make yourself into a completely universal person, if only because you can only travel in one direction in time. But if you find a work of art that would appeal equally to your friends, to people in Nepal, and to the ancient Greeks, you're probably onto something.

My main point here is not how to have good taste, but that there can even be such a thing. And I think I've shown that. There is such a

[6] It's a bit more complicated, because sometimes artists unconsciously use tricks by imitating art that does.

thing as good art. It's art that interests its human audience, and since humans have a lot in common, what interests them is not random. Since there's such a thing as good art, there's also such a thing as good taste, which is the ability to recognize it.

If we were talking about the taste of apples, I'd agree that taste is just personal preference. Some people like certain kinds of apples and others like other kinds, but how can you say that one is right and the other wrong? [7]

The thing is, art isn't apples. Art is man-made. It comes with a lot of cultural baggage, and in addition the people who make it often try to trick us. Most people's judgement of art is dominated by these extraneous factors; they're like someone trying to judge the taste of apples in a dish made of equal parts apples and jalapeno peppers. All they're tasting is the peppers. So it turns out you can pick out some people and say that they have better taste than others: they're the ones who actually taste art like apples.

Or to put it more prosaically, they're the people who (a) are hard to trick, and (b) don't just like whatever they grew up with. If you could find people who'd eliminated all such influences on their judgement, you'd probably still see variation in what they liked. But because humans have so much in common, you'd also find they agreed on a lot. They'd nearly all prefer the ceiling of the Sistine Chapel to a blank canvas.

Making It

I wrote this essay because I was tired of hearing "taste is subjective" and wanted to kill it once and for all. Anyone who makes things knows intuitively that's not true. When you're trying to make art, the temptation to be lazy is as great as in any other kind of work. Of course it matters to do a good job. And yet you can see how great a hold "taste

[7] I phrased this in terms of the taste of apples because if people can see the apples, they can be fooled. When I was a kid most apples were a variety called Red Delicious that had been bred to look appealing in stores, but which didn't taste very good.

is subjective" has even in the art world by how nervous it makes people to talk about art being good or bad. Those whose jobs require them to judge art, like curators, mostly resort to euphemisms like "significant" or "important" or (getting dangerously close) "realized." [8]

I don't have any illusions that being able to talk about art being good or bad will cause the people who talk about it to have anything more useful to say. Indeed, one of the reasons "taste is subjective" found such a receptive audience is that, historically, the things people have said about good taste have generally been such nonsense.

It's not for the people who talk about art that I want to free the idea of good art, but for those who make it. Right now, ambitious kids going to art school run smack into a brick wall. They arrive hoping one day to be as good as the famous artists they've seen in books, and the first thing they learn is that the concept of good has been retired. Instead everyone is just supposed to explore their own personal vision. [9]

When I was in art school, we were looking one day at a slide of some great fifteenth century painting, and one of the students asked "Why don't artists paint like that now?" The room suddenly got quiet. Though rarely asked out loud, this question lurks uncomfortably in the back of every art student's mind. It was as if someone had brought up the topic of lung cancer in a meeting within Philip Morris.

"Well," the professor replied, "we're interested in different questions now." He was a pretty nice guy, but at the time I couldn't help wish-

[8]To be fair, curators are in a difficult position. If they're dealing with recent art, they have to include things in shows that they think are bad. That's because the test for what gets included in shows is basically the market price, and for recent art that is largely determined by successful businessmen and their wives. So it's not always intellectual dishonesty that makes curators and dealers use neutral-sounding language.

[9]What happens in practice is that everyone gets really good at *talking* about art. As the art itself gets more random, the effort that would have gone into the work goes instead into the intellectual sounding theory behind it. "My work represents an exploration of gender and sexuality in an urban context," etc. Different people win at that game.

ing I could send him back to fifteenth century Florence to explain in person to Leonardo & Co. how we had moved beyond their early, limited concept of art. Just imagine that conversation.

In fact, one of the reasons artists in fifteenth century Florence made such great things was that they believed you could make great things. [10] They were intensely competitive and were always trying to outdo one another, like mathematicians or physicists today—maybe like anyone who has ever done anything really well.

The idea that you could make great things was not just a useful illusion. They were actually right. So the most important consequence of realizing there can be good art is that it frees artists to try to make it. To the ambitious kids arriving at art school this year hoping one day to make great things, I say: don't believe it when they tell you this is a naive and outdated ambition. There is such a thing as good art, and if you try to make it, there are people who will notice.

Thanks to Trevor Blackwell, Jessica Livingston, and Robert Morris for reading drafts of this, and to Paul Watson for permission to use the image at the top.

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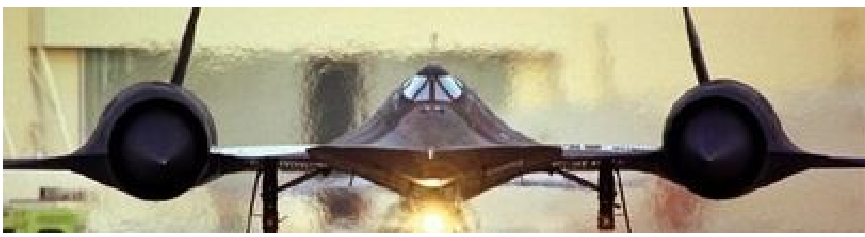
[10]There were several other reasons, including that Florence was then the richest and most sophisticated city in the world, and that they lived in a time before photography had (a) killed portraiture as a source of income and (b) made brand the dominant factor in the sale of art.

Incidentally, I'm not saying that good art = fifteenth century European art. I'm not saying we should make what they made, but that we should work like they worked. There are fields now in which many people work with the same energy and honesty that fifteenth century artists did, but art is not one of them.

Chapter 8

Taste for Makers



February 2002

"…Copernicus' aesthetic objections to [equants] provided one essential motive for his rejection of the Ptolemaic system…."

- Thomas Kuhn, *The Copernican Revolution*

"All of us had been trained by Kelly Johnson and believed fanatically in his insistence that an airplane that looked beautiful would fly the same way."

- Ben Rich, *Skunk Works*

"Beauty is the first test: there is no permanent place in this world for ugly mathematics."

- G. H. Hardy, *A Mathematician's Apology*

I was talking recently to a friend who teaches at MIT. His field is hot now and every year he is inundated by applications from would-be graduate students. "A lot of them seem smart," he said. "What I can't tell is whether they have any kind of taste."

Taste. You don't hear that word much now. And yet we still need the underlying concept, whatever we call it. What my friend meant was that he wanted students who were not just good technicians, but who could use their technical knowledge to design beautiful things.

Mathematicians call good work "beautiful," and so, either now or in the past, have scientists, engineers, musicians, architects, designers, writers, and painters. Is it just a coincidence that they used the same word, or is there some overlap in what they meant? If there is an overlap, can we use one field's discoveries about beauty to help us in another?

For those of us who design things, these are not just theoretical questions. If there is such a thing as beauty, we need to be able to recognize it. We need good taste to make good things. Instead of treating beauty as an airy abstraction, to be either blathered about or avoided depending on how one feels about airy abstractions, let's try considering it as a practical question: *how do you make good stuff?*

If you mention taste nowadays, a lot of people will tell you that "taste is subjective." They believe this because it really feels that way to them. When they like something, they have no idea why. It could be because it's beautiful, or because their mother had one, or because they saw a movie star with one in a magazine, or because they know it's expensive. Their thoughts are a tangle of unexamined impulses.

Most of us are encouraged, as children, to leave this tangle unexamined. If you make fun of your little brother for coloring people green in his coloring book, your mother is likely to tell you something like "you like to do it your way and he likes to do it his way."

Your mother at this point is not trying to teach you important truths about aesthetics. She's trying to get the two of you to stop bickering.

Like many of the half-truths adults tell us, this one contradicts other things they tell us. After dinning into you that taste is merely a matter of personal preference, they take you to the museum and tell you that you should pay attention because Leonardo is a great artist.

What goes through the kid's head at this point? What does he think "great artist" means? After having been told for years that everyone just likes to do things their own way, he is unlikely to head straight for the conclusion that a great artist is someone whose work is *better* than the others'. A far more likely theory, in his Ptolemaic model of the universe, is that a great artist is something that's good for you, like broccoli, because someone said so in a book.

Saying that taste is just personal preference is a good way to prevent disputes. The trouble is, it's not true. You feel this when you start to design things.

Whatever job people do, they naturally want to do better. Football players like to win games. CEOs like to increase earnings. It's a matter of pride, and a real pleasure, to get better at your job. But if your job is to design things, and there is no such thing as beauty, then there is *no way to get better at your job.* If taste is just personal preference, then everyone's is already perfect: you like whatever you like, and that's it.

As in any job, as you continue to design things, you'll get better at it. Your tastes will change. And, like anyone who gets better at their job, you'll know you're getting better. If so, your old tastes were not merely different, but worse. Poof goes the axiom that taste can't be wrong.

Relativism is fashionable at the moment, and that may hamper you from thinking about taste, even as yours grows. But if you come out of the closet and admit, at least to yourself, that there is such a thing as good and bad design, then you can start to study good design in detail. How has your taste changed? When you made mistakes, what caused you to make them? What have other people learned about design?

Once you start to examine the question, it's surprising how much different fields' ideas of beauty have in common. The same principles of good design crop up again and again.

Good design is simple. You hear this from math to painting. In math it means that a shorter proof tends to be a better one. Where axioms are concerned, especially, less is more. It means much the same thing in programming. For architects and designers it means that beauty should depend on a few carefully chosen structural elements rather than a profusion of superficial ornament. (Ornament is not in itself bad, only when it's camouflage on insipid form.) Similarly, in painting, a still life of a few carefully observed and solidly modelled objects will tend to be more interesting than a stretch of flashy but mindlessly repetitive painting of, say, a lace collar. In writing it means: say what you mean and say it briefly.

It seems strange to have to emphasize simplicity. You'd think simple would be the default. Ornate is more work. But something seems to come over people when they try to be creative. Beginning writers adopt a pompous tone that doesn't sound anything like the way they speak. Designers trying to be artistic resort to swooshes and curlicues. Painters discover that they're expressionists. It's all evasion. Underneath the long words or the "expressive" brush strokes, there is not much going on, and that's frightening.

When you're forced to be simple, you're forced to face the real problem. When you can't deliver ornament, you have to deliver substance.

Good design is timeless. In math, every proof is timeless unless it contains a mistake. So what does Hardy mean when he says there is no permanent place for ugly mathematics? He means the same thing Kelly Johnson did: if something is ugly, it can't be the best solution. There must be a better one, and eventually someone will discover it.

Aiming at timelessness is a way to make yourself find the best answer: if you can imagine someone surpassing you, you should do it yourself. Some of the greatest masters did this so well that they left little room for those who came after. Every engraver since Durer has had to live in his shadow.

Aiming at timelessness is also a way to evade the grip of fashion. Fashions almost by definition change with time, so if you can make something that will still look good far into the future, then its appeal must derive more from merit and less from fashion.

Strangely enough, if you want to make something that will appeal to future generations, one way to do it is to try to appeal to past generations. It's hard to guess what the future will be like, but we can be sure it will be like the past in caring nothing for present fashions. So if you can make something that appeals to people today and would also have appealed to people in 1500, there is a good chance it will appeal to people in 2500.

Good design solves the right problem. The typical stove has four burners arranged in a square, and a dial to control each. How do you arrange the dials? The simplest answer is to put them in a row. But this is a simple answer to the wrong question. The dials are for humans to use, and if you put them in a row, the unlucky human will have to stop and think each time about which dial matches which burner. Better to arrange the dials in a square like the burners.

A lot of bad design is industrious, but misguided. In the mid twentieth century there was a vogue for setting text in sans-serif fonts. These fonts *are* closer to the pure, underlying letterforms. But in text that's not the problem you're trying to solve. For legibility it's more important that letters be easy to tell apart. It may look Victorian, but a Times Roman lowercase g is easy to tell from a lowercase y.

Problems can be improved as well as solutions. In software, an intractable problem can usually be replaced by an equivalent one that's easy to solve. Physics progressed faster as the problem became predicting observable behavior, instead of reconciling it with scripture.

Good design is suggestive. Jane Austen's novels contain almost no description; instead of telling you how everything looks, she tells her story so well that you envision the scene for yourself. Likewise, a painting that suggests is usually more engaging than one that tells. Everyone makes up their own story about the Mona Lisa.

In architecture and design, this principle means that a building or object should let you use it how you want: a good building, for example, will serve as a backdrop for whatever life people want to lead in it, instead of making them live as if they were executing a program written by the architect.

In software, it means you should give users a few basic elements that they can combine as they wish, like Lego. In math it means a proof that becomes the basis for a lot of new work is preferable to a proof that was difficult, but doesn't lead to future discoveries; in the sciences generally, citation is considered a rough indicator of merit.

Good design is often slightly funny. This one may not always be true. But Durer's engravings and Saarinen's womb chair and the Pantheon and the original Porsche 911 all seem to me slightly funny. Godel's incompleteness theorem seems like a practical joke.

I think it's because humor is related to strength. To have a sense of humor is to be strong: to keep one's sense of humor is to shrug off misfortunes, and to lose one's sense of humor is to be wounded by them. And so the mark– or at least the prerogative– of strength is not to take oneself too seriously. The confident will often, like swallows, seem to be making fun of the whole process slightly, as Hitchcock does in his films or Bruegel in his paintings– or Shakespeare, for that matter.

Good design may not have to be funny, but it's hard to imagine something that could be called humorless also being good design.

Good design is hard. If you look at the people who've done great work, one thing they all seem to have in common is that they worked very hard. If you're not working hard, you're probably wasting your time.

Hard problems call for great efforts. In math, difficult proofs require ingenious solutions, and those tend to be interesting. Ditto in engineering.

When you have to climb a mountain you toss everything unnecessary out of your pack. And so an architect who has to build on a difficult site, or a small budget, will find that he is forced to produce an elegant design. Fashions and flourishes get knocked aside by the difficult business of solving the problem at all.

Not every kind of hard is good. There is good pain and bad pain. You want the kind of pain you get from going running, not the kind you get from stepping on a nail. A difficult problem could be good for a designer, but a fickle client or unreliable materials would not be.

In art, the highest place has traditionally been given to paintings of people. There is something to this tradition, and not just because pictures of faces get to press buttons in our brains that other pictures don't. We are so good at looking at faces that we force anyone who draws them to work hard to satisfy us. If you draw a tree and you change the angle of a branch five degrees, no one will know. When you change the angle of someone's eye five degrees, people notice.

When Bauhaus designers adopted Sullivan's "form follows function," what they meant was, form *should* follow function. And if function is hard enough, form is forced to follow it, because there is no effort to spare for error. Wild animals are beautiful because they have hard lives.

Good design looks easy. Like great athletes, great designers make it look easy. Mostly this is an illusion. The easy, conversational tone of good writing comes only on the eighth rewrite.

In science and engineering, some of the greatest discoveries seem so simple that you say to yourself, I could have thought of that. The discoverer is entitled to reply, why didn't you?

Some Leonardo heads are just a few lines. You look at them and you think, all you have to do is get eight or ten lines in the right place and you've made this beautiful portrait. Well, yes, but you have to get them in *exactly* the right place. The slightest error will make the whole thing collapse.

Line drawings are in fact the most difficult visual medium, because they demand near perfection. In math terms, they are a closed-form solution; lesser artists literally solve the same problems by successive approximation. One of the reasons kids give up drawing at ten or so is that they decide to start drawing like grownups, and one of the first things they try is a line drawing of a face. Smack!

In most fields the appearance of ease seems to come with practice. Perhaps what practice does is train your unconscious mind to handle tasks that used to require conscious thought. In some cases you literally train your body. An expert pianist can play notes faster than the brain can send signals to his hand. Likewise an artist, after a while, can make visual perception flow in through his eye and out through his hand as automatically as someone tapping his foot to a beat.

When people talk about being in "the zone," I think what they mean is that the spinal cord has the situation under control. Your spinal cord is less hesitant, and it frees conscious thought for the hard problems.

Good design uses symmetry. I think symmetry may just be one way to achieve simplicity, but it's important enough to be mentioned on its own. Nature uses it a lot, which is a good sign.

There are two kinds of symmetry, repetition and recursion. Recursion means repetition in subelements, like the pattern of veins in a leaf.

Symmetry is unfashionable in some fields now, in reaction to excesses in the past. Architects started consciously making buildings asymmetric in Victorian times and by the 1920s asymmetry was an explicit premise of modernist architecture. Even these buildings only tended to be asymmetric about major axes, though; there were hundreds of minor symmetries.

In writing you find symmetry at every level, from the phrases in a sentence to the plot of a novel. You find the same in music and art. Mosaics (and some Cezannes) get extra visual punch by making the whole picture out of the same atoms. Compositional symmetry yields some of the most memorable paintings, especially when two halves react to one another, as in the *Creation of Adam* or *American Gothic.*

In math and engineering, recursion, especially, is a big win. Inductive proofs are wonderfully short. In software, a problem that can be solved by recursion is nearly always best solved that way. The Eiffel

Tower looks striking partly because it is a recursive solution, a tower on a tower.

The danger of symmetry, and repetition especially, is that it can be used as a substitute for thought.

Good design resembles nature. It's not so much that resembling nature is intrinsically good as that nature has had a long time to work on the problem. It's a good sign when your answer resembles nature's.

It's not cheating to copy. Few would deny that a story should be like life. Working from life is a valuable tool in painting too, though its role has often been misunderstood. The aim is not simply to make a record. The point of painting from life is that it gives your mind something to chew on: when your eyes are looking at something, your hand will do more interesting work.

Imitating nature also works in engineering. Boats have long had spines and ribs like an animal's ribcage. In some cases we may have to wait for better technology: early aircraft designers were mistaken to design aircraft that looked like birds, because they didn't have materials or power sources light enough (the Wrights' engine weighed 152 lbs. and generated only 12 hp.) or control systems sophisticated enough for machines that flew like birds, but I could imagine little unmanned reconnaissance planes flying like birds in fifty years.

Now that we have enough computer power, we can imitate nature's method as well as its results. Genetic algorithms may let us create things too complex to design in the ordinary sense.

Good design is redesign. It's rare to get things right the first time. Experts expect to throw away some early work. They plan for plans to change.

It takes confidence to throw work away. You have to be able to think, *there's more where that came from.* When people first start drawing, for example, they're often reluctant to redo parts that aren't right; they feel they've been lucky to get that far, and if they try to redo something, it will turn out worse. Instead they convince themselves that the drawing is not that bad, really– in fact, maybe they meant it to look that way.

Dangerous territory, that; if anything you should cultivate dissatisfaction. In Leonardo's drawings there are often five or six attempts to get a line right. The distinctive back of the Porsche 911 only appeared in the redesign of an awkward prototype. In Wright's early plans for the Guggenheim, the right half was a ziggurat; he inverted it to get the present shape.

Mistakes are natural. Instead of treating them as disasters, make them easy to acknowledge and easy to fix. Leonardo more or less invented the sketch, as a way to make drawing bear a greater weight of exploration. Open-source software has fewer bugs because it admits the possibility of bugs.

It helps to have a medium that makes change easy. When oil paint replaced tempera in the fifteenth century, it helped painters to deal with difficult subjects like the human figure because, unlike tempera, oil can be blended and overpainted.

Good design can copy. Attitudes to copying often make a round trip. A novice imitates without knowing it; next he tries consciously to be original; finally, he decides it's more important to be right than original.

Unknowing imitation is almost a recipe for bad design. If you don't know where your ideas are coming from, you're probably imitating an imitator. Raphael so pervaded mid-nineteenth century taste that almost anyone who tried to draw was imitating him, often at several removes. It was this, more than Raphael's own work, that bothered the Pre-Raphaelites.

The ambitious are not content to imitate. The second phase in the growth of taste is a conscious attempt at originality.

I think the greatest masters go on to achieve a kind of selflessness.

They just want to get the right answer, and if part of the right answer has already been discovered by someone else, that's no reason not to use it. They're confident enough to take from anyone without feeling that their own vision will be lost in the process.

Good design is often strange. Some of the very best work has an uncanny quality: Euler's Formula, Bruegel's *Hunters in the Snow,* the SR-71, Lisp. They're not just beautiful, but strangely beautiful.

I'm not sure why. It may just be my own stupidity. A can-opener must seem miraculous to a dog. Maybe if I were smart enough it would seem the most natural thing in the world that ei*pi = -1. It is after all necessarily true.

Most of the qualities I've mentioned are things that can be cultivated, but I don't think it works to cultivate strangeness. The best you can do is not squash it if it starts to appear. Einstein didn't try to make relativity strange. He tried to make it true, and the truth turned out to be strange.

At an art school where I once studied, the students wanted most of all to develop a personal style. But if you just try to make good things, you'll inevitably do it in a distinctive way, just as each person walks in a distinctive way. Michelangelo was not trying to paint like Michelangelo. He was just trying to paint well; he couldn't help painting like Michelangelo.

The only style worth having is the one you can't help. And this is especially true for strangeness. There is no shortcut to it. The Northwest Passage that the Mannerists, the Romantics, and two generations of American high school students have searched for does not seem to exist. The only way to get there is to go through good and come out the other side.

Good design happens in chunks. The inhabitants of fifteenth century Florence included Brunelleschi, Ghiberti, Donatello, Masaccio, Filippo Lippi, Fra Angelico, Verrocchio, Botticelli, Leonardo, and Michelangelo. Milan at the time was as big as Florence. How many fifteenth century Milanese artists can you name?

Something was happening in Florence in the fifteenth century. And it can't have been heredity, because it isn't happening now. You have to assume that whatever inborn ability Leonardo and Michelangelo had, there were people born in Milan with just as much. What happened to the Milanese Leonardo?

There are roughly a thousand times as many people alive in the US right now as lived in Florence during the fifteenth century. A thousand Leonardos and a thousand Michelangelos walk among us. If DNA ruled, we should be greeted daily by artistic marvels. We aren't, and the reason is that to make Leonardo you need more than his innate ability. You also need Florence in 1450.

Nothing is more powerful than a community of talented people working on related problems. Genes count for little by comparison: being a genetic Leonardo was not enough to compensate for having been born near Milan instead of Florence. Today we move around more, but great work still comes disproportionately from a few hotspots: the Bauhaus, the Manhattan Project, the *New Yorker,* Lockheed's Skunk Works, Xerox Parc.

At any given time there are a few hot topics and a few groups doing great work on them, and it's nearly impossible to do good work yourself if you're too far removed from one of these centers. You can push or pull these trends to some extent, but you can't break away from them. (Maybe *you* can, but the Milanese Leonardo couldn't.)

Good design is often daring. At every period of history, people have believed things that were just ridiculous, and believed them so strongly that you risked ostracism or even violence by saying otherwise.

If our own time were any different, that would be remarkable. As far as I can tell it isn't.

This problem afflicts not just every era, but in some degree every field. Much Renaissance art was in its time considered shockingly secular: according to Vasari, Botticelli repented and gave up painting, and Fra Bartolommeo and Lorenzo di Credi actually burned some of their work. Einstein's theory of relativity offended many contemporary physicists, and was not fully accepted for decades– in France, not until the 1950s.

Today's experimental error is tomorrow's new theory. If you want to discover great new things, then instead of turning a blind eye to the places where conventional wisdom and truth don't quite meet, you should pay particular attention to them.

As a practical matter, I think it's easier to see ugliness than to imagine beauty. Most of the people who've made beautiful things seem to have done it by fixing something that they thought ugly. Great work usually seems to happen because someone sees something and thinks, *I could do better than that.* Giotto saw traditional Byzantine madonnas painted according to a formula that had satisfied everyone for centuries, and to him they looked wooden and unnatural. Copernicus was so troubled by a hack that all his contemporaries could tolerate that he felt there must be a better solution.

Intolerance for ugliness is not in itself enough. You have to understand a field well before you develop a good nose for what needs fixing. You have to do your homework. But as you become expert in a field, you'll start to hear little voices saying, *What a hack! There must be a better way.* Don't ignore those voices. Cultivate them. The recipe for great work is: very exacting taste, plus the ability to gratify it.

Sullivan actually said "form ever follows function," but I think the usual misquotation is closer to what modernist architects meant.

Stephen G. Brush, "Why was Relativity Accepted?" _Phys. Perspect. 1 (1999) 184-214.

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Work

Chapter 9

Before the Startup

October 2014

(This essay is derived from a guest lecture in Sam Altman's startup class at Stanford. It's intended for college students, but much of it is applicable to potential founders at other ages.)

One of the advantages of having kids is that when you have to give advice, you can ask yourself "what would I tell my own kids?" My kids are little, but I can imagine what I'd tell them about startups if they were in college, and that's what I'm going to tell you.

Startups are very counterintuitive. I'm not sure why. Maybe it's just because knowledge about them hasn't permeated our culture yet. But whatever the reason, starting a startup is a task where you can't always trust your instincts.

It's like skiing in that way. When you first try skiing and you want to slow down, your instinct is to lean back. But if you lean back on skis you fly down the hill out of control. So part of learning to ski is learning to suppress that impulse. Eventually you get new habits, but at first it takes a conscious effort. At first there's a list of things you're trying to remember as you start down the hill.

Startups are as unnatural as skiing, so there's a similar list for startups. Here I'm going to give you the first part of it — the things to remember if you want to prepare yourself to start a startup.

Counterintuitive

The first item on it is the fact I already mentioned: that startups are so weird that if you trust your instincts, you'll make a lot of mistakes.

If you know nothing more than this, you may at least pause before making them.

When I was running Y Combinator I used to joke that our function was to tell founders things they would ignore. It's really true. Batch after batch, the YC partners warn founders about mistakes they're about to make, and the founders ignore them, and then come back a year later and say "I wish we'd listened."

Why do the founders ignore the partners' advice? Well, that's the thing about counterintuitive ideas: they contradict your intuitions. They seem wrong. So of course your first impulse is to disregard them. And in fact my joking description is not merely the curse of Y Combinator but part of its raison d'etre. If founders' instincts already gave them the right answers, they wouldn't need us. You only need other people to give you advice that surprises you. That's why there are a lot of ski instructors and not many running instructors. [1]

You can, however, trust your instincts about people. And in fact one of the most common mistakes young founders make is not to do that enough. They get involved with people who seem impressive, but about whom they feel some misgivings personally. Later when things blow up they say "I knew there was something off about him, but I ignored it because he seemed so impressive."

If you're thinking about getting involved with someone — as a cofounder, an employee, an investor, or an acquirer — and you have misgivings about them, trust your gut. If someone seems slippery, or bogus, or a jerk, don't ignore it.

This is one case where it pays to be self-indulgent. Work with people you genuinely like, and you've known long enough to be sure.

Expertise

[1] Some founders listen more than others, and this tends to be a predictor of success. One of the things I remember about the Airbnbs during YC is how intently they listened.

The second counterintuitive point is that it's not that important to know a lot about startups. The way to succeed in a startup is not to be an expert on startups, but to be an expert on your users and the problem you're solving for them. Mark Zuckerberg didn't succeed because he was an expert on startups. He succeeded despite being a complete noob at startups, because he understood his users really well.

If you don't know anything about, say, how to raise an angel round, don't feel bad on that account. That sort of thing you can learn when you need to, and forget after you've done it.

In fact, I worry it's not merely unnecessary to learn in great detail about the mechanics of startups, but possibly somewhat dangerous. If I met an undergrad who knew all about convertible notes and employee agreements and (God forbid) class FF stock, I wouldn't think "here is someone who is way ahead of their peers." It would set off alarms. Because another of the characteristic mistakes of young founders is to go through the motions of starting a startup. They make up some plausible-sounding idea, raise money at a good valuation, rent a cool office, hire a bunch of people. From the outside that seems like what startups do. But the next step after rent a cool office and hire a bunch of people is: gradually realize how completely fucked they are, because while imitating all the outward forms of a startup they have neglected the one thing that's actually essential: making something people want.

Game

We saw this happen so often that we made up a name for it: playing house. Eventually I realized why it was happening. The reason young founders go through the motions of starting a startup is because that's what they've been trained to do for their whole lives up to that point. Think about what you have to do to get into college, for example. Extracurricular activities, check. Even in college classes most of the work is as artificial as running laps.

I'm not attacking the educational system for being this way. There will always be a certain amount of fakeness in the work you do when you're being taught something, and if you measure their performance it's inevitable that people will exploit the difference to the point where much of what you're measuring is artifacts of the fakeness.

I confess I did it myself in college. I found that in a lot of classes there might only be 20 or 30 ideas that were the right shape to make good exam questions. The way I studied for exams in these classes was not (except incidentally) to master the material taught in the class, but to make a list of potential exam questions and work out the answers in advance. When I walked into the final, the main thing I'd be feeling was curiosity about which of my questions would turn up on the exam. It was like a game.

It's not surprising that after being trained for their whole lives to play such games, young founders' first impulse on starting a startup is to try to figure out the tricks for winning at this new game. Since fundraising appears to be the measure of success for startups (another classic noob mistake), they always want to know what the tricks are for convincing investors. We tell them the best way to convince investors is to make a startup that's actually doing well, meaning growing fast, and then simply tell investors so. Then they want to know what the tricks are for growing fast. And we have to tell them the best way to do that is simply to make something people want.

So many of the conversations YC partners have with young founders begin with the founder asking "How do we…" and the partner replying "Just…"

Why do the founders always make things so complicated? The reason, I realized, is that they're looking for the trick.

So this is the third counterintuitive thing to remember about startups: starting a startup is where gaming the system stops working. Gaming the system may continue to work if you go to work for a big company. Depending on how broken the company is, you can succeed by sucking up to the right people, giving the impression of productivity, and so on. [2] But that doesn't work with startups. There is no boss to trick, only users, and all users care about is whether your product does what they want. Startups are as impersonal as physics. You have to make something people want, and you prosper only to the extent you do.

The dangerous thing is, faking does work to some degree on investors. If you're super good at sounding like you know what you're talking about, you can fool investors for at least one and perhaps even two rounds of funding. But it's not in your interest to. The company is ultimately doomed. All you're doing is wasting your own time riding it down.

So stop looking for the trick. There are tricks in startups, as there are in any domain, but they are an order of magnitude less important than solving the real problem. A founder who knows nothing about fundraising but has made something users love will have an easier time raising money than one who knows every trick in the book but has a flat usage graph. And more importantly, the founder who has made something users love is the one who will go on to succeed after raising the money.

Though in a sense it's bad news in that you're deprived of one of your most powerful weapons, I think it's exciting that gaming the system stops working when you start a startup. It's exciting that there even exist parts of the world where you win by doing good work. Imagine how depressing the world would be if it were all like school and big companies, where you either have to spend a lot of time on bullshit things or lose to people who do. [3] I would have been delighted if I'd realized in college that there were parts of the real world where gaming the system mattered less than others, and a few where it hardly

[2]In fact, this is one of the reasons startups are possible. If big companies weren't plagued by internal inefficiencies, they'd be proportionately more effective, leaving less room for startups.

[3]In a startup you have to spend a lot of time on schleps, but this sort of work is merely unglamorous, not bogus.

mattered at all. But there are, and this variation is one of the most important things to consider when you're thinking about your future. How do you win in each type of work, and what would you like to win by doing? [4]

All-Consuming

That brings us to our fourth counterintuitive point: startups are all-consuming. If you start a startup, it will take over your life to a degree you cannot imagine. And if your startup succeeds, it will take over your life for a long time: for several years at the very least, maybe for a decade, maybe for the rest of your working life. So there is a real opportunity cost here.

Larry Page may seem to have an enviable life, but there are aspects of it that are unenviable. Basically at 25 he started running as fast as he could and it must seem to him that he hasn't stopped to catch his breath since. Every day new shit happens in the Google empire that only the CEO can deal with, and he, as CEO, has to deal with it. If he goes on vacation for even a week, a whole week's backlog of shit accumulates. And he has to bear this uncomplainingly, partly because as the company's daddy he can never show fear or weakness, and partly because billionaires get less than zero sympathy if they talk about having difficult lives. Which has the strange side effect that the difficulty of being a successful startup founder is concealed from almost everyone except those who've done it.

Y Combinator has now funded several companies that can be called big successes, and in every single case the founders say the same thing. It never gets any easier. The nature of the problems change. You're worrying about construction delays at your London office instead of the broken air conditioner in your studio apartment. But the total volume of worry never decreases; if anything it increases.

Starting a successful startup is similar to having kids in that it's like

[4] What should you do if your true calling is gaming the system? Management consulting.

a button you push that changes your life irrevocably. And while it's truly wonderful having kids, there are a lot of things that are easier to do before you have them than after. Many of which will make you a better parent when you do have kids. And since you can delay pushing the button for a while, most people in rich countries do.

Yet when it comes to startups, a lot of people seem to think they're supposed to start them while they're still in college. Are you crazy? And what are the universities thinking? They go out of their way to ensure their students are well supplied with contraceptives, and yet they're setting up entrepreneurship programs and startup incubators left and right.

To be fair, the universities have their hand forced here. A lot of incoming students are interested in startups. Universities are, at least de facto, expected to prepare them for their careers. So students who want to start startups hope universities can teach them about startups. And whether universities can do this or not, there's some pressure to claim they can, lest they lose applicants to other universities that do.

Can universities teach students about startups? Yes and no. They can teach students about startups, but as I explained before, this is not what you need to know. What you need to learn about are the needs of your own users, and you can't do that until you actually start the company. [5] So starting a startup is intrinsically something you can only really learn by doing it. And it's impossible to do that in college, for the reason I just explained: startups take over your life. You can't start a startup for real as a student, because if you start a startup for real you're not a student anymore. You may be nominally a student for a bit, but you won't even be that for long. [6]

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[5]The company may not be incorporated, but if you start to get significant numbers of users, you've started it, whether you realize it yet or not.

[6]It shouldn't be that surprising that colleges can't teach students how to be good startup founders, because they can't teach them how to be good employees either.

The way universities "teach" students how to be employees is to hand off the task to companies via internship programs. But you couldn't do the equivalent thing for startups, because by definition if the students did well they would never come back.

Given this dichotomy, which of the two paths should you take? Be a real student and not start a startup, or start a real startup and not be a student? I can answer that one for you. Do not start a startup in college. How to start a startup is just a subset of a bigger problem you're trying to solve: how to have a good life. And though starting a startup can be part of a good life for a lot of ambitious people, age 20 is not the optimal time to do it. Starting a startup is like a brutally fast depth-first search. Most people should still be searching breadth-first at 20.

You can do things in your early 20s that you can't do as well before or after, like plunge deeply into projects on a whim and travel super cheaply with no sense of a deadline. For unambitious people, this sort of thing is the dreaded "failure to launch," but for the ambitious ones it can be an incomparably valuable sort of exploration. If you start a startup at 20 and you're sufficiently successful, you'll never get to do it. [7]

Mark Zuckerberg will never get to bum around a foreign country. He can do other things most people can't, like charter jets to fly him to foreign countries. But success has taken a lot of the serendipity out of his life. Facebook is running him as much as he's running Facebook. And while it can be very cool to be in the grip of a project you consider your life's work, there are advantages to serendipity too, especially early in life. Among other things it gives you more options to choose your life's work from.

There's not even a tradeoff here. You're not sacrificing anything if you forgo starting a startup at 20, because you're more likely to succeed if you wait. In the unlikely case that you're 20 and one of your side projects takes off like Facebook did, you'll face a choice of running with it or not, and it may be reasonable to run with it. But the usual

[7] Charles Darwin was 22 when he received an invitation to travel aboard the HMS Beagle as a naturalist. It was only because he was otherwise unoccupied, to a degree that alarmed his family, that he could accept it. And yet if he hadn't we probably would not know his name.

way startups take off is for the founders to make them take off, and it's gratuitously stupid to do that at 20.

Try

Should you do it at any age? I realize I've made startups sound pretty hard. If I haven't, let me try again: starting a startup is really hard. What if it's too hard? How can you tell if you're up to this challenge?

The answer is the fifth counterintuitive point: you can't tell. Your life so far may have given you some idea what your prospects might be if you tried to become a mathematician, or a professional football player. But unless you've had a very strange life you haven't done much that was like being a startup founder. Starting a startup will change you a lot. So what you're trying to estimate is not just what you are, but what you could grow into, and who can do that?

For the past 9 years it was my job to predict whether people would have what it took to start successful startups. It was easy to tell how smart they were, and most people reading this will be over that threshold. The hard part was predicting how tough and ambitious they would become. There may be no one who has more experience at trying to predict that, so I can tell you how much an expert can know about it, and the answer is: not much. I learned to keep a completely open mind about which of the startups in each batch would turn out to be the stars.

The founders sometimes think they know. Some arrive feeling sure they will ace Y Combinator just as they've aced every one of the (few, artificial, easy) tests they've faced in life so far. Others arrive wondering how they got in, and hoping YC doesn't discover whatever mistake caused it to accept them. But there is little correlation between founders' initial attitudes and how well their companies do.

I've read that the same is true in the military — that the swaggering recruits are no more likely to turn out to be really tough than the quiet ones. And probably for the same reason: that the tests involved are so different from the ones in their previous lives.

If you're absolutely terrified of starting a startup, you probably shouldn't do it. But if you're merely unsure whether you're up to it, the only way to find out is to try. Just not now.

Ideas

So if you want to start a startup one day, what should you do in college? There are only two things you need initially: an idea and cofounders. And the m.o. for getting both is the same. Which leads to our sixth and last counterintuitive point: that the way to get startup ideas is not to try to think of startup ideas.

I've written a whole essay on this, so I won't repeat it all here. But the short version is that if you make a conscious effort to think of startup ideas, the ideas you come up with will not merely be bad, but bad and plausible-sounding, meaning you'll waste a lot of time on them before realizing they're bad.

The way to come up with good startup ideas is to take a step back. Instead of making a conscious effort to think of startup ideas, turn your mind into the type that startup ideas form in without any conscious effort. In fact, so unconsciously that you don't even realize at first that they're startup ideas.

This is not only possible, it's how Apple, Yahoo, Google, and Facebook all got started. None of these companies were even meant to be companies at first. They were all just side projects. The best startups almost have to start as side projects, because great ideas tend to be such outliers that your conscious mind would reject them as ideas for companies.

Ok, so how do you turn your mind into the type that startup ideas form in unconsciously? (1) Learn a lot about things that matter, then (2) work on problems that interest you (3) with people you like and respect. The third part, incidentally, is how you get cofounders at the same time as the idea.

The first time I wrote that paragraph, instead of "learn a lot about things that matter," I wrote "become good at some technology." But that prescription, though sufficient, is too narrow. What was special about Brian Chesky and Joe Gebbia was not that they were experts in technology. They were good at design, and perhaps even more importantly, they were good at organizing groups and making projects happen. So you don't have to work on technology per se, so long as you work on problems demanding enough to stretch you.

What kind of problems are those? That is very hard to answer in the general case. History is full of examples of young people who were working on important problems that no one else at the time thought were important, and in particular that their parents didn't think were important. On the other hand, history is even fuller of examples of parents who thought their kids were wasting their time and who were right. So how do you know when you're working on real stuff? [8]

I know how *I* know. Real problems are interesting, and I am self-indulgent in the sense that I always want to work on interesting things, even if no one else cares about them (in fact, especially if no one else cares about them), and find it very hard to make myself work on boring things, even if they're supposed to be important.

My life is full of case after case where I worked on something just because it seemed interesting, and it turned out later to be useful in some worldly way. Y Combinator itself was something I only did because it seemed interesting. So I seem to have some sort of internal compass that helps me out. But I don't know what other people have in their heads. Maybe if I think more about this I can come up with heuristics for recognizing genuinely interesting problems, but for the moment the best I can offer is the hopelessly question-begging advice that if you have a taste for genuinely interesting problems, indulging it energetically is the best way to prepare yourself for a startup. And indeed,

[8] Parents can sometimes be especially conservative in this department. There are some whose definition of important problems includes only those on the critical path to med school.

probably also the best way to live. [9]

But although I can't explain in the general case what counts as an interesting problem, I can tell you about a large subset of them. If you think of technology as something that's spreading like a sort of fractal stain, every moving point on the edge represents an interesting problem. So one guaranteed way to turn your mind into the type that has good startup ideas is to get yourself to the leading edge of some technology — to cause yourself, as Paul Buchheit put it, to "live in the future." When you reach that point, ideas that will seem to other people uncannily prescient will seem obvious to you. You may not realize they're startup ideas, but you'll know they're something that ought to exist.

For example, back at Harvard in the mid 90s a fellow grad student of my friends Robert and Trevor wrote his own voice over IP software. He didn't mean it to be a startup, and he never tried to turn it into one. He just wanted to talk to his girlfriend in Taiwan without paying for long distance calls, and since he was an expert on networks it seemed obvious to him that the way to do it was turn the sound into packets and ship it over the Internet. He never did any more with his software than talk to his girlfriend, but this is exactly the way the best startups get started.

So strangely enough the optimal thing to do in college if you want to be a successful startup founder is not some sort of new, vocational version of college focused on "entrepreneurship." It's the classic version of college as education for its own sake. If you want to start a startup after college, what you should do in college is learn powerful things. And if you have genuine intellectual curiosity, that's what you'll naturally tend to do if you just follow your own inclinations. [10]

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[9] I did manage to think of a heuristic for detecting whether you have a taste for interesting ideas: whether you find known boring ideas intolerable. Could you endure studying literary theory, or working in middle management at a large company?

[10] In fact, if your goal is to start a startup, you can stick even more closely to the ideal of a liberal education than past generations have. Back when students focused mainly on getting a job after college, they thought at least a little about how the

The component of entrepreneurship that really matters is domain expertise. The way to become Larry Page was to become an expert on search. And the way to become an expert on search was to be driven by genuine curiosity, not some ulterior motive.

At its best, starting a startup is merely an ulterior motive for curiosity. And you'll do it best if you introduce the ulterior motive toward the end of the process.

So here is the ultimate advice for young would-be startup founders, boiled down to two words: just learn.

Thanks to Sam Altman, Paul Buchheit, John Collison, Patrick Collison, Jessica Livingston, Robert Morris, Geoff Ralston, and Fred Wilson for reading drafts of this.

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courses they took might look to an employer. And perhaps even worse, they might shy away from taking a difficult class lest they get a low grade, which would harm their all-important GPA. Good news: users don't care what your GPA was. And I've never heard of investors caring either. Y Combinator certainly never asks what classes you took in college or what grades you got in them.

Chapter 10

The Power of the Marginal

June 2006

(This essay is derived from talks at Usenix 2006 and Railsconf 2006.)

A couple years ago my friend Trevor and I went to look at the Apple garage. As we stood there, he said that as a kid growing up in Saskatchewan he'd been amazed at the dedication Jobs and Wozniak must have had to work in a garage.

"Those guys must have been freezing!"

That's one of California's hidden advantages: the mild climate means there's lots of marginal space. In cold places that margin gets trimmed off. There's a sharper line between outside and inside, and only projects that are officially sanctioned — by organizations, or parents, or wives, or at least by oneself — get proper indoor space. That raises the activation energy for new ideas. You can't just tinker. You have to justify.

Some of Silicon Valley's most famous companies began in garages: Hewlett-Packard in 1938, Apple in 1976, Google in 1998. In Apple's case the garage story is a bit of an urban legend. Woz says all they did there was assemble some computers, and that he did all the actual design of the Apple I and Apple II in his apartment or his cube at HP. [1] This was apparently too marginal even for Apple's PR people.

By conventional standards, Jobs and Wozniak were marginal people too. Obviously they were smart, but they can't have looked good on

[1] The facts about Apple's early history are from an interview with Steve Wozniak in Jessica Livingston's *Founders at Work*.

paper. They were at the time a pair of college dropouts with about three years of school between them, and hippies to boot. Their previous business experience consisted of making "blue boxes" to hack into the phone system, a business with the rare distinction of being both illegal and unprofitable.

Outsiders

Now a startup operating out of a garage in Silicon Valley would feel part of an exalted tradition, like the poet in his garret, or the painter who can't afford to heat his studio and thus has to wear a beret indoors. But in 1976 it didn't seem so cool. The world hadn't yet realized that starting a computer company was in the same category as being a writer or a painter. It hadn't been for long. Only in the preceding couple years had the dramatic fall in the cost of hardware allowed outsiders to compete.

In 1976, everyone looked down on a company operating out of a garage, including the founders. One of the first things Jobs did when they got some money was to rent office space. He wanted Apple to seem like a real company.

They already had something few real companies ever have: a fabulously well designed product. You'd think they'd have had more confidence. But I've talked to a lot of startup founders, and it's always this way. They've built something that's going to change the world, and they're worried about some nit like not having proper business cards.

That's the paradox I want to explore: great new things often come from the margins, and yet the people who discover them are looked down on by everyone, including themselves.

It's an old idea that new things come from the margins. I want to examine its internal structure. Why do great ideas come from the margins? What kind of ideas? And is there anything we can do to encourage the process?

Insiders

One reason so many good ideas come from the margin is simply that there's so much of it. There have to be more outsiders than insiders, if insider means anything. If the number of outsiders is huge it will always seem as if a lot of ideas come from them, even if few do per capita. But I think there's more going on than this. There are real disadvantages to being an insider, and in some kinds of work they can outweigh the advantages.

Imagine, for example, what would happen if the government decided to commission someone to write an official Great American Novel. First there'd be a huge ideological squabble over who to choose. Most of the best writers would be excluded for having offended one side or the other. Of the remainder, the smart ones would refuse such a job, leaving only a few with the wrong sort of ambition. The committee would choose one at the height of his career — that is, someone whose best work was behind him — and hand over the project with copious free advice about how the book should show in positive terms the strength and diversity of the American people, etc, etc.

The unfortunate writer would then sit down to work with a huge weight of expectation on his shoulders. Not wanting to blow such a public commission, he'd play it safe. This book had better command respect, and the way to ensure that would be to make it a tragedy. Audiences have to be enticed to laugh, but if you kill people they feel obliged to take you seriously. As everyone knows, America plus tragedy equals the Civil War, so that's what it would have to be about. When finally completed twelve years later, the book would be a 900-page pastiche of existing popular novels — roughly *Gone with the Wind* plus *Roots*. But its bulk and celebrity would make it a bestseller for a few months, until blown out of the water by a talk-show host's autobiography. The book would be made into a movie and thereupon forgotten, except by the more waspish sort of reviewers, among whom it would be a byword for bogusness like Milli Vanilli or *Battlefield Earth*.

Maybe I got a little carried away with this example. And yet is this not at each point the way such a project would play out? The government knows better than to get into the novel business, but in other fields where they have a natural monopoly, like nuclear waste dumps, aircraft carriers, and regime change, you'd find plenty of projects isomorphic to this one — and indeed, plenty that were less successful.

This little thought experiment suggests a few of the disadvantages of insider projects: the selection of the wrong kind of people, the excessive scope, the inability to take risks, the need to seem serious, the weight of expectations, the power of vested interests, the undiscerning audience, and perhaps most dangerous, the tendency of such work to become a duty rather than a pleasure.

Tests

A world with outsiders and insiders implies some kind of test for distinguishing between them. And the trouble with most tests for selecting elites is that there are two ways to pass them: to be good at what they try to measure, and to be good at hacking the test itself.

So the first question to ask about a field is how honest its tests are, because this tells you what it means to be an outsider. This tells you how much to trust your instincts when you disagree with authorities, whether it's worth going through the usual channels to become one yourself, and perhaps whether you want to work in this field at all.

Tests are least hackable when there are consistent standards for quality, and the people running the test really care about its integrity. Admissions to PhD programs in the hard sciences are fairly honest, for example. The professors will get whoever they admit as their own grad students, so they try hard to choose well, and they have a fair amount of data to go on. Whereas undergraduate admissions seem to be much more hackable.

One way to tell whether a field has consistent standards is the overlap between the leading practitioners and the people who teach the subject in universities. At one end of the scale you have fields like math and physics, where nearly all the teachers are among the best practitioners. In the middle are medicine, law, history, architecture, and computer science, where many are. At the bottom are business, literature, and the visual arts, where there's almost no overlap between the teachers and the leading practitioners. It's this end that gives rise to phrases like "those who can't do, teach."

Incidentally, this scale might be helpful in deciding what to study in college. When I was in college the rule seemed to be that you should study whatever you were most interested in. But in retrospect you're probably better off studying something moderately interesting with someone who's good at it than something very interesting with someone who isn't. You often hear people say that you shouldn't major in business in college, but this is actually an instance of a more general rule: don't learn things from teachers who are bad at them.

How much you should worry about being an outsider depends on the quality of the insiders. If you're an amateur mathematician and think you've solved a famous open problem, better go back and check. When I was in grad school, a friend in the math department had the job of replying to people who sent in proofs of Fermat's last theorem and so on, and it did not seem as if he saw it as a valuable source of tips — more like manning a mental health hotline. Whereas if the stuff you're writing seems different from what English professors are interested in, that's not necessarily a problem.

Anti-Tests

Where the method of selecting the elite is thoroughly corrupt, most of the good people will be outsiders. In art, for example, the image of the poor, misunderstood genius is not just one possible image of a great artist: it's the *standard* image. I'm not saying it's correct, incidentally, but it is telling how well this image has stuck. You couldn't make a rap like that stick to math or medicine. [2]

[2]As usual the popular image is several decades behind reality. Now the misunderstood artist is not a chain-smoking drunk who pours his soul into big, messy canvases that philistines see and say "that's not art" because it isn't a picture of anything. The philistines have now been trained that anything hung on a wall is art. Now the misunderstood artist is a coffee-drinking vegan cartoonist whose work

If it's corrupt enough, a test becomes an anti-test, filtering out the people it should select by making them to do things only the wrong people would do. Popularity in high school seems to be such a test. There are plenty of similar ones in the grownup world. For example, rising up through the hierarchy of the average big company demands an attention to politics few thoughtful people could spare. [3] Someone like Bill Gates can grow a company under him, but it's hard to imagine him having the patience to climb the corporate ladder at General Electric — or Microsoft, actually.

It's kind of strange when you think about it, because lord-of-the-flies schools and bureaucratic companies are both the default. There are probably a lot of people who go from one to the other and never realize the whole world doesn't work this way.

I think that's one reason big companies are so often blindsided by startups. People at big companies don't realize the extent to which they live in an environment that is one large, ongoing test for the wrong qualities.

If you're an outsider, your best chances for beating insiders are obviously in fields where corrupt tests select a lame elite. But there's a catch: if the tests are corrupt, your victory won't be recognized, at least in your lifetime. You may feel you don't need that, but history suggests it's dangerous to work in fields with corrupt tests. You may beat the insiders, and yet not do as good work, on an absolute scale, as you would in a field that was more honest.

Standards in art, for example, were almost as corrupt in the first half of the eighteenth century as they are today. This was the era of those fluffy idealized portraits of countesses with their lapdogs. Chardin decided to skip all that and paint ordinary things as he saw them. He's now considered the best of that period — and yet not the equal of

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they see and say "that's not art" because it looks like stuff they've seen in the Sunday paper.

[3] In fact this would do fairly well as a definition of politics: what determines rank in the absence of objective tests.

Leonardo or Bellini or Memling, who all had the additional encouragement of honest standards.

It can be worth participating in a corrupt contest, however, if it's followed by another that isn't corrupt. For example, it would be worth competing with a company that can spend more than you on marketing, as long as you can survive to the next round, when customers compare your actual products. Similarly, you shouldn't be discouraged by the comparatively corrupt test of college admissions, because it's followed immediately by less hackable tests. [4]

Risk

Even in a field with honest tests, there are still advantages to being an outsider. The most obvious is that outsiders have nothing to lose. They can do risky things, and if they fail, so what? Few will even notice.

The eminent, on the other hand, are weighed down by their eminence. Eminence is like a suit: it impresses the wrong people, and it constrains the wearer.

Outsiders should realize the advantage they have here. Being able to take risks is hugely valuable. Everyone values safety too much, both the obscure and the eminent. No one wants to look like a fool. But it's very useful to be able to. If most of your ideas aren't stupid, you're probably being too conservative. You're not bracketing the problem.

Lord Acton said we should judge talent at its best and character at its worst. For example, if you write one great book and ten bad ones, you still count as a great writer — or at least, a better writer than someone who wrote eleven that were merely good. Whereas if you're a quiet, law-abiding citizen most of the time but occasionally cut someone up and bury them in your backyard, you're a bad guy.

[4] In high school you're led to believe your whole future depends on where you go to college, but it turns out only to buy you a couple years. By your mid-twenties the people worth impressing already judge you more by what you've done than where you went to school.

Almost everyone makes the mistake of treating ideas as if they were indications of character rather than talent — as if having a stupid idea made you stupid. There's a huge weight of tradition advising us to play it safe. "Even a fool is thought wise if he keeps silent," says the Old Testament (Proverbs 17:28).

Well, that may be fine advice for a bunch of goatherds in Bronze Age Palestine. There conservatism would be the order of the day. But times have changed. It might still be reasonable to stick with the Old Testament in political questions, but materially the world now has a lot more state. Tradition is less of a guide, not just because things change faster, but because the space of possibilities is so large. The more complicated the world gets, the more valuable it is to be willing to look like a fool.

Delegation

And yet the more successful people become, the more heat they get if they screw up — or even seem to screw up. In this respect, as in many others, the eminent are prisoners of their own success. So the best way to understand the advantages of being an outsider may be to look at the disadvantages of being an insider.

If you ask eminent people what's wrong with their lives, the first thing they'll complain about is the lack of time. A friend of mine at Google is fairly high up in the company and went to work for them long before they went public. In other words, he's now rich enough not to have to work. I asked him if he could still endure the annoyances of having a job, now that he didn't have to. And he said that there weren't really any annoyances, except — and he got a wistful look when he said this — that he got *so much email.*

The eminent feel like everyone wants to take a bite out of them. The problem is so widespread that people pretending to be eminent do it by pretending to be overstretched.

The lives of the eminent become scheduled, and that's not good for thinking. One of the great advantages of being an outsider is long, uninterrupted blocks of time. That's what I remember about grad school: apparently endless supplies of time, which I spent worrying about, but not writing, my dissertation. Obscurity is like health food — unpleasant, perhaps, but good for you. Whereas fame tends to be like the alcohol produced by fermentation. When it reaches a certain concentration, it kills off the yeast that produced it.

The eminent generally respond to the shortage of time by turning into managers. They don't have time to work. They're surrounded by junior people they're supposed to help or supervise. The obvious solution is to have the junior people do the work. Some good stuff happens this way, but there are problems it doesn't work so well for: the kind where it helps to have everything in one head.

For example, it recently emerged that the famous glass artist Dale Chihuly hasn't actually blown glass for 27 years. He has assistants do the work for him. But one of the most valuable sources of ideas in the visual arts is the resistance of the medium. That's why oil paintings look so different from watercolors. In principle you could make any mark in any medium; in practice the medium steers you. And if you're no longer doing the work yourself, you stop learning from this.

So if you want to beat those eminent enough to delegate, one way to do it is to take advantage of direct contact with the medium. In the arts it's obvious how: blow your own glass, edit your own films, stage your own plays. And in the process pay close attention to accidents and to new ideas you have on the fly. This technique can be generalized to any sort of work: if you're an outsider, don't be ruled by plans. Planning is often just a weakness forced on those who delegate.

Is there a general rule for finding problems best solved in one head? Well, you can manufacture them by taking any project usually done by multiple people and trying to do it all yourself. Wozniak's work was a classic example: he did everything himself, hardware and software, and the result was miraculous. He claims not one bug was ever found in the Apple II, in either hardware or software.

Another way to find good problems to solve in one head is to focus on the grooves in the chocolate bar — the places where tasks are divided when they're split between several people. If you want to beat delegation, focus on a vertical slice: for example, be both writer and editor, or both design buildings and construct them.

One especially good groove to span is the one between tools and things made with them. For example, programming languages and applications are usually written by different people, and this is responsible for a lot of the worst flaws in programming languages. I think every language should be designed simultaneously with a large application written in it, the way C was with Unix.

Techniques for competing with delegation translate well into business, because delegation is endemic there. Instead of avoiding it as a drawback of senility, many companies embrace it as a sign of maturity. In big companies software is often designed, implemented, and sold by three separate types of people. In startups one person may have to do all three. And though this feels stressful, it's one reason startups win. The needs of customers and the means of satisfying them are all in one head.

Focus

The very skill of insiders can be a weakness. Once someone is good at something, they tend to spend all their time doing that. This kind of focus is very valuable, actually. Much of the skill of experts is the ability to ignore false trails. But focus has drawbacks: you don't learn from other fields, and when a new approach arrives, you may be the last to notice.

For outsiders this translates into two ways to win. One is to work on a variety of things. Since you can't derive as much benefit (yet) from a narrow focus, you may as well cast a wider net and derive what benefit you can from similarities between fields. Just as you can compete with delegation by working on larger vertical slices, you can compete with specialization by working on larger horizontal slices — by both writing and illustrating your book, for example.

The second way to compete with focus is to see what focus overlooks. In particular, new things. So if you're not good at anything yet, consider working on something so new that no one else is either. It won't have any prestige yet, if no one is good at it, but you'll have it all to yourself.

The potential of a new medium is usually underestimated, precisely because no one has yet explored its possibilities. Before Durer tried making engravings, no one took them very seriously. Engraving was for making little devotional images — basically fifteenth century baseball cards of saints. Trying to make masterpieces in this medium must have seemed to Durer's contemporaries the way that, say, making masterpieces in comics might seem to the average person today.

In the computer world we get not new mediums but new platforms: the minicomputer, the microprocessor, the web-based application. At first they're always dismissed as being unsuitable for real work. And yet someone always decides to try anyway, and it turns out you can do more than anyone expected. So in the future when you hear people say of a new platform: yeah, it's popular and cheap, but not ready yet for real work, jump on it.

As well as being more comfortable working on established lines, insiders generally have a vested interest in perpetuating them. The professor who made his reputation by discovering some new idea is not likely to be the one to discover its replacement. This is particularly true with companies, who have not only skill and pride anchoring them to the status quo, but money as well. The Achilles heel of successful companies is their inability to cannibalize themselves. Many innovations consist of replacing something with a cheaper alternative, and companies just don't want to see a path whose immediate effect is to cut an existing source of revenue.

So if you're an outsider you should actively seek out contrarian projects. Instead of working on things the eminent have made prestigious, work on things that could steal that prestige.

The really juicy new approaches are not the ones insiders reject as impossible, but those they ignore as undignified. For example, after Wozniak designed the Apple II he offered it first to his employer, HP. They passed. One of the reasons was that, to save money, he'd designed the Apple II to use a TV as a monitor, and HP felt they couldn't produce anything so declasse.

Less

Wozniak used a TV as a monitor for the simple reason that he couldn't afford a monitor. Outsiders are not merely free but compelled to make things that are cheap and lightweight. And both are good bets for growth: cheap things spread faster, and lightweight things evolve faster.

The eminent, on the other hand, are almost forced to work on a large scale. Instead of garden sheds they must design huge art museums. One reason they work on big things is that they can: like our hypothetical novelist, they're flattered by such opportunities. They also know that big projects will by their sheer bulk impress the audience. A garden shed, however lovely, would be easy to ignore; a few might even snicker at it. You can't snicker at a giant museum, no matter how much you dislike it. And finally, there are all those people the eminent have working for them; they have to choose projects that can keep them all busy.

Outsiders are free of all this. They can work on small things, and there's something very pleasing about small things. Small things can be perfect; big ones always have something wrong with them. But there's a magic in small things that goes beyond such rational explanations. All kids know it. Small things have more personality.

Plus making them is more fun. You can do what you want; you don't have to satisfy committees. And perhaps most important, small things can be done fast. The prospect of seeing the finished project hangs in the air like the smell of dinner cooking. If you work fast, maybe you could have it done tonight.

Working on small things is also a good way to learn. The most important kinds of learning happen one project at a time. ("Next time, I won't...") The faster you cycle through projects, the faster you'll evolve.

Plain materials have a charm like small scale. And in addition there's the challenge of making do with less. Every designer's ears perk up at the mention of that game, because it's a game you can't lose. Like the JV playing the varsity, if you even tie, you win. So paradoxically there are cases where fewer resources yield better results, because the designers' pleasure at their own ingenuity more than compensates. [5]

So if you're an outsider, take advantage of your ability to make small and inexpensive things. Cultivate the pleasure and simplicity of that kind of work; one day you'll miss it.

Responsibility

When you're old and eminent, what will you miss about being young and obscure? What people seem to miss most is the lack of responsibilities.

Responsibility is an occupational disease of eminence. In principle you could avoid it, just as in principle you could avoid getting fat as you get old, but few do. I sometimes suspect that responsibility is a trap and that the most virtuous route would be to shirk it, but regardless it's certainly constraining.

When you're an outsider you're constrained too, of course. You're short of money, for example. But that constrains you in different ways. How does responsibility constrain you? The worst thing is that it allows you not to focus on real work. Just as the most dangerous forms

[5] Managers are presumably wondering, how can I make this miracle happen? How can I make the people working for me do more with less? Unfortunately the constraint probably has to be self-imposed. If you're *expected* to do more with less, then you're being starved, not eating virtuously.

of procrastination are those that seem like work, the danger of responsibilities is not just that they can consume a whole day, but that they can do it without setting off the kind of alarms you'd set off if you spent a whole day sitting on a park bench.

A lot of the pain of being an outsider is being aware of one's own procrastination. But this is actually a good thing. You're at least close enough to work that the smell of it makes you hungry.

As an outsider, you're just one step away from getting things done. A huge step, admittedly, and one that most people never seem to make, but only one step. If you can summon up the energy to get started, you can work on projects with an intensity (in both senses) that few insiders can match. For insiders work turns into a duty, laden with responsibilities and expectations. It's never so pure as it was when they were young.

Work like a dog being taken for a walk, instead of an ox being yoked to the plow. That's what they miss.

Audience

A lot of outsiders make the mistake of doing the opposite; they admire the eminent so much that they copy even their flaws. Copying is a good way to learn, but copy the right things. When I was in college I imitated the pompous diction of famous professors. But this wasn't what *made* them eminent — it was more a flaw their eminence had allowed them to sink into. Imitating it was like pretending to have gout in order to seem rich.

Half the distinguishing qualities of the eminent are actually disadvantages. Imitating these is not only a waste of time, but will make you seem a fool to your models, who are often well aware of it.

What are the genuine advantages of being an insider? The greatest is an audience. It often seems to outsiders that the great advantage of insiders is money — that they have the resources to do what they want. But so do people who inherit money, and that doesn't seem to help, not as much as an audience. It's good for morale to know people want to see what you're making; it draws work out of you.

If I'm right that the defining advantage of insiders is an audience, then we live in exciting times, because just in the last ten years the Internet has made audiences a lot more liquid. Outsiders don't have to content themselves anymore with a proxy audience of a few smart friends. Now, thanks to the Internet, they can start to grow themselves actual audiences. This is great news for the marginal, who retain the advantages of outsiders while increasingly being able to siphon off what had till recently been the prerogative of the elite.

Though the Web has been around for more than ten years, I think we're just beginning to see its democratizing effects. Outsiders are still learning how to steal audiences. But more importantly, audiences are still learning how to be stolen — they're still just beginning to realize how much deeper bloggers can dig than journalists, how much more interesting a democratic news site can be than a front page controlled by editors, and how much funnier a bunch of kids with webcams can be than mass-produced sitcoms.

The big media companies shouldn't worry that people will post their copyrighted material on YouTube. They should worry that people will post their own stuff on YouTube, and audiences will watch that instead.

Hacking

If I had to condense the power of the marginal into one sentence it would be: just try hacking something together. That phrase draws in most threads I've mentioned here. Hacking something together means deciding what to do as you're doing it, not a subordinate executing the vision of his boss. It implies the result won't be pretty, because it will be made quickly out of inadequate materials. It may work, but it won't be the sort of thing the eminent would want to put their name on. Something hacked together means something that barely solves the problem, or maybe doesn't solve the problem at all, but another you discovered en route. But that's ok, because the main value of that initial version is not the thing itself, but what it leads to. Insiders who daren't walk through the mud in their nice clothes will never make it to the solid ground on the other side.

The word "try" is an especially valuable component. I disagree here with Yoda, who said there is no try. There is try. It implies there's no punishment if you fail. You're driven by curiosity instead of duty. That means the wind of procrastination will be in your favor: instead of avoiding this work, this will be what you do as a way of avoiding other work. And when you do it, you'll be in a better mood. The more the work depends on imagination, the more that matters, because most people have more ideas when they're happy.

If I could go back and redo my twenties, that would be one thing I'd do more of: just try hacking things together. Like many people that age, I spent a lot of time worrying about what I should do. I also spent some time trying to build stuff. I should have spent less time worrying and more time building. If you're not sure what to do, make something.

Raymond Chandler's advice to thriller writers was "When in doubt, have a man come through a door with a gun in his hand." He followed that advice. Judging from his books, he was often in doubt. But though the result is occasionally cheesy, it's never boring. In life, as in books, action is underrated.

Fortunately the number of things you can just hack together keeps increasing. People fifty years ago would be astonished that one could just hack together a movie, for example. Now you can even hack together distribution. Just make stuff and put it online.

Inappropriate

If you really want to score big, the place to focus is the margin of the margin: the territories only recently captured from the insiders. That's where you'll find the juiciest projects still undone, either because they seemed too risky, or simply because there were too few insiders to explore everything.

This is why I spend most of my time writing essays lately. The writing of essays used to be limited to those who could get them published. In principle you could have written them and just shown them to your friends; in practice that didn't work. [6] An essayist needs the resistance of an audience, just as an engraver needs the resistance of the plate.

Up till a few years ago, writing essays was the ultimate insider's game. Domain experts were allowed to publish essays about their field, but the pool allowed to write on general topics was about eight people who went to the right parties in New York. Now the reconquista has overrun this territory, and, not surprisingly, found it sparsely cultivated. There are so many essays yet unwritten. They tend to be the naughtier ones; the insiders have pretty much exhausted the motherhood and apple pie topics.

This leads to my final suggestion: a technique for determining when you're on the right track. You're on the right track when people complain that you're unqualified, or that you've done something inappropriate. If people are complaining, that means you're doing something rather than sitting around, which is the first step. And if they're driven to such empty forms of complaint, that means you've probably done something good.

If you make something and people complain that it doesn't *work*, that's a problem. But if the worst thing they can hit you with is your own status as an outsider, that implies that in every other respect you've succeeded. Pointing out that someone is unqualified is as desperate as resorting to racial slurs. It's just a legitimate sounding way of saying: we don't like your type around here.

But the best thing of all is when people call what you're doing inap-

[6] Without the prospect of publication, the closest most people come to writing essays is to write in a journal. I find I never get as deeply into subjects as I do in proper essays. As the name implies, you don't go back and rewrite journal entries over and over for two weeks.

propriate. I've been hearing this word all my life and I only recently realized that it is, in fact, the sound of the homing beacon. "Inappropriate" is the null criticism. It's merely the adjective form of "I don't like it."

So that, I think, should be the highest goal for the marginal. Be inappropriate. When you hear people saying that, you're golden. And they, incidentally, are busted.

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Chapter 11

The Bus Ticket Theory of Genius

November 2019

Everyone knows that to do great work you need both natural ability and determination. But there's a third ingredient that's not as well understood: an obsessive interest in a particular topic.

To explain this point I need to burn my reputation with some group of people, and I'm going to choose bus ticket collectors. There are people who collect old bus tickets. Like many collectors, they have an obsessive interest in the minutiae of what they collect. They can keep track of distinctions between different types of bus tickets that would be hard for the rest of us to remember. Because we don't care enough. What's the point of spending so much time thinking about old bus tickets?

Which leads us to the second feature of this kind of obsession: there is no point. A bus ticket collector's love is disinterested. They're not doing it to impress us or to make themselves rich, but for its own sake.

When you look at the lives of people who've done great work, you see a consistent pattern. They often begin with a bus ticket collector's obsessive interest in something that would have seemed pointless to most of their contemporaries. One of the most striking features of Darwin's book about his voyage on the Beagle is the sheer depth of his interest in natural history. His curiosity seems infinite. Ditto for Ramanujan, sitting by the hour working out on his slate what happens to series.

It's a mistake to think they were "laying the groundwork" for the discoveries they made later. There's too much intention in that metaphor. Like bus ticket collectors, they were doing it because they liked it.

But there is a difference between Ramanujan and a bus ticket collector. Series matter, and bus tickets don't.

If I had to put the recipe for genius into one sentence, that might be it: to have a disinterested obsession with something that matters.

Aren't I forgetting about the other two ingredients? Less than you might think. An obsessive interest in a topic is both a proxy for ability and a substitute for determination. Unless you have sufficient mathematical aptitude, you won't find series interesting. And when you're obsessively interested in something, you don't need as much determination: you don't need to push yourself as hard when curiosity is pulling you.

An obsessive interest will even bring you luck, to the extent anything can. Chance, as Pasteur said, favors the prepared mind, and if there's one thing an obsessed mind is, it's prepared.

The disinterestedness of this kind of obsession is its most important feature. Not just because it's a filter for earnestness, but because it helps you discover new ideas.

The paths that lead to new ideas tend to look unpromising. If they looked promising, other people would already have explored them. How do the people who do great work discover these paths that others overlook? The popular story is that they simply have better vision: because they're so talented, they see paths that others miss. But if you look at the way great discoveries are made, that's not what happens. Darwin didn't pay closer attention to individual species than other people because he saw that this would lead to great discoveries, and they didn't. He was just really, really interested in such things.

Darwin couldn't turn it off. Neither could Ramanujan. They didn't discover the hidden paths that they did because they seemed promising, but because they couldn't help it. That's what allowed them to follow paths that someone who was merely ambitious would have ignored.

What rational person would decide that the way to write great novels was to begin by spending several years creating an imaginary elvish language, like Tolkien, or visiting every household in southwestern Britain, like Trollope? No one, including Tolkien and Trollope.

The bus ticket theory is similar to Carlyle's famous definition of genius as an infinite capacity for taking pains. But there are two differences. The bus ticket theory makes it clear that the source of this infinite capacity for taking pains is not infinite diligence, as Carlyle seems to have meant, but the sort of infinite interest that collectors have. It also adds an important qualification: an infinite capacity for taking pains about something that matters.

So what matters? You can never be sure. It's precisely because no one can tell in advance which paths are promising that you can discover new ideas by working on what you're interested in.

But there are some heuristics you can use to guess whether an obsession might be one that matters. For example, it's more promising if you're creating something, rather than just consuming something someone else creates. It's more promising if something you're interested in is difficult, especially if it's more difficult for other people than it is for you. And the obsessions of talented people are more likely to be promising. When talented people become interested in random things, they're not truly random.

But you can never be sure. In fact, here's an interesting idea that's also rather alarming if it's true: it may be that to do great work, you also have to waste a lot of time.

In many different areas, reward is proportionate to risk. If that rule holds here, then the way to find paths that lead to truly great work is to be willing to expend a lot of effort on things that turn out to be every bit as unpromising as they seem.

I'm not sure if this is true. On one hand, it seems surprisingly difficult to waste your time so long as you're working hard on something interesting. So much of what you do ends up being useful. But on the other hand, the rule about the relationship between risk and reward is so powerful that it seems to hold wherever risk occurs. Newton's case, at least, suggests that the risk/reward rule holds here. He's famous for one particular obsession of his that turned out to be unprecedentedly fruitful: using math to describe the world. But he had two other obsessions, alchemy and theology, that seem to have been complete wastes of time. He ended up net ahead. His bet on what we now call physics paid off so well that it more than compensated for the other two. But were the other two necessary, in the sense that he had to take big risks to make such big discoveries? I don't know.

Here's an even more alarming idea: might one make all bad bets? It probably happens quite often. But we don't know how often, because these people don't become famous.

It's not merely that the returns from following a path are hard to predict. They change dramatically over time. 1830 was a really good time to be obsessively interested in natural history. If Darwin had been born in 1709 instead of 1809, we might never have heard of him.

What can one do in the face of such uncertainty? One solution is to hedge your bets, which in this case means to follow the obviously promising paths instead of your own private obsessions. But as with any hedge, you're decreasing reward when you decrease risk. If you forgo working on what you like in order to follow some more conventionally ambitious path, you might miss something wonderful that you'd otherwise have discovered. That too must happen all the time, perhaps even more often than the genius whose bets all fail.

The other solution is to let yourself be interested in lots of different things. You don't decrease your upside if you switch between equally genuine interests based on which seems to be working so far. But there is a danger here too: if you work on too many different projects, you might not get deeply enough into any of them.

One interesting thing about the bus ticket theory is that it may help explain why different types of people excel at different kinds of work. Interest is much more unevenly distributed than ability. If natural ability is all you need to do great work, and natural ability is evenly distributed, you have to invent elaborate theories to explain the skewed distributions we see among those who actually do great work in various fields. But it may be that much of the skew has a simpler explanation: different people are interested in different things.

The bus ticket theory also explains why people are less likely to do great work after they have children. Here interest has to compete not just with external obstacles, but with another interest, and one that for most people is extremely powerful. It's harder to find time for work after you have kids, but that's the easy part. The real change is that you don't want to.

But the most exciting implication of the bus ticket theory is that it suggests ways to encourage great work. If the recipe for genius is simply natural ability plus hard work, all we can do is hope we have a lot of ability, and work as hard as we can. But if interest is a critical ingredient in genius, we may be able, by cultivating interest, to cultivate genius.

For example, for the very ambitious, the bus ticket theory suggests that the way to do great work is to relax a little. Instead of gritting your teeth and diligently pursuing what all your peers agree is the most promising line of research, maybe you should try doing something just for fun. And if you're stuck, that may be the vector along which to break out.

I've always liked Hamming's famous double-barrelled question: what are the most important problems in your field, and why aren't you working on one of them? It's a great way to shake yourself up. But it may be overfitting a bit. It might be at least as useful to ask yourself: if you could take a year off to work on something that probably wouldn't be important but would be really interesting, what would it be?

The bus ticket theory also suggests a way to avoid slowing down as you get older. Perhaps the reason people have fewer new ideas as they get older is not simply that they're losing their edge. It may also be because once you become established, you can no longer mess about with irresponsible side projects the way you could when you were young and no one cared what you did.

The solution to that is obvious: remain irresponsible. It will be hard, though, because the apparently random projects you take up to stave off decline will read to outsiders as evidence of it. And you yourself won't know for sure that they're wrong. But it will at least be more fun to work on what you want.

It may even be that we can cultivate a habit of intellectual bus ticket collecting in kids. The usual plan in education is to start with a broad, shallow focus, then gradually become more specialized. But I've done the opposite with my kids. I know I can count on their school to handle the broad, shallow part, so I take them deep.

When they get interested in something, however random, I encourage them to go preposterously, bus ticket collectorly, deep. I don't do this because of the bus ticket theory. I do it because I want them to feel the joy of learning, and they're never going to feel that about something I'm making them learn. It has to be something they're interested in. I'm just following the path of least resistance; depth is a byproduct. But if in trying to show them the joy of learning I also end up training them to go deep, so much the better.

Will it have any effect? I have no idea. But that uncertainty may be the most interesting point of all. There is so much more to learn about how to do great work. As old as human civilization feels, it's really still very young if we haven't nailed something so basic. It's exciting to think there are still discoveries to make about discovery. If that's the sort of thing you're interested in.

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[1]There are other types of collecting that illustrate this point better than bus tickets, but they're also more popular. It seemed just as well to use an inferior example rather than offend more people by telling them their hobby doesn't matter.

[2]I worried a little about using the word "disinterested," since some people mistakenly believe it means not interested. But anyone who expects to be a genius will have to know the meaning of such a basic word, so I figure they may as well start now.

[3]Think how often genius must have been nipped in the bud by people being told, or telling themselves, to stop messing about and be responsible. Ramanujan's mother was a huge enabler. Imagine if she hadn't been. Imagine if his parents had made him go out and get a job instead of sitting around at home doing math.

On the other hand, anyone quoting the preceding paragraph to justify not getting a job is probably mistaken.

[4]1709 Darwin is to time what the Milanese Leonardo is to space.

[5]"An infinite capacity for taking pains" is a paraphrase of what Carlyle wrote. What he wrote, in his *History of Frederick the Great*, was "… it is the fruit of 'genius' (which means transcendent capacity of taking trouble, first of all)…." Since the paraphrase seems the name of the idea at this point, I kept it.

Carlyle's *History* was published in 1858. In 1785 Hérault de Séchelles quoted Buffon as saying "Le génie n'est qu'une plus grande aptitude à la patience." (Genius is only a greater aptitude for patience.)

[6]Trollope was establishing the system of postal routes. He himself sensed the obsessiveness with which he pursued this goal.

> It is amusing to watch how a passion will grow upon a man. During those two years it was the ambition of my life to cover the country with rural letter-carriers.

Even Newton occasionally sensed the degree of his obsessiveness. After computing pi to 15 digits, he wrote in a letter to a friend:

> I am ashamed to tell you to how many figures I carried these computations, having no other business at the time.

Incidentally, Ramanujan was also a compulsive calculator. As Kanigel writes in his excellent biography:

> One Ramanujan scholar, B. M. Wilson, later told how Ramanujan's research into number theory was often "preceded by a table of numerical results, carried usually to a length from which most of us would shrink."

[7]Working to understand the natural world counts as creating rather than con-

Thanks to Marc Andreessen, Trevor Blackwell, Patrick Collison, Kevin Lacker, Jessica Livingston, Jackie McDonough, Robert Morris, Lisa Randall, Zak Stone, and my 7 year old for reading drafts of this.

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suming.

Newton tripped over this distinction when he chose to work on theology. His beliefs did not allow him to see it, but chasing down paradoxes in nature is fruitful in a way that chasing down paradoxes in sacred texts is not.

[8]How much of people's propensity to become interested in a topic is inborn? My experience so far suggests the answer is: most of it. Different kids get interested in different things, and it's hard to make a child interested in something they wouldn't otherwise be. Not in a way that sticks. The most you can do on behalf of a topic is to make sure it gets a fair showing — to make it clear to them, for example, that there's more to math than the dull drills they do in school. After that it's up to the child.

Chapter 12

Good and Bad Procrastination

December 2005

The most impressive people I know are all terrible procrastinators. So could it be that procrastination isn't always bad?

Most people who write about procrastination write about how to cure it. But this is, strictly speaking, impossible. There are an infinite number of things you could be doing. No matter what you work on, you're not working on everything else. So the question is not how to avoid procrastination, but how to procrastinate well.

There are three variants of procrastination, depending on what you do instead of working on something: you could work on (a) nothing, (b) something less important, or (c) something more important. That last type, I'd argue, is good procrastination.

That's the "absent-minded professor," who forgets to shave, or eat, or even perhaps look where he's going while he's thinking about some interesting question. His mind is absent from the everyday world because it's hard at work in another.

That's the sense in which the most impressive people I know are all procrastinators. They're type-C procrastinators: they put off working on small stuff to work on big stuff.

What's "small stuff?" Roughly, work that has zero chance of being mentioned in your obituary. It's hard to say at the time what will turn out to be your best work (will it be your magnum opus on Sumerian temple architecture, or the detective thriller you wrote under a pseudonym?), but there's a whole class of tasks you can safely rule out:

shaving, doing your laundry, cleaning the house, writing thank-you notes—anything that might be called an errand.

Good procrastination is avoiding errands to do real work.

Good in a sense, at least. The people who want you to do the errands won't think it's good. But you probably have to annoy them if you want to get anything done. The mildest seeming people, if they want to do real work, all have a certain degree of ruthlessness when it comes to avoiding errands.

Some errands, like replying to letters, go away if you ignore them (perhaps taking friends with them). Others, like mowing the lawn, or filing tax returns, only get worse if you put them off. In principle it shouldn't work to put off the second kind of errand. You're going to have to do whatever it is eventually. Why not (as past-due notices are always saying) do it now?

The reason it pays to put off even those errands is that real work needs two things errands don't: big chunks of time, and the right mood. If you get inspired by some project, it can be a net win to blow off everything you were supposed to do for the next few days to work on it. Yes, those errands may cost you more time when you finally get around to them. But if you get a lot done during those few days, you will be net more productive.

In fact, it may not be a difference in degree, but a difference in kind. There may be types of work that can only be done in long, uninterrupted stretches, when inspiration hits, rather than dutifully in scheduled little slices. Empirically it seems to be so. When I think of the people I know who've done great things, I don't imagine them dutifully crossing items off to-do lists. I imagine them sneaking off to work on some new idea.

Conversely, forcing someone to perform errands synchronously is bound to limit their productivity. The cost of an interruption is not just the time it takes, but that it breaks the time on either side in half. You probably only have to interrupt someone a couple times a day before they're unable to work on hard problems at all.

I've wondered a lot about why startups are most productive at the very beginning, when they're just a couple guys in an apartment. The main reason may be that there's no one to interrupt them yet. In theory it's good when the founders finally get enough money to hire people to do some of the work for them. But it may be better to be overworked than interrupted. Once you dilute a startup with ordinary office workers—with type-B procrastinators—the whole company starts to resonate at their frequency. They're interrupt-driven, and soon you are too.

Errands are so effective at killing great projects that a lot of people use them for that purpose. Someone who has decided to write a novel, for example, will suddenly find that the house needs cleaning. People who fail to write novels don't do it by sitting in front of a blank page for days without writing anything. They do it by feeding the cat, going out to buy something they need for their apartment, meeting a friend for coffee, checking email. "I don't have time to work," they say. And they don't; they've made sure of that.

(There's also a variant where one has no place to work. The cure is to visit the places where famous people worked, and see how unsuitable they were.)

I've used both these excuses at one time or another. I've learned a lot of tricks for making myself work over the last 20 years, but even now I don't win consistently. Some days I get real work done. Other days are eaten up by errands. And I know it's usually my fault: I *let* errands eat up the day, to avoid facing some hard problem.

The most dangerous form of procrastination is unacknowledged type-B procrastination, because it doesn't feel like procrastination. You're "getting things done." Just the wrong things.

Any advice about procrastination that concentrates on crossing things off your to-do list is not only incomplete, but positively misleading, if it doesn't consider the possibility that the to-do list is itself a form of type-B procrastination. In fact, possibility is too weak a word. Nearly everyone's is. Unless you're working on the biggest things you could be working on, you're type-B procrastinating, no matter how much you're getting done.

In his famous essay You and Your Research (which I recommend to anyone ambitious, no matter what they're working on), Richard Hamming suggests that you ask yourself three questions:

1. What are the most important problems in your field?

2. Are you working on one of them?

3. Why not?

Hamming was at Bell Labs when he started asking such questions. In principle anyone there ought to have been able to work on the most important problems in their field. Perhaps not everyone can make an equally dramatic mark on the world; I don't know; but whatever your capacities, there are projects that stretch them. So Hamming's exercise can be generalized to:

> What's the best thing you could be working on, and why aren't you?

Most people will shy away from this question. I shy away from it myself; I see it there on the page and quickly move on to the next sentence. Hamming used to go around actually asking people this, and it didn't make him popular. But it's a question anyone ambitious should face.

The trouble is, you may end up hooking a very big fish with this bait. To do good work, you need to do more than find good projects. Once you've found them, you have to get yourself to work on them, and that can be hard. The bigger the problem, the harder it is to get yourself to work on it.

Of course, the main reason people find it difficult to work on a particular problem is that they don't enjoy it. When you're young, especially, you often find yourself working on stuff you don't really like– because it seems impressive, for example, or because you've been assigned to work on it. Most grad students are stuck working on big problems they don't really like, and grad school is thus synonymous with procrastination.

But even when you like what you're working on, it's easier to get yourself to work on small problems than big ones. Why? Why is it so hard to work on big problems? One reason is that you may not get any reward in the forseeable future. If you work on something you can finish in a day or two, you can expect to have a nice feeling of accomplishment fairly soon. If the reward is indefinitely far in the future, it seems less real.

Another reason people don't work on big projects is, ironically, fear of wasting time. What if they fail? Then all the time they spent on it will be wasted. (In fact it probably won't be, because work on hard projects almost always leads somewhere.)

But the trouble with big problems can't be just that they promise no immediate reward and might cause you to waste a lot of time. If that were all, they'd be no worse than going to visit your in-laws. There's more to it than that. Big problems are *terrifying*. There's an almost physical pain in facing them. It's like having a vacuum cleaner hooked up to your imagination. All your initial ideas get sucked out immediately, and you don't have any more, and yet the vacuum cleaner is still sucking.

You can't look a big problem too directly in the eye. You have to approach it somewhat obliquely. But you have to adjust the angle just right: you have to be facing the big problem directly enough that you catch some of the excitement radiating from it, but not so much that it paralyzes you. You can tighten the angle once you get going, just as a sailboat can sail closer to the wind once it gets underway.

If you want to work on big things, you seem to have to trick yourself into doing it. You have to work on small things that could grow into big things, or work on successively larger things, or split the moral load with collaborators. It's not a sign of weakness to depend on such tricks. The very best work has been done this way.

When I talk to people who've managed to make themselves work on big things, I find that all blow off errands, and all feel guilty about it. I don't think they should feel guilty. There's more to do than anyone could. So someone doing the best work they can is inevitably going to leave a lot of errands undone. It seems a mistake to feel bad about that.

I think the way to "solve" the problem of procrastination is to let delight pull you instead of making a to-do list push you. Work on an ambitious project you really enjoy, and sail as close to the wind as you can, and you'll leave the right things undone.

Thanks to Trevor Blackwell, Jessica Livingston, and Robert Morris for reading drafts of this.

Chapter 13

Maker's Schedule, Manager's Schedule

"...the mere consciousness of an engagement will sometimes worry a whole day."

– Charles Dickens

July 2009

One reason programmers dislike meetings so much is that they're on a different type of schedule from other people. Meetings cost them more.

There are two types of schedule, which I'll call the manager's schedule and the maker's schedule. The manager's schedule is for bosses. It's embodied in the traditional appointment book, with each day cut into one hour intervals. You can block off several hours for a single task if you need to, but by default you change what you're doing every hour.

When you use time that way, it's merely a practical problem to meet with someone. Find an open slot in your schedule, book them, and you're done.

Most powerful people are on the manager's schedule. It's the schedule of command. But there's another way of using time that's common among people who make things, like programmers and writers. They generally prefer to use time in units of half a day at least. You can't write or program well in units of an hour. That's barely enough time to get started.

When you're operating on the maker's schedule, meetings are a disaster. A single meeting can blow a whole afternoon, by breaking it into two pieces each too small to do anything hard in. Plus you have to remember to go to the meeting. That's no problem for someone on the manager's schedule. There's always something coming on the next hour; the only question is what. But when someone on the maker's schedule has a meeting, they have to think about it.

For someone on the maker's schedule, having a meeting is like throwing an exception. It doesn't merely cause you to switch from one task to another; it changes the mode in which you work.

I find one meeting can sometimes affect a whole day. A meeting commonly blows at least half a day, by breaking up a morning or afternoon. But in addition there's sometimes a cascading effect. If I know the afternoon is going to be broken up, I'm slightly less likely to start something ambitious in the morning. I know this may sound oversensitive, but if you're a maker, think of your own case. Don't your spirits rise at the thought of having an entire day free to work, with no appointments at all? Well, that means your spirits are correspondingly depressed when you don't. And ambitious projects are by definition close to the limits of your capacity. A small decrease in morale is enough to kill them off.

Each type of schedule works fine by itself. Problems arise when they meet. Since most powerful people operate on the manager's schedule, they're in a position to make everyone resonate at their frequency if they want to. But the smarter ones restrain themselves, if they know that some of the people working for them need long chunks of time to work in.

Our case is an unusual one. Nearly all investors, including all VCs I know, operate on the manager's schedule. But Y Combinator runs on the maker's schedule. Rtm and Trevor and I do because we always have, and Jessica does too, mostly, because she's gotten into sync with us.

I wouldn't be surprised if there start to be more companies like us. I suspect founders may increasingly be able to resist, or at least postpone, turning into managers, just as a few decades ago they started to be able to resist switching from jeans to suits.

How do we manage to advise so many startups on the maker's schedule? By using the classic device for simulating the manager's schedule within the maker's: office hours. Several times a week I set aside a chunk of time to meet founders we've funded. These chunks of time are at the end of my working day, and I wrote a signup program that ensures all the appointments within a given set of office hours are clustered at the end. Because they come at the end of my day these meetings are never an interruption. (Unless their working day ends at the same time as mine, the meeting presumably interrupts theirs, but since they made the appointment it must be worth it to them.) During busy periods, office hours sometimes get long enough that they compress the day, but they never interrupt it.

When we were working on our own startup, back in the 90s, I evolved another trick for partitioning the day. I used to program from dinner till about 3 am every day, because at night no one could interrupt me. Then I'd sleep till about 11 am, and come in and work until dinner on what I called "business stuff." I never thought of it in these terms, but in effect I had two workdays each day, one on the manager's schedule and one on the maker's.

When you're operating on the manager's schedule you can do something you'd never want to do on the maker's: you can have speculative meetings. You can meet someone just to get to know one another. If you have an empty slot in your schedule, why not? Maybe it will turn out you can help one another in some way.

Business people in Silicon Valley (and the whole world, for that matter) have speculative meetings all the time. They're effectively free if you're on the manager's schedule. They're so common that there's distinctive language for proposing them: saying that you want to "grab coffee," for example.

Speculative meetings are terribly costly if you're on the maker's schedule, though. Which puts us in something of a bind. Everyone assumes that, like other investors, we run on the manager's schedule. So they introduce us to someone they think we ought to meet, or send us an email proposing we grab coffee. At this point we have two options, neither of them good: we can meet with them, and lose half a day's work; or we can try to avoid meeting them, and probably offend them.

Till recently we weren't clear in our own minds about the source of the problem. We just took it for granted that we had to either blow our schedules or offend people. But now that I've realized what's going on, perhaps there's a third option: to write something explaining the two types of schedule. Maybe eventually, if the conflict between the manager's schedule and the maker's schedule starts to be more widely understood, it will become less of a problem.

Those of us on the maker's schedule are willing to compromise. We know we have to have some number of meetings. All we ask from those on the manager's schedule is that they understand the cost.

Thanks to Sam Altman, Trevor Blackwell, Paul Buchheit, Jessica Livingston, and Robert Morris for reading drafts of this.

Related:

Startups

Chapter 14

How to Start a Startup

March 2005

(This essay is derived from a talk at the Harvard Computer Society.)

You need three things to create a successful startup: to start with good people, to make something customers actually want, and to spend as little money as possible. Most startups that fail do it because they fail at one of these. A startup that does all three will probably succeed.

And that's kind of exciting, when you think about it, because all three are doable. Hard, but doable. And since a startup that succeeds ordinarily makes its founders rich, that implies getting rich is doable too. Hard, but doable.

If there is one message I'd like to get across about startups, that's it. There is no magically difficult step that requires brilliance to solve.

The Idea

In particular, you don't need a brilliant idea to start a startup around. The way a startup makes money is to offer people better technology than they have now. But what people have now is often so bad that it doesn't take brilliance to do better.

Google's plan, for example, was simply to create a search site that didn't suck. They had three new ideas: index more of the Web, use links to rank search results, and have clean, simple web pages with unintrusive keyword-based ads. Above all, they were determined to make a site that was good to use. No doubt there are great technical tricks within Google, but the overall plan was straightforward. And while they probably have bigger ambitions now, this alone brings them a billion dollars a year. [1]

There are plenty of other areas that are just as backward as search was before Google. I can think of several heuristics for generating ideas for startups, but most reduce to this: look at something people are trying to do, and figure out how to do it in a way that doesn't suck.

For example, dating sites currently suck far worse than search did before Google. They all use the same simple-minded model. They seem to have approached the problem by thinking about how to do database matches instead of how dating works in the real world. An undergrad could build something better as a class project. And yet there's a lot of money at stake. Online dating is a valuable business now, and it might be worth a hundred times as much if it worked.

An idea for a startup, however, is only a beginning. A lot of would-be startup founders think the key to the whole process is the initial idea, and from that point all you have to do is execute. Venture capitalists know better. If you go to VC firms with a brilliant idea that you'll tell them about if they sign a nondisclosure agreement, most will tell you to get lost. That shows how much a mere idea is worth. The market price is less than the inconvenience of signing an NDA.

Another sign of how little the initial idea is worth is the number of startups that change their plan en route. Microsoft's original plan was to make money selling programming languages, of all things. Their current business model didn't occur to them until IBM dropped it in their lap five years later.

Ideas for startups are worth something, certainly, but the trouble is, they're not transferrable. They're not something you could hand to someone else to execute. Their value is mainly as starting points: as questions for the people who had them to continue thinking about.

What matters is not ideas, but the people who have them. Good people can fix bad ideas, but good ideas can't save bad people.

[1] Google's revenues are about two billion a year, but half comes from ads on other sites.

People

What do I mean by good people? One of the best tricks I learned during our startup was a rule for deciding who to hire. Could you describe the person as an animal? It might be hard to translate that into another language, but I think everyone in the US knows what it means. It means someone who takes their work a little too seriously; someone who does what they do so well that they pass right through professional and cross over into obsessive.

What it means specifically depends on the job: a salesperson who just won't take no for an answer; a hacker who will stay up till 4:00 AM rather than go to bed leaving code with a bug in it; a PR person who will cold-call *New York Times* reporters on their cell phones; a graphic designer who feels physical pain when something is two millimeters out of place.

Almost everyone who worked for us was an animal at what they did. The woman in charge of sales was so tenacious that I used to feel sorry for potential customers on the phone with her. You could sense them squirming on the hook, but you knew there would be no rest for them till they'd signed up.

If you think about people you know, you'll find the animal test is easy to apply. Call the person's image to mind and imagine the sentence "so-and-so is an animal." If you laugh, they're not. You don't need or perhaps even want this quality in big companies, but you need it in a startup.

For programmers we had three additional tests. Was the person genuinely smart? If so, could they actually get things done? And finally, since a few good hackers have unbearable personalities, could we stand to have them around?

That last test filters out surprisingly few people. We could bear any amount of nerdiness if someone was truly smart. What we couldn't stand were people with a lot of attitude. But most of those weren't truly smart, so our third test was largely a restatement of the first.

When nerds are unbearable it's usually because they're trying too hard to seem smart. But the smarter they are, the less pressure they feel to act smart. So as a rule you can recognize genuinely smart people by their ability to say things like "I don't know," "Maybe you're right," and "I don't understand x well enough."

This technique doesn't always work, because people can be influenced by their environment. In the MIT CS department, there seems to be a tradition of acting like a brusque know-it-all. I'm told it derives ultimately from Marvin Minsky, in the same way the classic airline pilot manner is said to derive from Chuck Yeager. Even genuinely smart people start to act this way there, so you have to make allowances.

It helped us to have Robert Morris, who is one of the readiest to say "I don't know" of anyone I've met. (At least, he was before he became a professor at MIT.) No one dared put on attitude around Robert, because he was obviously smarter than they were and yet had zero attitude himself.

Like most startups, ours began with a group of friends, and it was through personal contacts that we got most of the people we hired. This is a crucial difference between startups and big companies. Being friends with someone for even a couple days will tell you more than companies could ever learn in interviews. [2]

It's no coincidence that startups start around universities, because that's where smart people meet. It's not what people learn in classes at MIT and Stanford that has made technology companies spring up around them. They could sing campfire songs in the classes so long as admissions worked the same.

[2] One advantage startups have over established companies is that there are no discrimination laws about starting businesses. For example, I would be reluctant to start a startup with a woman who had small children, or was likely to have them soon. But you're not allowed to ask prospective employees if they plan to have kids soon. Believe it or not, under current US law, you're not even allowed to discriminate on the basis of intelligence. Whereas when you're starting a company, you can discriminate on any basis you want about who you start it with.

If you start a startup, there's a good chance it will be with people you know from college or grad school. So in theory you ought to try to make friends with as many smart people as you can in school, right? Well, no. Don't make a conscious effort to schmooze; that doesn't work well with hackers.

What you should do in college is work on your own projects. Hackers should do this even if they don't plan to start startups, because it's the only real way to learn how to program. In some cases you may collaborate with other students, and this is the best way to get to know good hackers. The project may even grow into a startup. But once again, I wouldn't aim too directly at either target. Don't force things; just work on stuff you like with people you like.

Ideally you want between two and four founders. It would be hard to start with just one. One person would find the moral weight of starting a company hard to bear. Even Bill Gates, who seems to be able to bear a good deal of moral weight, had to have a co-founder. But you don't want so many founders that the company starts to look like a group photo. Partly because you don't need a lot of people at first, but mainly because the more founders you have, the worse disagreements you'll have. When there are just two or three founders, you know you have to resolve disputes immediately or perish. If there are seven or eight, disagreements can linger and harden into factions. You don't want mere voting; you need unanimity.

In a technology startup, which most startups are, the founders should include technical people. During the Internet Bubble there were a number of startups founded by business people who then went looking for hackers to create their product for them. This doesn't work well. Business people are bad at deciding what to do with technology, because they don't know what the options are, or which kinds of problems are hard and which are easy. And when business people try to hire hackers, they can't tell which ones are good. Even other hackers have a hard time doing that. For business people it's roulette.

Do the founders of a startup have to include business people? That depends. We thought so when we started ours, and we asked several people who were said to know about this mysterious thing called "business" if they would be the president. But they all said no, so I had to do it myself. And what I discovered was that business was no great mystery. It's not something like physics or medicine that requires extensive study. You just try to get people to pay you for stuff.

I think the reason I made such a mystery of business was that I was disgusted by the idea of doing it. I wanted to work in the pure, intellectual world of software, not deal with customers' mundane problems. People who don't want to get dragged into some kind of work often develop a protective incompetence at it. Paul Erdos was particularly good at this. By seeming unable even to cut a grapefruit in half (let alone go to the store and buy one), he forced other people to do such things for him, leaving all his time free for math. Erdos was an extreme case, but most husbands use the same trick to some degree.

Once I was forced to discard my protective incompetence, I found that business was neither so hard nor so boring as I feared. There are esoteric areas of business that are quite hard, like tax law or the pricing of derivatives, but you don't need to know about those in a startup. All you need to know about business to run a startup are commonsense things people knew before there were business schools, or even universities.

If you work your way down the Forbes 400 making an x next to the name of each person with an MBA, you'll learn something important about business school. After Warren Buffett, you don't hit another MBA till number 22, Phil Knight, the CEO of Nike. There are only 5 MBAs in the top 50. What you notice in the Forbes 400 are a lot of people with technical backgrounds. Bill Gates, Steve Jobs, Larry Ellison, Michael Dell, Jeff Bezos, Gordon Moore. The rulers of the technology business tend to come from technology, not business. So if you want to invest two years in something that will help you succeed in business, the evidence suggests you'd do better to learn how to hack than get an MBA. [3]

There is one reason you might want to include business people in a startup, though: because you have to have at least one person willing and able to focus on what customers want. Some believe only business people can do this– that hackers can implement software, but not design it. That's nonsense. There's nothing about knowing how to program that prevents hackers from understanding users, or about not knowing how to program that magically enables business people to understand them.

If you can't understand users, however, you should either learn how or find a co-founder who can. That is the single most important issue for technology startups, and the rock that sinks more of them than anything else.

What Customers Want

It's not just startups that have to worry about this. I think most businesses that fail do it because they don't give customers what they want. Look at restaurants. A large percentage fail, about a quarter in the first year. But can you think of one restaurant that had really good food and went out of business?

Restaurants with great food seem to prosper no matter what. A restaurant with great food can be expensive, crowded, noisy, dingy, out of the way, and even have bad service, and people will keep coming. It's true that a restaurant with mediocre food can sometimes attract customers through gimmicks. But that approach is very risky. It's more straightforward just to make the food good.

It's the same with technology. You hear all kinds of reasons why startups fail. But can you think of one that had a massively popular product and still failed?

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[3] Learning to hack is a lot cheaper than business school, because you can do it mostly on your own. For the price of a Linux box, a copy of K&R, and a few hours of advice from your neighbor's fifteen year old son, you'll be well on your way.

In nearly every failed startup, the real problem was that customers didn't want the product. For most, the cause of death is listed as "ran out of funding," but that's only the immediate cause. Why couldn't they get more funding? Probably because the product was a dog, or never seemed likely to be done, or both.

When I was trying to think of the things every startup needed to do, I almost included a fourth: get a version 1 out as soon as you can. But I decided not to, because that's implicit in making something customers want. The only way to make something customers want is to get a prototype in front of them and refine it based on their reactions.

The other approach is what I call the "Hail Mary" strategy. You make elaborate plans for a product, hire a team of engineers to develop it (people who do this tend to use the term "engineer" for hackers), and then find after a year that you've spent two million dollars to develop something no one wants. This was not uncommon during the Bubble, especially in companies run by business types, who thought of software development as something terrifying that therefore had to be carefully planned.

We never even considered that approach. As a Lisp hacker, I come from the tradition of rapid prototyping. I would not claim (at least, not here) that this is the right way to write every program, but it's certainly the right way to write software for a startup. In a startup, your initial plans are almost certain to be wrong in some way, and your first priority should be to figure out where. The only way to do that is to try implementing them.

Like most startups, we changed our plan on the fly. At first we expected our customers to be Web consultants. But it turned out they didn't like us, because our software was easy to use and we hosted the site. It would be too easy for clients to fire them. We also thought we'd be able to sign up a lot of catalog companies, because selling online was a natural extension of their existing business. But in 1996 that was a hard sell. The middle managers we talked to at catalog companies saw the Web not as an opportunity, but as something that meant more work for them.

We did get a few of the more adventurous catalog companies. Among them was Frederick's of Hollywood, which gave us valuable experience dealing with heavy loads on our servers. But most of our users were small, individual merchants who saw the Web as an opportunity to build a business. Some had retail stores, but many only existed online. And so we changed direction to focus on these users. Instead of concentrating on the features Web consultants and catalog companies would want, we worked to make the software easy to use.

I learned something valuable from that. It's worth trying very, very hard to make technology easy to use. Hackers are so used to computers that they have no idea how horrifying software seems to normal people. Stephen Hawking's editor told him that every equation he included in his book would cut sales in half. When you work on making technology easier to use, you're riding that curve up instead of down. A 10% improvement in ease of use doesn't just increase your sales 10%. It's more likely to double your sales.

How do you figure out what customers want? Watch them. One of the best places to do this was at trade shows. Trade shows didn't pay as a way of getting new customers, but they were worth it as market research. We didn't just give canned presentations at trade shows. We used to show people how to build real, working stores. Which meant we got to watch as they used our software, and talk to them about what they needed.

No matter what kind of startup you start, it will probably be a stretch for you, the founders, to understand what users want. The only kind of software you can build without studying users is the sort for which you are the typical user. But this is just the kind that tends to be open source: operating systems, programming languages, editors, and so on. So if you're developing technology for money, you're probably not going to be developing it for people like you. Indeed, you can use this as a way to generate ideas for startups: what do people who are not like you want from technology?

When most people think of startups, they think of companies like Apple or Google. Everyone knows these, because they're big consumer brands. But for every startup like that, there are twenty more that operate in niche markets or live quietly down in the infrastructure. So if you start a successful startup, odds are you'll start one of those.

Another way to say that is, if you try to start the kind of startup that has to be a big consumer brand, the odds against succeeding are steeper. The best odds are in niche markets. Since startups make money by offering people something better than they had before, the best opportunities are where things suck most. And it would be hard to find a place where things suck more than in corporate IT departments. You would not believe the amount of money companies spend on software, and the crap they get in return. This imbalance equals opportunity.

If you want ideas for startups, one of the most valuable things you could do is find a middle-sized non-technology company and spend a couple weeks just watching what they do with computers. Most good hackers have no more idea of the horrors perpetrated in these places than rich Americans do of what goes on in Brazilian slums.

Start by writing software for smaller companies, because it's easier to sell to them. It's worth so much to sell stuff to big companies that the people selling them the crap they currently use spend a lot of time and money to do it. And while you can outhack Oracle with one frontal lobe tied behind your back, you can't outsell an Oracle salesman. So if you want to win through better technology, aim at smaller customers. [4]

They're the more strategically valuable part of the market anyway. In technology, the low end always eats the high end. It's easier to make an inexpensive product more powerful than to make a powerful product cheaper. So the products that start as cheap, simple options tend to gradually grow more powerful till, like water rising in a room, they

[4] Corollary: Avoid starting a startup to sell things to the biggest company of all, the government. Yes, there are lots of opportunities to sell them technology. But let someone else start those startups.

squash the “high-end” products against the ceiling. Sun did this to mainframes, and Intel is doing it to Sun. Microsoft Word did it to desktop publishing software like Interleaf and Framemaker. Mass-market digital cameras are doing it to the expensive models made for professionals. Avid did it to the manufacturers of specialized video editing systems, and now Apple is doing it to Avid. *Henry Ford* did it to the car makers that preceded him. If you build the simple, inexpensive option, you’ll not only find it easier to sell at first, but you’ll also be in the best position to conquer the rest of the market.

It’s very dangerous to let anyone fly under you. If you have the cheapest, easiest product, you’ll own the low end. And if you don’t, you’re in the crosshairs of whoever does.

Raising Money

To make all this happen, you’re going to need money. Some startups have been self-funding– Microsoft for example– but most aren’t. I think it’s wise to take money from investors. To be self-funding, you have to start as a consulting company, and it’s hard to switch from that to a product company.

Financially, a startup is like a pass/fail course. The way to get rich from a startup is to maximize the company’s chances of succeeding, not to maximize the amount of stock you retain. So if you can trade stock for something that improves your odds, it’s probably a smart move.

To most hackers, getting investors seems like a terrifying and mysterious process. Actually it’s merely tedious. I’ll try to give an outline of how it works.

The first thing you’ll need is a few tens of thousands of dollars to pay your expenses while you develop a prototype. This is called seed capital. Because so little money is involved, raising seed capital is comparatively easy– at least in the sense of getting a quick yes or no.

Usually you get seed money from individual rich people called “angels." Often they're people who themselves got rich from technology. At the seed stage, investors don't expect you to have an elaborate business plan. Most know that they're supposed to decide quickly. It's not unusual to get a check within a week based on a half-page agreement.

We started Viaweb with $10,000 of seed money from our friend Julian. But he gave us a lot more than money. He's a former CEO and also a corporate lawyer, so he gave us a lot of valuable advice about business, and also did all the legal work of getting us set up as a company. Plus he introduced us to one of the two angel investors who supplied our next round of funding.

Some angels, especially those with technology backgrounds, may be satisfied with a demo and a verbal description of what you plan to do. But many will want a copy of your business plan, if only to remind themselves what they invested in.

Our angels asked for one, and looking back, I'm amazed how much worry it caused me. "Business plan" has that word "business" in it, so I figured it had to be something I'd have to read a book about business plans to write. Well, it doesn't. At this stage, all most investors expect is a brief description of what you plan to do and how you're going to make money from it, and the resumes of the founders. If you just sit down and write out what you've been saying to one another, that should be fine. It shouldn't take more than a couple hours, and you'll probably find that writing it all down gives you more ideas about what to do.

For the angel to have someone to make the check out to, you're going to have to have some kind of company. Merely incorporating yourselves isn't hard. The problem is, for the company to exist, you have to decide who the founders are, and how much stock they each have. If there are two founders with the same qualifications who are both equally committed to the business, that's easy. But if you have a number of people who are expected to contribute in varying degrees, arranging the proportions of stock can be hard. And once you've done it, it tends to be set in stone.

I have no tricks for dealing with this problem. All I can say is, try hard to do it right. I do have a rule of thumb for recognizing when you have, though. When everyone feels they're getting a slightly bad deal, that they're doing more than they should for the amount of stock they have, the stock is optimally apportioned.

There is more to setting up a company than incorporating it, of course: insurance, business license, unemployment compensation, various things with the IRS. I'm not even sure what the list is, because we, ah, skipped all that. When we got real funding near the end of 1996, we hired a great CFO, who fixed everything retroactively. It turns out that no one comes and arrests you if you don't do everything you're supposed to when starting a company. And a good thing too, or a lot of startups would never get started. [5]

It can be dangerous to delay turning yourself into a company, because one or more of the founders might decide to split off and start another company doing the same thing. This does happen. So when you set up the company, as well as as apportioning the stock, you should get all the founders to sign something agreeing that everyone's ideas belong to this company, and that this company is going to be everyone's only job.

[If this were a movie, ominous music would begin here.]

While you're at it, you should ask what else they've signed. One of the worst things that can happen to a startup is to run into intellectual property problems. We did, and it came closer to killing us than any competitor ever did.

As we were in the middle of getting bought, we discovered that one of our people had, early on, been bound by an agreement that said all his ideas belonged to the giant company that was paying for him to go to grad school. In theory, that could have meant someone else owned

[5] A friend who started a company in Germany told me they do care about the paperwork there, and that there's more of it. Which helps explain why there are not more startups in Germany.

big chunks of our software. So the acquisition came to a screeching halt while we tried to sort this out. The problem was, since we'd been about to be acquired, we'd allowed ourselves to run low on cash. Now we needed to raise more to keep going. But it's hard to raise money with an IP cloud over your head, because investors can't judge how serious it is.

Our existing investors, knowing that we needed money and had nowhere else to get it, at this point attempted certain gambits which I will not describe in detail, except to remind readers that the word "angel" is a metaphor. The founders thereupon proposed to walk away from the company, after giving the investors a brief tutorial on how to administer the servers themselves. And while this was happening, the acquirers used the delay as an excuse to welch on the deal.

Miraculously it all turned out ok. The investors backed down; we did another round of funding at a reasonable valuation; the giant company finally gave us a piece of paper saying they didn't own our software; and six months later we were bought by Yahoo for much more than the earlier acquirer had agreed to pay. So we were happy in the end, though the experience probably took several years off my life.

Don't do what we did. Before you consummate a startup, ask everyone about their previous IP history.

Once you've got a company set up, it may seem presumptuous to go knocking on the doors of rich people and asking them to invest tens of thousands of dollars in something that is really just a bunch of guys with some ideas. But when you look at it from the rich people's point of view, the picture is more encouraging. Most rich people are looking for good investments. If you really think you have a chance of succeeding, you're doing them a favor by letting them invest. Mixed with any annoyance they might feel about being approached will be the thought: are these guys the next Google?

Usually angels are financially equivalent to founders. They get the same kind of stock and get diluted the same amount in future rounds.

How much stock should they get? That depends on how ambitious you feel. When you offer x percent of your company for y dollars, you're implicitly claiming a certain value for the whole company. Venture investments are usually described in terms of that number. If you give an investor new shares equal to 5% of those already outstanding in return for $100,000, then you've done the deal at a pre-money valuation of $2 million.

How do you decide what the value of the company should be? There is no rational way. At this stage the company is just a bet. I didn't realize that when we were raising money. Julian thought we ought to value the company at several million dollars. I thought it was preposterous to claim that a couple thousand lines of code, which was all we had at the time, were worth several million dollars. Eventually we settled on one million, because Julian said no one would invest in a company with a valuation any lower. [6]

What I didn't grasp at the time was that the valuation wasn't just the value of the code we'd written so far. It was also the value of our ideas, which turned out to be right, and of all the future work we'd do, which turned out to be a lot.

The next round of funding is the one in which you might deal with actual venture capital firms. But don't wait till you've burned through your last round of funding to start approaching them. VCs are slow to make up their minds. They can take months. You don't want to be running out of money while you're trying to negotiate with them.

Getting money from an actual VC firm is a bigger deal than getting money from angels. The amounts of money involved are larger, millions usually. So the deals take longer, dilute you more, and impose more onerous conditions.

Sometimes the VCs want to install a new CEO of their own choosing.

[6] At the seed stage our valuation was in principle $100,000, because Julian got 10% of the company. But this is a very misleading number, because the money was the least important of the things Julian gave us.

Usually the claim is that you need someone mature and experienced, with a business background. Maybe in some cases this is true. And yet Bill Gates was young and inexperienced and had no business background, and he seems to have done ok. Steve Jobs got booted out of his own company by someone mature and experienced, with a business background, who then proceeded to ruin the company. So I think people who are mature and experienced, with a business background, may be overrated. We used to call these guys "newscasters," because they had neat hair and spoke in deep, confident voices, and generally didn't know much more than they read on the teleprompter.

We talked to a number of VCs, but eventually we ended up financing our startup entirely with angel money. The main reason was that we feared a brand-name VC firm would stick us with a newscaster as part of the deal. That might have been ok if he was content to limit himself to talking to the press, but what if he wanted to have a say in running the company? That would have led to disaster, because our software was so complex. We were a company whose whole m.o. was to win through better technology. The strategic decisions were mostly decisions about technology, and we didn't need any help with those.

This was also one reason we didn't go public. Back in 1998 our CFO tried to talk me into it. In those days you could go public as a dogfood portal, so as a company with a real product and real revenues, we might have done well. But I feared it would have meant taking on a newscaster– someone who, as they say, "can talk Wall Street's language."

I'm happy to see Google is bucking that trend. They didn't talk Wall Street's language when they did their IPO, and Wall Street didn't buy. And now Wall Street is collectively kicking itself. They'll pay attention next time. Wall Street learns new languages fast when money is involved.

You have more leverage negotiating with VCs than you realize. The reason is other VCs. I know a number of VCs now, and when you talk to them you realize that it's a seller's market. Even now there is too much money chasing too few good deals.

VCs form a pyramid. At the top are famous ones like Sequoia and Kleiner Perkins, but beneath those are a huge number you've never heard of. What they all have in common is that a dollar from them is worth one dollar. Most VCs will tell you that they don't just provide money, but connections and advice. If you're talking to Vinod Khosla or John Doerr or Mike Moritz, this is true. But such advice and connections can come very expensive. And as you go down the food chain the VCs get rapidly dumber. A few steps down from the top you're basically talking to bankers who've picked up a few new vocabulary words from reading *Wired*. (Does your product use *XML?*) So I'd advise you to be skeptical about claims of experience and connections. Basically, a VC is a source of money. I'd be inclined to go with whoever offered the most money the soonest with the least strings attached.

You may wonder how much to tell VCs. And you should, because some of them may one day be funding your competitors. I think the best plan is not to be overtly secretive, but not to tell them everything either. After all, as most VCs say, they're more interested in the people than the ideas. The main reason they want to talk about your idea is to judge you, not the idea. So as long as you seem like you know what you're doing, you can probably keep a few things back from them. [7]

Talk to as many VCs as you can, even if you don't want their money, because a) they may be on the board of someone who will buy you, and b) if you seem impressive, they'll be discouraged from investing in your competitors. The most efficient way to reach VCs, especially if you only want them to know about you and don't want their money, is at the conferences that are occasionally organized for startups to present to them.

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[7] The same goes for companies that seem to want to acquire you. There will be a few that are only pretending to in order to pick your brains. But you can never tell for sure which these are, so the best approach is to seem entirely open, but to fail to mention a few critical technical secrets.

Not Spending It

When and if you get an infusion of real money from investors, what should you do with it? Not spend it, that's what. In nearly every startup that fails, the proximate cause is running out of money. Usually there is something deeper wrong. But even a proximate cause of death is worth trying hard to avoid.

During the Bubble many startups tried to "get big fast." Ideally this meant getting a lot of customers fast. But it was easy for the meaning to slide over into hiring a lot of people fast.

Of the two versions, the one where you get a lot of customers fast is of course preferable. But even that may be overrated. The idea is to get there first and get all the users, leaving none for competitors. But I think in most businesses the advantages of being first to market are not so overwhelmingly great. Google is again a case in point. When they appeared it seemed as if search was a mature market, dominated by big players who'd spent millions to build their brands: Yahoo, Lycos, Excite, Infoseek, Altavista, Inktomi. Surely 1998 was a little late to arrive at the party.

But as the founders of Google knew, brand is worth next to nothing in the search business. You can come along at any point and make something better, and users will gradually seep over to you. As if to emphasize the point, Google never did any advertising. They're like dealers; they sell the stuff, but they know better than to use it themselves.

The competitors Google buried would have done better to spend those millions improving their software. Future startups should learn from that mistake. Unless you're in a market where products are as undifferentiated as cigarettes or vodka or laundry detergent, spending a lot on brand advertising is a sign of breakage. And few if any Web businesses are so undifferentiated. The dating sites are running big ad campaigns right now, which is all the more evidence they're ripe for the picking. (Fee, fie, fo, fum, I smell a company run by marketing guys.)

We were compelled by circumstances to grow slowly, and in retrospect it was a good thing. The founders all learned to do every job in the company. As well as writing software, I had to do sales and customer support. At sales I was not very good. I was persistent, but I didn't have the smoothness of a good salesman. My message to potential customers was: you'd be stupid not to sell online, and if you sell online you'd be stupid to use anyone else's software. Both statements were true, but that's not the way to convince people.

I was great at customer support though. Imagine talking to a customer support person who not only knew everything about the product, but would apologize abjectly if there was a bug, and then fix it immediately, while you were on the phone with them. Customers loved us. And we loved them, because when you're growing slow by word of mouth, your first batch of users are the ones who were smart enough to find you by themselves. There is nothing more valuable, in the early stages of a startup, than smart users. If you listen to them, they'll tell you exactly how to make a winning product. And not only will they give you this advice for free, they'll pay you.

We officially launched in early 1996. By the end of that year we had about 70 users. Since this was the era of "get big fast," I worried about how small and obscure we were. But in fact we were doing exactly the right thing. Once you get big (in users or employees) it gets hard to change your product. That year was effectively a laboratory for improving our software. By the end of it, we were so far ahead of our competitors that they never had a hope of catching up. And since all the hackers had spent many hours talking to users, we understood online commerce way better than anyone else.

That's the key to success as a startup. There is nothing more important than understanding your business. You might think that anyone in a business must, ex officio, understand it. Far from it. Google's secret weapon was simply that they understood search. I was working for Yahoo when Google appeared, and Yahoo didn't understand search. I know because I once tried to convince the powers that be that we had to make search better, and I got in reply what was then the party line about it: that Yahoo was no longer a mere "search engine." Search was now only a small percentage of our page views, less than one month's growth, and now that we were established as a "media company," or "portal," or whatever we were, search could safely be allowed to wither and drop off, like an umbilical cord.

Well, a small fraction of page views they may be, but they are an important fraction, because they are the page views that Web sessions start with. I think Yahoo gets that now.

Google understands a few other things most Web companies still don't. The most important is that you should put users before advertisers, even though the advertisers are paying and users aren't. One of my favorite bumper stickers reads "if the people lead, the leaders will follow." Paraphrased for the Web, this becomes "get all the users, and the advertisers will follow." More generally, design your product to please users first, and then think about how to make money from it. If you don't put users first, you leave a gap for competitors who do.

To make something users love, you have to understand them. And the bigger you are, the harder that is. So I say "get big slow." The slower you burn through your funding, the more time you have to learn.

The other reason to spend money slowly is to encourage a culture of cheapness. That's something Yahoo did understand. David Filo's title was "Chief Yahoo," but he was proud that his unofficial title was "Cheap Yahoo." Soon after we arrived at Yahoo, we got an email from Filo, who had been crawling around our directory hierarchy, asking if it was really necessary to store so much of our data on expensive RAID drives. I was impressed by that. Yahoo's market cap then was already in the billions, and they were still worrying about wasting a few gigs of disk space.

When you get a couple million dollars from a VC firm, you tend to feel rich. It's important to realize you're not. A rich company is one with large revenues. This money isn't revenue. It's money investors have given you in the hope you'll be able to generate revenues. So despite those millions in the bank, you're still poor.

For most startups the model should be grad student, not law firm. Aim for cool and cheap, not expensive and impressive. For us the test of whether a startup understood this was whether they had Aeron chairs. The Aeron came out during the Bubble and was very popular with startups. Especially the type, all too common then, that was like a bunch of kids playing house with money supplied by VCs. We had office chairs so cheap that the arms all fell off. This was slightly embarrassing at the time, but in retrospect the grad-studenty atmosphere of our office was another of those things we did right without knowing it.

Our offices were in a wooden triple-decker in Harvard Square. It had been an apartment until about the 1970s, and there was still a claw-footed bathtub in the bathroom. It must once have been inhabited by someone fairly eccentric, because a lot of the chinks in the walls were stuffed with aluminum foil, as if to protect against cosmic rays. When eminent visitors came to see us, we were a bit sheepish about the low production values. But in fact that place was the perfect space for a startup. We felt like our role was to be impudent underdogs instead of corporate stuffed shirts, and that is exactly the spirit you want.

An apartment is also the right kind of place for developing software. Cube farms suck for that, as you've probably discovered if you've tried it. Ever notice how much easier it is to hack at home than at work? So why not make work more like home?

When you're looking for space for a startup, don't feel that it has to look professional. Professional means doing good work, not elevators and glass walls. I'd advise most startups to avoid corporate space at first and just rent an apartment. You want to live at the office in a startup, so why not have a place designed to be lived in as your office?

Besides being cheaper and better to work in, apartments tend to be in better locations than office buildings. And for a startup location is very important. The key to productivity is for people to come back to work after dinner. Those hours after the phone stops ringing are by far the best for getting work done. Great things happen when a group of employees go out to dinner together, talk over ideas, and then come back to their offices to implement them. So you want to be in a place where there are a lot of restaurants around, not some dreary office park that's a wasteland after 6:00 PM. Once a company shifts over into the model where everyone drives home to the suburbs for dinner, however late, you've lost something extraordinarily valuable. God help you if you actually start in that mode.

If I were going to start a startup today, there are only three places I'd consider doing it: on the Red Line near Central, Harvard, or Davis Squares (Kendall is too sterile); in Palo Alto on University or California Aves; and in Berkeley immediately north or south of campus. These are the only places I know that have the right kind of vibe.

The most important way to not spend money is by not hiring people. I may be an extremist, but I think hiring people is the worst thing a company can do. To start with, people are a recurring expense, which is the worst kind. They also tend to cause you to grow out of your space, and perhaps even move to the sort of uncool office building that will make your software worse. But worst of all, they slow you down: instead of sticking your head in someone's office and checking out an idea with them, eight people have to have a meeting about it. So the fewer people you can hire, the better.

During the Bubble a lot of startups had the opposite policy. They wanted to get "staffed up" as soon as possible, as if you couldn't get anything done unless there was someone with the corresponding job title. That's big company thinking. Don't hire people to fill the gaps in some a priori org chart. The only reason to hire someone is to do something you'd like to do but can't.

If hiring unnecessary people is expensive and slows you down, why do nearly all companies do it? I think the main reason is that people like the idea of having a lot of people working for them. This weakness often extends right up to the CEO. If you ever end up running a company, you'll find the most common question people ask is how many employees you have. This is their way of weighing you. It's not just random people who ask this; even reporters do. And they're going to be a lot more impressed if the answer is a thousand than if it's ten.

This is ridiculous, really. If two companies have the same revenues, it's the one with fewer employees that's more impressive. When people used to ask me how many people our startup had, and I answered "twenty," I could see them thinking that we didn't count for much. I used to want to add "but our main competitor, whose ass we regularly kick, has a hundred and forty, so can we have credit for the larger of the two numbers?"

As with office space, the number of your employees is a choice between seeming impressive, and being impressive. Any of you who were nerds in high school know about this choice. Keep doing it when you start a company.

Should You?

But should you start a company? Are you the right sort of person to do it? If you are, is it worth it?

More people are the right sort of person to start a startup than realize it. That's the main reason I wrote this. There could be ten times more startups than there are, and that would probably be a good thing.

I was, I now realize, exactly the right sort of person to start a startup. But the idea terrified me at first. I was forced into it because I was a Lisp hacker. The company I'd been consulting for seemed to be running into trouble, and there were not a lot of other companies using Lisp. Since I couldn't bear the thought of programming in another language (this was 1995, remember, when "another language" meant C++) the only option seemed to be to start a new company using Lisp.

I realize this sounds far-fetched, but if you're a Lisp hacker you'll know what I mean. And if the idea of starting a startup frightened me so much that I only did it out of necessity, there must be a lot of people who would be good at it but who are too intimidated to try.

So who should start a startup? Someone who is a good hacker, between about 23 and 38, and who wants to solve the money problem in one shot instead of getting paid gradually over a conventional working life.

I can't say precisely what a good hacker is. At a first rate university this might include the top half of computer science majors. Though of course you don't have to be a CS major to be a hacker; I was a philosophy major in college.

It's hard to tell whether you're a good hacker, especially when you're young. Fortunately the process of starting startups tends to select them automatically. What drives people to start startups is (or should be) looking at existing technology and thinking, don't these guys realize they should be doing x, y, and z? And that's also a sign that one is a good hacker.

I put the lower bound at 23 not because there's something that doesn't happen to your brain till then, but because you need to see what it's like in an existing business before you try running your own. The business doesn't have to be a startup. I spent a year working for a software company to pay off my college loans. It was the worst year of my adult life, but I learned, without realizing it at the time, a lot of valuable lessons about the software business. In this case they were mostly negative lessons: don't have a lot of meetings; don't have chunks of code that multiple people own; don't have a sales guy running the company; don't make a high-end product; don't let your code get too big; don't leave finding bugs to QA people; don't go too long between releases; don't isolate developers from users; don't move from Cambridge to Route 128; and so on. [8] But negative lessons are just as

[8] I was as bad an employee as this place was a company. I apologize to anyone who had to work with me there.

valuable as positive ones. Perhaps even more valuable: it's hard to repeat a brilliant performance, but it's straightforward to avoid errors. [9]

The other reason it's hard to start a company before 23 is that people won't take you seriously. VCs won't trust you, and will try to reduce you to a mascot as a condition of funding. Customers will worry you're going to flake out and leave them stranded. Even you yourself, unless you're very unusual, will feel your age to some degree; you'll find it awkward to be the boss of someone much older than you, and if you're 21, hiring only people younger rather limits your options.

Some people could probably start a company at 18 if they wanted to. Bill Gates was 19 when he and Paul Allen started Microsoft. (Paul Allen was 22, though, and that probably made a difference.) So if you're thinking, I don't care what he says, I'm going to start a company now, you may be the sort of person who could get away with it.

The other cutoff, 38, has a lot more play in it. One reason I put it there is that I don't think many people have the physical stamina much past that age. I used to work till 2:00 or 3:00 AM every night, seven days a week. I don't know if I could do that now.

Also, startups are a big risk financially. If you try something that blows up and leaves you broke at 26, big deal; a lot of 26 year olds are broke. By 38 you can't take so many risks– especially if you have kids.

My final test may be the most restrictive. Do you actually want to start a startup? What it amounts to, economically, is compressing your working life into the smallest possible space. Instead of working at an ordinary rate for 40 years, you work like hell for four. And maybe end up with nothing– though in that case it probably won't take four years.

During this time you'll do little but work, because when you're not working, your competitors will be. My only leisure activities were running, which I needed to do to keep working anyway, and about fifteen

[9] You could probably write a book about how to succeed in business by doing everything in exactly the opposite way from the DMV.

minutes of reading a night. I had a girlfriend for a total of two months during that three year period. Every couple weeks I would take a few hours off to visit a used bookshop or go to a friend's house for dinner. I went to visit my family twice. Otherwise I just worked.

Working was often fun, because the people I worked with were some of my best friends. Sometimes it was even technically interesting. But only about 10% of the time. The best I can say for the other 90% is that some of it is funnier in hindsight than it seemed then. Like the time the power went off in Cambridge for about six hours, and we made the mistake of trying to start a gasoline powered generator inside our offices. I won't try that again.

I don't think the amount of bullshit you have to deal with in a startup is more than you'd endure in an ordinary working life. It's probably less, in fact; it just seems like a lot because it's compressed into a short period. So mainly what a startup buys you is time. That's the way to think about it if you're trying to decide whether to start one. If you're the sort of person who would like to solve the money problem once and for all instead of working for a salary for 40 years, then a startup makes sense.

For a lot of people the conflict is between startups and graduate school. Grad students are just the age, and just the sort of people, to start software startups. You may worry that if you do you'll blow your chances of an academic career. But it's possible to be part of a startup and stay in grad school, especially at first. Two of our three original hackers were in grad school the whole time, and both got their degrees. There are few sources of energy so powerful as a procrastinating grad student.

If you do have to leave grad school, in the worst case it won't be for too long. If a startup fails, it will probably fail quickly enough that you can return to academic life. And if it succeeds, you may find you no longer have such a burning desire to be an assistant professor.

If you want to do it, do it. Starting a startup is not the great mystery it seems from outside. It's not something you have to know about "business" to do. Build something users love, and spend less than you make. How hard is that?

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Chapter 15

How to Get Startup Ideas

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The way to get startup ideas is not to try to think of startup ideas. It's to look for problems, preferably problems you have yourself.

The very best startup ideas tend to have three things in common: they're something the founders themselves want, that they themselves can build, and that few others realize are worth doing. Microsoft, Apple, Yahoo, Google, and Facebook all began this way.

Problems

Why is it so important to work on a problem you have? Among other things, it ensures the problem really exists. It sounds obvious to say you should only work on problems that exist. And yet by far the most common mistake startups make is to solve problems no one has.

I made it myself. In 1995 I started a company to put art galleries online. But galleries didn't want to be online. It's not how the art business works. So why did I spend 6 months working on this stupid idea? Because I didn't pay attention to users. I invented a model of the world that didn't correspond to reality, and worked from that. I didn't notice my model was wrong until I tried to convince users to pay for what we'd built. Even then I took embarrassingly long to catch on. I was attached to my model of the world, and I'd spent a lot of time on the software. They had to want it!

Why do so many founders build things no one wants? Because they begin by trying to think of startup ideas. That m.o. is doubly dangerous: it doesn't merely yield few good ideas; it yields bad ideas that sound plausible enough to fool you into working on them.

At YC we call these "made-up" or "sitcom" startup ideas. Imagine one of the characters on a TV show was starting a startup. The writers would have to invent something for it to do. But coming up with good startup ideas is hard. It's not something you can do for the asking. So (unless they got amazingly lucky) the writers would come up with an idea that sounded plausible, but was actually bad.

For example, a social network for pet owners. It doesn't sound obviously mistaken. Millions of people have pets. Often they care a lot about their pets and spend a lot of money on them. Surely many of these people would like a site where they could talk to other pet owners. Not all of them perhaps, but if just 2 or 3 percent were regular visitors, you could have millions of users. You could serve them targeted offers, and maybe charge for premium features. [1]

The danger of an idea like this is that when you run it by your friends with pets, they don't say "I would *never* use this." They say "Yeah, maybe I could see using something like that." Even when the startup launches, it will sound plausible to a lot of people. They don't want to use it themselves, at least not right now, but they could imagine other people wanting it. Sum that reaction across the entire population, and you have zero users. [2]

[1]This form of bad idea has been around as long as the web. It was common in the 1990s, except then people who had it used to say they were going to create a portal for x instead of a social network for x. Structurally the idea is stone soup: you post a sign saying "this is the place for people interested in x," and all those people show up and you make money from them. What lures founders into this sort of idea are statistics about the millions of people who might be interested in each type of x. What they forget is that any given person might have 20 affinities by this standard, and no one is going to visit 20 different communities regularly.

[2]I'm not saying, incidentally, that I know for sure a social network for pet owners is a bad idea. I know it's a bad idea the way I know randomly generated DNA would not produce a viable organism. The set of plausible sounding startup ideas is many times larger than the set of good ones, and many of the good ones don't even sound that plausible. So if all you know about a startup idea is that it sounds plausible, you have to assume it's bad.

Well

When a startup launches, there have to be at least some users who really need what they're making — not just people who could see themselves using it one day, but who want it urgently. Usually this initial group of users is small, for the simple reason that if there were something that large numbers of people urgently needed and that could be built with the amount of effort a startup usually puts into a version one, it would probably already exist. Which means you have to compromise on one dimension: you can either build something a large number of people want a small amount, or something a small number of people want a large amount. Choose the latter. Not all ideas of that type are good startup ideas, but nearly all good startup ideas are of that type.

Imagine a graph whose x axis represents all the people who might want what you're making and whose y axis represents how much they want it. If you invert the scale on the y axis, you can envision companies as holes. Google is an immense crater: hundreds of millions of people use it, and they need it a lot. A startup just starting out can't expect to excavate that much volume. So you have two choices about the shape of hole you start with. You can either dig a hole that's broad but shallow, or one that's narrow and deep, like a well.

Made-up startup ideas are usually of the first type. Lots of people are mildly interested in a social network for pet owners.

Nearly all good startup ideas are of the second type. Microsoft was a well when they made Altair Basic. There were only a couple thousand Altair owners, but without this software they were programming in machine language. Thirty years later Facebook had the same shape. Their first site was exclusively for Harvard students, of which there are only a few thousand, but those few thousand users wanted it a lot.

When you have an idea for a startup, ask yourself: who wants this right now? Who wants this so much that they'll use it even when it's a crappy version one made by a two-person startup they've never heard of? If you can't answer that, the idea is probably bad. [3]

You don't need the narrowness of the well per se. It's depth you need; you get narrowness as a byproduct of optimizing for depth (and speed). But you almost always do get it. In practice the link between depth and narrowness is so strong that it's a good sign when you know that an idea will appeal strongly to a specific group or type of user.

But while demand shaped like a well is almost a necessary condition for a good startup idea, it's not a sufficient one. If Mark Zuckerberg had built something that could only ever have appealed to Harvard students, it would not have been a good startup idea. Facebook was a good idea because it started with a small market there was a fast path out of. Colleges are similar enough that if you build a facebook that works at Harvard, it will work at any college. So you spread rapidly through all the colleges. Once you have all the college students, you get everyone else simply by letting them in.

Similarly for Microsoft: Basic for the Altair; Basic for other machines; other languages besides Basic; operating systems; applications; IPO.

Self

How do you tell whether there's a path out of an idea? How do you tell whether something is the germ of a giant company, or just a niche product? Often you can't. The founders of Airbnb didn't realize at first how big a market they were tapping. Initially they had a much narrower idea. They were going to let hosts rent out space on their floors during conventions. They didn't foresee the expansion of this idea; it forced itself upon them gradually. All they knew at first is that they were onto something. That's probably as much as Bill Gates or Mark Zuckerberg knew at first.

[3] More precisely, the users' need has to give them sufficient activation energy to start using whatever you make, which can vary a lot. For example, the activation energy for enterprise software sold through traditional channels is very high, so you'd have to be a *lot* better to get users to switch. Whereas the activation energy required to switch to a new search engine is low. Which in turn is why search engines are so much better than enterprise software.

Occasionally it's obvious from the beginning when there's a path out of the initial niche. And sometimes I can see a path that's not immediately obvious; that's one of our specialties at YC. But there are limits to how well this can be done, no matter how much experience you have. The most important thing to understand about paths out of the initial idea is the meta-fact that these are hard to see.

So if you can't predict whether there's a path out of an idea, how do you choose between ideas? The truth is disappointing but interesting: if you're the right sort of person, you have the right sort of hunches. If you're at the leading edge of a field that's changing fast, when you have a hunch that something is worth doing, you're more likely to be right.

In *Zen and the Art of Motorcycle Maintenance*, Robert Pirsig says:

> You want to know how to paint a perfect painting? It's easy. Make yourself perfect and then just paint naturally.

I've wondered about that passage since I read it in high school. I'm not sure how useful his advice is for painting specifically, but it fits this situation well. Empirically, the way to have good startup ideas is to become the sort of person who has them.

Being at the leading edge of a field doesn't mean you have to be one of the people pushing it forward. You can also be at the leading edge as a user. It was not so much because he was a programmer that Facebook seemed a good idea to Mark Zuckerberg as because he used computers so much. If you'd asked most 40 year olds in 2004 whether they'd like to publish their lives semi-publicly on the Internet, they'd have been horrified at the idea. But Mark already lived online; to him it seemed natural.

Paul Buchheit says that people at the leading edge of a rapidly changing field "live in the future." Combine that with Pirsig and you get:

> Live in the future, then build what's missing.

That describes the way many if not most of the biggest startups got started. Neither Apple nor Yahoo nor Google nor Facebook were even supposed to be companies at first. They grew out of things their founders built because there seemed a gap in the world.

If you look at the way successful founders have had their ideas, it's generally the result of some external stimulus hitting a prepared mind. Bill Gates and Paul Allen hear about the Altair and think "I bet we could write a Basic interpreter for it." Drew Houston realizes he's forgotten his USB stick and thinks "I really need to make my files live online." Lots of people heard about the Altair. Lots forgot USB sticks. The reason those stimuli caused those founders to start companies was that their experiences had prepared them to notice the opportunities they represented.

The verb you want to be using with respect to startup ideas is not "think up" but "notice." At YC we call ideas that grow naturally out of the founders' own experiences "organic" startup ideas. The most successful startups almost all begin this way.

That may not have been what you wanted to hear. You may have expected recipes for coming up with startup ideas, and instead I'm telling you that the key is to have a mind that's prepared in the right way. But disappointing though it may be, this is the truth. And it is a recipe of a sort, just one that in the worst case takes a year rather than a weekend.

If you're not at the leading edge of some rapidly changing field, you can get to one. For example, anyone reasonably smart can probably get to an edge of programming (e.g. building mobile apps) in a year. Since a successful startup will consume at least 3-5 years of your life, a year's preparation would be a reasonable investment. Especially if you're also looking for a cofounder. [4]

[4]This gets harder as you get older. While the space of ideas doesn't have dangerous local maxima, the space of careers does. There are fairly high walls between most of the paths people take through life, and the older you get, the higher the walls become.

You don't have to learn programming to be at the leading edge of a domain that's changing fast. Other domains change fast. But while learning to hack is not necessary, it is for the forseeable future sufficient. As Marc Andreessen put it, software is eating the world, and this trend has decades left to run.

Knowing how to hack also means that when you have ideas, you'll be able to implement them. That's not absolutely necessary (Jeff Bezos couldn't) but it's an advantage. It's a big advantage, when you're considering an idea like putting a college facebook online, if instead of merely thinking "That's an interesting idea," you can think instead "That's an interesting idea. I'll try building an initial version tonight." It's even better when you're both a programmer and the target user, because then the cycle of generating new versions and testing them on users can happen inside one head.

Noticing

Once you're living in the future in some respect, the way to notice startup ideas is to look for things that seem to be missing. If you're really at the leading edge of a rapidly changing field, there will be things that are obviously missing. What won't be obvious is that they're startup ideas. So if you want to find startup ideas, don't merely turn on the filter "What's missing?" Also turn off every other filter, particularly "Could this be a big company?" There's plenty of time to apply that test later. But if you're thinking about that initially, it may not only filter out lots of good ideas, but also cause you to focus on bad ones.

Most things that are missing will take some time to see. You almost have to trick yourself into seeing the ideas around you.

But you *know* the ideas are out there. This is not one of those problems where there might not be an answer. It's impossibly unlikely that this is the exact moment when technological progress stops. You can be sure people are going to build things in the next few years that will make you think "What did I do before x?"

And when these problems get solved, they will probably seem flamingly obvious in retrospect. What you need to do is turn off the filters that usually prevent you from seeing them. The most powerful is simply taking the current state of the world for granted. Even the most radically open-minded of us mostly do that. You couldn't get from your bed to the front door if you stopped to question everything.

But if you're looking for startup ideas you can sacrifice some of the efficiency of taking the status quo for granted and start to question things. Why is your inbox overflowing? Because you get a lot of email, or because it's hard to get email out of your inbox? Why do you get so much email? What problems are people trying to solve by sending you email? Are there better ways to solve them? And why is it hard to get emails out of your inbox? Why do you keep emails around after you've read them? Is an inbox the optimal tool for that?

Pay particular attention to things that chafe you. The advantage of taking the status quo for granted is not just that it makes life (locally) more efficient, but also that it makes life more tolerable. If you knew about all the things we'll get in the next 50 years but don't have yet, you'd find present day life pretty constraining, just as someone from the present would if they were sent back 50 years in a time machine. When something annoys you, it could be because you're living in the future.

When you find the right sort of problem, you should probably be able to describe it as *obvious*, at least to you. When we started Viaweb, all the online stores were built by hand, by web designers making individual HTML pages. It was obvious to us as programmers that these sites would have to be generated by software. [5]

Which means, strangely enough, that coming up with startup ideas is a question of seeing the obvious. That suggests how weird this process is: you're trying to see things that are obvious, and yet that you hadn't

[5] It was also obvious to us that the web was going to be a big deal. Few non-programmers grasped that in 1995, but the programmers had seen what GUIs had done for desktop computers.

seen.

Since what you need to do here is loosen up your own mind, it may be best not to make too much of a direct frontal attack on the problem — i.e. to sit down and try to think of ideas. The best plan may be just to keep a background process running, looking for things that seem to be missing. Work on hard problems, driven mainly by curiosity, but have a second self watching over your shoulder, taking note of gaps and anomalies. [6]

Give yourself some time. You have a lot of control over the rate at which you turn yours into a prepared mind, but you have less control over the stimuli that spark ideas when they hit it. If Bill Gates and Paul Allen had constrained themselves to come up with a startup idea in one month, what if they'd chosen a month before the Altair appeared? They probably would have worked on a less promising idea. Drew Houston did work on a less promising idea before Dropbox: an SAT prep startup. But Dropbox was a much better idea, both in the absolute sense and also as a match for his skills. [7]

A good way to trick yourself into noticing ideas is to work on projects that seem like they'd be cool. If you do that, you'll naturally tend to build things that are missing. It wouldn't seem as interesting to build something that already existed.

Just as trying to think up startup ideas tends to produce bad ones, working on things that could be dismissed as "toys" often produces good ones. When something is described as a toy, that means it has

[6]Maybe it would work to have this second self keep a journal, and each night to make a brief entry listing the gaps and anomalies you'd noticed that day. Not startup ideas, just the raw gaps and anomalies.

[7]Sam Altman points out that taking time to come up with an idea is not merely a better strategy in an absolute sense, but also like an undervalued stock in that so few founders do it.

There's comparatively little competition for the best ideas, because few founders are willing to put in the time required to notice them. Whereas there is a great deal of competition for mediocre ideas, because when people make up startup ideas, they tend to make up the same ones.

everything an idea needs except being important. It's cool; users love it; it just doesn't matter. But if you're living in the future and you build something cool that users love, it may matter more than outsiders think. Microcomputers seemed like toys when Apple and Microsoft started working on them. I'm old enough to remember that era; the usual term for people with their own microcomputers was "hobbyists." BackRub seemed like an inconsequential science project. The Facebook was just a way for undergrads to stalk one another.

At YC we're excited when we meet startups working on things that we could imagine know-it-alls on forums dismissing as toys. To us that's positive evidence an idea is good.

If you can afford to take a long view (and arguably you can't afford not to), you can turn "Live in the future and build what's missing" into something even better:

> Live in the future and build what seems interesting.

School

That's what I'd advise college students to do, rather than trying to learn about "entrepreneurship." "Entrepreneurship" is something you learn best by doing it. The examples of the most successful founders make that clear. What you should be spending your time on in college is ratcheting yourself into the future. College is an incomparable opportunity to do that. What a waste to sacrifice an opportunity to solve the hard part of starting a startup — becoming the sort of person who can have organic startup ideas — by spending time learning about the easy part. Especially since you won't even really learn about it, any more than you'd learn about sex in a class. All you'll learn is the words for things.

The clash of domains is a particularly fruitful source of ideas. If you know a lot about programming and you start learning about some other field, you'll probably see problems that software could solve. In fact, you're doubly likely to find good problems in another domain:

(a) the inhabitants of that domain are not as likely as software people to have already solved their problems with software, and (b) since you come into the new domain totally ignorant, you don't even know what the status quo is to take it for granted.

So if you're a CS major and you want to start a startup, instead of taking a class on entrepreneurship you're better off taking a class on, say, genetics. Or better still, go work for a biotech company. CS majors normally get summer jobs at computer hardware or software companies. But if you want to find startup ideas, you might do better to get a summer job in some unrelated field. [8]

Or don't take any extra classes, and just build things. It's no coincidence that Microsoft and Facebook both got started in January. At Harvard that is (or was) Reading Period, when students have no classes to attend because they're supposed to be studying for finals. [9]

But don't feel like you have to build things that will become startups. That's premature optimization. Just build things. Preferably with other students. It's not just the classes that make a university such a good place to crank oneself into the future. You're also surrounded by other people trying to do the same thing. If you work together with them on projects, you'll end up producing not just organic ideas, but organic ideas with organic founding teams — and that, empirically, is the best combination.

Beware of research. If an undergrad writes something all his friends start using, it's quite likely to represent a good startup idea. Whereas a PhD dissertation is extremely unlikely to. For some reason, the more a project has to count as research, the less likely it is to be something

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[8] For the computer hardware and software companies, summer jobs are the first phase of the recruiting funnel. But if you're good you can skip the first phase. If you're good you'll have no trouble getting hired by these companies when you graduate, regardless of how you spent your summers.

[9] The empirical evidence suggests that if colleges want to help their students start startups, the best thing they can do is leave them alone in the right way.

that could be turned into a startup. [10] I think the reason is that the subset of ideas that count as research is so narrow that it's unlikely that a project that satisfied that constraint would also satisfy the orthogonal constraint of solving users' problems. Whereas when students (or professors) build something as a side-project, they automatically gravitate toward solving users' problems — perhaps even with an additional energy that comes from being freed from the constraints of research.

Competition

Because a good idea should seem obvious, when you have one you'll tend to feel that you're late. Don't let that deter you. Worrying that you're late is one of the signs of a good idea. Ten minutes of searching the web will usually settle the question. Even if you find someone else working on the same thing, you're probably not too late. It's exceptionally rare for startups to be killed by competitors — so rare that you can almost discount the possibility. So unless you discover a competitor with the sort of lock-in that would prevent users from choosing you, don't discard the idea.

If you're uncertain, ask users. The question of whether you're too late is subsumed by the question of whether anyone urgently needs what you plan to make. If you have something that no competitor does and that some subset of users urgently need, you have a beachhead. [11]

The question then is whether that beachhead is big enough. Or more importantly, who's in it: if the beachhead consists of people doing something lots more people will be doing in the future, then it's probably big enough no matter how small it is. For example, if you're building something differentiated from competitors by the fact that it works on phones, but it only works on the newest phones, that's probably a big enough beachhead.

Err on the side of doing things where you'll face competitors. Inexpe-

[10] I'm speaking here of IT startups; in biotech things are different.

[11] This is an instance of a more general rule: focus on users, not competitors. The most important information about competitors is what you learn via users anyway.

rienced founders usually give competitors more credit than they deserve. Whether you succeed depends far more on you than on your competitors. So better a good idea with competitors than a bad one without.

You don't need to worry about entering a "crowded market" so long as you have a thesis about what everyone else in it is overlooking. In fact that's a very promising starting point. Google was that type of idea. Your thesis has to be more precise than "we're going to make an x that doesn't suck" though. You have to be able to phrase it in terms of something the incumbents are overlooking. Best of all is when you can say that they didn't have the courage of their convictions, and that your plan is what they'd have done if they'd followed through on their own insights. Google was that type of idea too. The search engines that preceded them shied away from the most radical implications of what they were doing — particularly that the better a job they did, the faster users would leave.

A crowded market is actually a good sign, because it means both that there's demand and that none of the existing solutions are good enough. A startup can't hope to enter a market that's obviously big and yet in which they have no competitors. So any startup that succeeds is either going to be entering a market with existing competitors, but armed with some secret weapon that will get them all the users (like Google), or entering a market that looks small but which will turn out to be big (like Microsoft). [12]

Filters

There are two more filters you'll need to turn off if you want to notice startup ideas: the unsexy filter and the schlep filter.

Most programmers wish they could start a startup by just writing some brilliant code, pushing it to a server, and having users pay them lots

[12] In practice most successful startups have elements of both. And you can describe each strategy in terms of the other by adjusting the boundaries of what you call the market. But it's useful to consider these two ideas separately.

of money. They'd prefer not to deal with tedious problems or get involved in messy ways with the real world. Which is a reasonable preference, because such things slow you down. But this preference is so widespread that the space of convenient startup ideas has been stripped pretty clean. If you let your mind wander a few blocks down the street to the messy, tedious ideas, you'll find valuable ones just sitting there waiting to be implemented.

The schlep filter is so dangerous that I wrote a separate essay about the condition it induces, which I called schlep blindness. I gave Stripe as an example of a startup that benefited from turning off this filter, and a pretty striking example it is. Thousands of programmers were in a position to see this idea; thousands of programmers knew how painful it was to process payments before Stripe. But when they looked for startup ideas they didn't see this one, because unconsciously they shrank from having to deal with payments. And dealing with payments is a schlep for Stripe, but not an intolerable one. In fact they might have had net less pain; because the fear of dealing with payments kept most people away from this idea, Stripe has had comparatively smooth sailing in other areas that are sometimes painful, like user acquisition. They didn't have to try very hard to make themselves heard by users, because users were desperately waiting for what they were building.

The unsexy filter is similar to the schlep filter, except it keeps you from working on problems you despise rather than ones you fear. We overcame this one to work on Viaweb. There were interesting things about the architecture of our software, but we weren't interested in ecommerce per se. We could see the problem was one that needed to be solved though.

Turning off the schlep filter is more important than turning off the unsexy filter, because the schlep filter is more likely to be an illusion. And even to the degree it isn't, it's a worse form of self-indulgence. Starting a successful startup is going to be fairly laborious no matter what. Even if the product doesn't entail a lot of schleps, you'll still have plenty dealing with investors, hiring and firing people, and so on. So if there's some idea you think would be cool but you're kept away from by fear of the schleps involved, don't worry: any sufficiently good idea will have as many.

The unsexy filter, while still a source of error, is not as entirely useless as the schlep filter. If you're at the leading edge of a field that's changing rapidly, your ideas about what's sexy will be somewhat correlated with what's valuable in practice. Particularly as you get older and more experienced. Plus if you find an idea sexy, you'll work on it more enthusiastically. [13]

Recipes

While the best way to discover startup ideas is to become the sort of person who has them and then build whatever interests you, sometimes you don't have that luxury. Sometimes you need an idea now. For example, if you're working on a startup and your initial idea turns out to be bad.

For the rest of this essay I'll talk about tricks for coming up with startup ideas on demand. Although empirically you're better off using the organic strategy, you could succeed this way. You just have to be more disciplined. When you use the organic method, you don't even notice an idea unless it's evidence that something is truly missing. But when you make a conscious effort to think of startup ideas, you have to replace this natural constraint with self-discipline. You'll see a lot more ideas, most of them bad, so you need to be able to filter them.

One of the biggest dangers of not using the organic method is the example of the organic method. Organic ideas feel like inspirations. There are a lot of stories about successful startups that began when the founders had what seemed a crazy idea but "just knew" it was promising. When you feel that about an idea you've had while trying

[13] I almost hesitate to raise that point though. Startups are businesses; the point of a business is to make money; and with that additional constraint, you can't expect you'll be able to spend all your time working on what interests you most.

to come up with startup ideas, you're probably mistaken.

When searching for ideas, look in areas where you have some expertise. If you're a database expert, don't build a chat app for teenagers (unless you're also a teenager). Maybe it's a good idea, but you can't trust your judgment about that, so ignore it. There have to be other ideas that involve databases, and whose quality you can judge. Do you find it hard to come up with good ideas involving databases? That's because your expertise raises your standards. Your ideas about chat apps are just as bad, but you're giving yourself a Dunning-Kruger pass in that domain.

The place to start looking for ideas is things you need. There *must* be things you need. [14]

One good trick is to ask yourself whether in your previous job you ever found yourself saying "Why doesn't someone make x? If someone made x we'd buy it in a second." If you can think of any x people said that about, you probably have an idea. You know there's demand, and people don't say that about things that are impossible to build.

More generally, try asking yourself whether there's something unusual about you that makes your needs different from most other people's. You're probably not the only one. It's especially good if you're different in a way people will increasingly be.

If you're changing ideas, one unusual thing about you is the idea you'd previously been working on. Did you discover any needs while working on it? Several well-known startups began this way. Hotmail began as something its founders wrote to talk about their previous startup idea while they were working at their day jobs. [15]

[14]The need has to be a strong one. You can retroactively describe any made-up idea as something you need. But do you really need that recipe site or local event aggregator as much as Drew Houston needed Dropbox, or Brian Chesky and Joe Gebbia needed Airbnb?

Quite often at YC I find myself asking founders "Would you use this thing yourself, if you hadn't written it?" and you'd be surprised how often the answer is no.

[15]Paul Buchheit points out that trying to sell something bad can be a source of

A particularly promising way to be unusual is to be young. Some of the most valuable new ideas take root first among people in their teens and early twenties. And while young founders are at a disadvantage in some respects, they're the only ones who really understand their peers. It would have been very hard for someone who wasn't a college student to start Facebook. So if you're a young founder (under 23 say), are there things you and your friends would like to do that current technology won't let you?

The next best thing to an unmet need of your own is an unmet need of someone else. Try talking to everyone you can about the gaps they find in the world. What's missing? What would they like to do that they can't? What's tedious or annoying, particularly in their work? Let the conversation get general; don't be trying too hard to find startup ideas. You're just looking for something to spark a thought. Maybe you'll notice a problem they didn't consciously realize they had, because you know how to solve it.

When you find an unmet need that isn't your own, it may be somewhat blurry at first. The person who needs something may not know exactly what they need. In that case I often recommend that founders act like consultants — that they do what they'd do if they'd been retained to solve the problems of this one user. People's problems are similar enough that nearly all the code you write this way will be reusable, and whatever isn't will be a small price to start out certain that you've reached the bottom of the well. [16]

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better ideas:

"The best technique I've found for dealing with YC companies that have bad ideas is to tell them to go sell the product ASAP (before wasting time building it). Not only do they learn that nobody wants what they are building, they very often come back with a real idea that they discovered in the process of trying to sell the bad idea."

[16] Here's a recipe that might produce the next Facebook, if you're college students. If you have a connection to one of the more powerful sororities at your school, approach the queen bees thereof and offer to be their personal IT consultants, building anything they could imagine needing in their social lives that didn't already exist. Anything that got built this way would be very promising, because such users are

One way to ensure you do a good job solving other people's problems is to make them your own. When Rajat Suri of E la Carte decided to write software for restaurants, he got a job as a waiter to learn how restaurants worked. That may seem like taking things to extremes, but startups are extreme. We love it when founders do such things.

In fact, one strategy I recommend to people who need a new idea is not merely to turn off their schlep and unsexy filters, but to seek out ideas that are unsexy or involve schleps. Don't try to start Twitter. Those ideas are so rare that you can't find them by looking for them. Make something unsexy that people will pay you for.

A good trick for bypassing the schlep and to some extent the unsexy filter is to ask what you wish someone else would build, so that you could use it. What would you pay for right now?

Since startups often garbage-collect broken companies and industries, it can be a good trick to look for those that are dying, or deserve to, and try to imagine what kind of company would profit from their demise. For example, journalism is in free fall at the moment. But there may still be money to be made from something like journalism. What sort of company might cause people in the future to say "this replaced journalism" on some axis?

But imagine asking that in the future, not now. When one company or industry replaces another, it usually comes in from the side. So don't look for a replacement for x; look for something that people will later say turned out to be a replacement for x. And be imaginative about the axis along which the replacement occurs. Traditional journalism, for example, is a way for readers to get information and to kill time, a way for writers to make money and to get attention, and a vehicle for several different types of advertising. It could be replaced on any of these axes (it has already started to be on most).

When startups consume incumbents, they usually start by serving

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not just the most demanding but also the perfect point to spread from.

I have no idea whether this would work.

some small but important market that the big players ignore. It's particularly good if there's an admixture of disdain in the big players' attitude, because that often misleads them. For example, after Steve Wozniak built the computer that became the Apple I, he felt obliged to give his then-employer Hewlett-Packard the option to produce it. Fortunately for him, they turned it down, and one of the reasons they did was that it used a TV for a monitor, which seemed intolerably déclassé to a high-end hardware company like HP was at the time. [17]

Are there groups of scruffy but sophisticated users like the early microcomputer "hobbyists" that are currently being ignored by the big players? A startup with its sights set on bigger things can often capture a small market easily by expending an effort that wouldn't be justified by that market alone.

Similarly, since the most successful startups generally ride some wave bigger than themselves, it could be a good trick to look for waves and ask how one could benefit from them. The prices of gene sequencing and 3D printing are both experiencing Moore's Law-like declines. What new things will we be able to do in the new world we'll have in a few years? What are we unconsciously ruling out as impossible that will soon be possible?

Organic

But talking about looking explicitly for waves makes it clear that such recipes are plan B for getting startup ideas. Looking for waves is essentially a way to simulate the organic method. If you're at the leading edge of some rapidly changing field, you don't have to look for waves; you are the wave.

Finding startup ideas is a subtle business, and that's why most people who try fail so miserably. It doesn't work well simply to try to think of startup ideas. If you do that, you get bad ones that sound dangerously plausible. The best approach is more indirect: if you have the right

[17] And the reason it used a TV for a monitor is that Steve Wozniak started out by solving his own problems. He, like most of his peers, couldn't afford a monitor.

sort of background, good startup ideas will seem obvious to you. But even then, not immediately. It takes time to come across situations where you notice something missing. And often these gaps won't seem to be ideas for companies, just things that would be interesting to build. Which is why it's good to have the time and the inclination to build things just because they're interesting.

Live in the future and build what seems interesting. Strange as it sounds, that's the real recipe.

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Chapter 16

Do Things that Don't Scale

July 2013

One of the most common types of advice we give at Y Combinator is to do things that don't scale. A lot of would-be founders believe that startups either take off or don't. You build something, make it available, and if you've made a better mousetrap, people beat a path to your door as promised. Or they don't, in which case the market must not exist. [1]

Actually startups take off because the founders make them take off. There may be a handful that just grew by themselves, but usually it takes some sort of push to get them going. A good metaphor would be the cranks that car engines had before they got electric starters. Once the engine was going, it would keep going, but there was a separate and laborious process to get it going.

Recruit

The most common unscalable thing founders have to do at the start is to recruit users manually. Nearly all startups have to. You can't wait for users to come to you. You have to go out and get them.

Stripe is one of the most successful startups we've funded, and the problem they solved was an urgent one. If anyone could have sat back and waited for users, it was Stripe. But in fact they're famous within YC for aggressive early user acquisition.

[1] Actually Emerson never mentioned mousetraps specifically. He wrote "If a man has good corn or wood, or boards, or pigs, to sell, or can make better chairs or knives, crucibles or church organs, than anybody else, you will find a broad hard-beaten road to his house, though it be in the woods."

Startups building things for other startups have a big pool of potential users in the other companies we've funded, and none took better advantage of it than Stripe. At YC we use the term "Collison installation" for the technique they invented. More diffident founders ask "Will you try our beta?" and if the answer is yes, they say "Great, we'll send you a link." But the Collison brothers weren't going to wait. When anyone agreed to try Stripe they'd say "Right then, give me your laptop" and set them up on the spot.

There are two reasons founders resist going out and recruiting users individually. One is a combination of shyness and laziness. They'd rather sit at home writing code than go out and talk to a bunch of strangers and probably be rejected by most of them. But for a startup to succeed, at least one founder (usually the CEO) will have to spend a lot of time on sales and marketing. [2]

The other reason founders ignore this path is that the absolute numbers seem so small at first. This can't be how the big, famous startups got started, they think. The mistake they make is to underestimate the power of compound growth. We encourage every startup to measure their progress by weekly growth rate. If you have 100 users, you need to get 10 more next week to grow 10% a week. And while 110 may not seem much better than 100, if you keep growing at 10% a week you'll be surprised how big the numbers get. After a year you'll have 14,000 users, and after 2 years you'll have 2 million.

You'll be doing different things when you're acquiring users a thousand at a time, and growth has to slow down eventually. But if the market exists you can usually start by recruiting users manually and then gradually switch to less manual methods. [3]

[2] Thanks to Sam Altman for suggesting I make this explicit. And no, you can't avoid doing sales by hiring someone to do it for you. You have to do sales yourself initially. Later you can hire a real salesperson to replace you.

[3] The reason this works is that as you get bigger, your size helps you grow. Patrick Collison wrote "At some point, there was a very noticeable change in how Stripe felt. It tipped from being this boulder we had to push to being a train car that in fact had its own momentum."

Airbnb is a classic example of this technique. Marketplaces are so hard to get rolling that you should expect to take heroic measures at first. In Airbnb's case, these consisted of going door to door in New York, recruiting new users and helping existing ones improve their listings. When I remember the Airbnbs during YC, I picture them with rolly bags, because when they showed up for tuesday dinners they'd always just flown back from somewhere.

Fragile

Airbnb now seems like an unstoppable juggernaut, but early on it was so fragile that about 30 days of going out and engaging in person with users made the difference between success and failure.

That initial fragility was not a unique feature of Airbnb. Almost all startups are fragile initially. And that's one of the biggest things inexperienced founders and investors (and reporters and know-it-alls on forums) get wrong about them. They unconsciously judge larval startups by the standards of established ones. They're like someone looking at a newborn baby and concluding "there's no way this tiny creature could ever accomplish anything."

It's harmless if reporters and know-it-alls dismiss your startup. They always get things wrong. It's even ok if investors dismiss your startup; they'll change their minds when they see growth. The big danger is that you'll dismiss your startup yourself. I've seen it happen. I often have to encourage founders who don't see the full potential of what they're building. Even Bill Gates made that mistake. He returned to Harvard for the fall semester after starting Microsoft. He didn't stay long, but he wouldn't have returned at all if he'd realized Microsoft was going to be even a fraction of the size it turned out to be. [4]

The question to ask about an early stage startup is not "is this company taking over the world?" but "how big could this company get if

[4] One of the more subtle ways in which YC can help founders is by calibrating their ambitions, because we know exactly how a lot of successful startups looked when they were just getting started.

the founders did the right things?" And the right things often seem both laborious and inconsequential at the time. Microsoft can't have seemed very impressive when it was just a couple guys in Albuquerque writing Basic interpreters for a market of a few thousand hobbyists (as they were then called), but in retrospect that was the optimal path to dominating microcomputer software. And I know Brian Chesky and Joe Gebbia didn't feel like they were en route to the big time as they were taking "professional" photos of their first hosts' apartments. They were just trying to survive. But in retrospect that too was the optimal path to dominating a big market.

How do you find users to recruit manually? If you build something to solve your own problems, then you only have to find your peers, which is usually straightforward. Otherwise you'll have to make a more deliberate effort to locate the most promising vein of users. The usual way to do that is to get some initial set of users by doing a comparatively untargeted launch, and then to observe which kind seem most enthusiastic, and seek out more like them. For example, Ben Silbermann noticed that a lot of the earliest Pinterest users were interested in design, so he went to a conference of design bloggers to recruit users, and that worked well. [5]

Delight

You should take extraordinary measures not just to acquire users, but also to make them happy. For as long as they could (which turned out to be surprisingly long), Wufoo sent each new user a hand-written thank you note. Your first users should feel that signing up with you was one of the best choices they ever made. And you in turn should be racking your brains to think of new ways to delight them.

Why do we have to teach startups this? Why is it counterintuitive for founders? Three reasons, I think.

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[5] If you're building something for which you can't easily get a small set of users to observe — e.g. enterprise software — and in a domain where you have no connections, you'll have to rely on cold calls and introductions. But should you even be working on such an idea?

One is that a lot of startup founders are trained as engineers, and customer service is not part of the training of engineers. You're supposed to build things that are robust and elegant, not be slavishly attentive to individual users like some kind of salesperson. Ironically, part of the reason engineering is traditionally averse to handholding is that its traditions date from a time when engineers were less powerful — when they were only in charge of their narrow domain of building things, rather than running the whole show. You can be ornery when you're Scotty, but not when you're Kirk.

Another reason founders don't focus enough on individual customers is that they worry it won't scale. But when founders of larval startups worry about this, I point out that in their current state they have nothing to lose. Maybe if they go out of their way to make existing users super happy, they'll one day have too many to do so much for. That would be a great problem to have. See if you can make it happen. And incidentally, when it does, you'll find that delighting customers scales better than you expected. Partly because you can usually find ways to make anything scale more than you would have predicted, and partly because delighting customers will by then have permeated your culture.

I have never once seen a startup lured down a blind alley by trying too hard to make their initial users happy.

But perhaps the biggest thing preventing founders from realizing how attentive they could be to their users is that they've never experienced such attention themselves. Their standards for customer service have been set by the companies they've been customers of, which are mostly big ones. Tim Cook doesn't send you a hand-written note after you buy a laptop. He can't. But you can. That's one advantage of being small: you can provide a level of service no big company can. [6]

[6] Garry Tan pointed out an interesting trap founders fall into in the beginning. They want so much to seem big that they imitate even the flaws of big companies, like indifference to individual users. This seems to them more "professional." Actually it's better to embrace the fact that you're small and use whatever advantages

Once you realize that existing conventions are not the upper bound on user experience, it's interesting in a very pleasant way to think about how far you could go to delight your users.

Experience

I was trying to think of a phrase to convey how extreme your attention to users should be, and I realized Steve Jobs had already done it: insanely great. Steve wasn't just using "insanely" as a synonym for "very." He meant it more literally — that one should focus on quality of execution to a degree that in everyday life would be considered pathological.

All the most successful startups we've funded have, and that probably doesn't surprise would-be founders. What novice founders don't get is what insanely great translates to in a larval startup. When Steve Jobs started using that phrase, Apple was already an established company. He meant the Mac (and its documentation and even packaging — such is the nature of obsession) should be insanely well designed and manufactured. That's not hard for engineers to grasp. It's just a more extreme version of designing a robust and elegant product.

What founders have a hard time grasping (and Steve himself might have had a hard time grasping) is what insanely great morphs into as you roll the time slider back to the first couple months of a startup's life. It's not the product that should be insanely great, but the experience of being your user. The product is just one component of that. For a big company it's necessarily the dominant one. But you can and should give users an insanely great experience with an early, incomplete, buggy product, if you make up the difference with attentiveness.

Can, perhaps, but should? Yes. Over-engaging with early users is not just a permissible technique for getting growth rolling. For most successful startups it's a necessary part of the feedback loop that makes the product good. Making a better mousetrap is not an atomic operation. Even if you start the way most successful startups have, by building

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that brings.

something you yourself need, the first thing you build is never quite right. And except in domains with big penalties for making mistakes, it's often better not to aim for perfection initially. In software, especially, it usually works best to get something in front of users as soon as it has a quantum of utility, and then see what they do with it. Perfectionism is often an excuse for procrastination, and in any case your initial model of users is always inaccurate, even if you're one of them. [7]

The feedback you get from engaging directly with your earliest users will be the best you ever get. When you're so big you have to resort to focus groups, you'll wish you could go over to your users' homes and offices and watch them use your stuff like you did when there were only a handful of them.

Fire

Sometimes the right unscalable trick is to focus on a deliberately narrow market. It's like keeping a fire contained at first to get it really hot before adding more logs.

That's what Facebook did. At first it was just for Harvard students. In that form it only had a potential market of a few thousand people, but because they felt it was really for them, a critical mass of them signed up. After Facebook stopped being for Harvard students, it remained for students at specific colleges for quite a while. When I interviewed Mark Zuckerberg at Startup School, he said that while it was a lot of work creating course lists for each school, doing that made students feel the site was their natural home.

Any startup that could be described as a marketplace usually has to start in a subset of the market, but this can work for other startups as well. It's always worth asking if there's a subset of the market in which

[7] Your user model almost couldn't be perfectly accurate, because users' needs often change in response to what you build for them. Build them a microcomputer, and suddenly they need to run spreadsheets on it, because the arrival of your new microcomputer causes someone to invent the spreadsheet.

you can get a critical mass of users quickly. [8]

Most startups that use the contained fire strategy do it unconsciously. They build something for themselves and their friends, who happen to be the early adopters, and only realize later that they could offer it to a broader market. The strategy works just as well if you do it unconsciously. The biggest danger of not being consciously aware of this pattern is for those who naively discard part of it. E.g. if you don't build something for yourself and your friends, or even if you do, but you come from the corporate world and your friends are not early adopters, you'll no longer have a perfect initial market handed to you on a platter.

Among companies, the best early adopters are usually other startups. They're more open to new things both by nature and because, having just been started, they haven't made all their choices yet. Plus when they succeed they grow fast, and you with them. It was one of many unforeseen advantages of the YC model (and specifically of making YC big) that B2B startups now have an instant market of hundreds of other startups ready at hand.

Meraki

For hardware startups there's a variant of doing things that don't scale that we call "pulling a Meraki." Although we didn't fund Meraki, the founders were Robert Morris's grad students, so we know their history. They got started by doing something that really doesn't scale: assembling their routers themselves.

Hardware startups face an obstacle that software startups don't. The minimum order for a factory production run is usually several hundred thousand dollars. Which can put you in a catch-22: without a product you can't generate the growth you need to raise the money to

[8] If you have to choose between the subset that will sign up quickest and those that will pay the most, it's usually best to pick the former, because those are probably the early adopters. They'll have a better influence on your product, and they won't make you expend as much effort on sales. And though they have less money, you don't need that much to maintain your target growth rate early on.

manufacture your product. Back when hardware startups had to rely on investors for money, you had to be pretty convincing to overcome this. The arrival of crowdfunding (or more precisely, preorders) has helped a lot. But even so I'd advise startups to pull a Meraki initially if they can. That's what Pebble did. The Pebbles assembled the first several hundred watches themselves. If they hadn't gone through that phase, they probably wouldn't have sold $10 million worth of watches when they did go on Kickstarter.

Like paying excessive attention to early customers, fabricating things yourself turns out to be valuable for hardware startups. You can tweak the design faster when you're the factory, and you learn things you'd never have known otherwise. Eric Migicovsky of Pebble said one of the things he learned was "how valuable it was to source good screws." Who knew?

Consult

Sometimes we advise founders of B2B startups to take over-engagement to an extreme, and to pick a single user and act as if they were consultants building something just for that one user. The initial user serves as the form for your mold; keep tweaking till you fit their needs perfectly, and you'll usually find you've made something other users want too. Even if there aren't many of them, there are probably adjacent territories that have more. As long as you can find just one user who really needs something and can act on that need, you've got a toehold in making something people want, and that's as much as any startup needs initially. [9]

Consulting is the canonical example of work that doesn't scale. But (like other ways of bestowing one's favors liberally) it's safe to do it so long as you're not being paid to. That's where companies cross the line. So long as you're a product company that's merely being extra

[9]Yes, I can imagine cases where you could end up making something that was really only useful for one user. But those are usually obvious, even to inexperienced founders. So if it's not obvious you'd be making something for a market of one, don't worry about that danger.

attentive to a customer, they're very grateful even if you don't solve all their problems. But when they start paying you specifically for that attentiveness — when they start paying you by the hour — they expect you to do everything.

Another consulting-like technique for recruiting initially lukewarm users is to use your software yourselves on their behalf. We did that at Viaweb. When we approached merchants asking if they wanted to use our software to make online stores, some said no, but they'd let us make one for them. Since we would do anything to get users, we did. We felt pretty lame at the time. Instead of organizing big strategic e-commerce partnerships, we were trying to sell luggage and pens and men's shirts. But in retrospect it was exactly the right thing to do, because it taught us how it would feel to merchants to use our software. Sometimes the feedback loop was near instantaneous: in the middle of building some merchant's site I'd find I needed a feature we didn't have, so I'd spend a couple hours implementing it and then resume building the site.

Manual

There's a more extreme variant where you don't just use your software, but are your software. When you only have a small number of users, you can sometimes get away with doing by hand things that you plan to automate later. This lets you launch faster, and when you do finally automate yourself out of the loop, you'll know exactly what to build because you'll have muscle memory from doing it yourself.

When manual components look to the user like software, this technique starts to have aspects of a practical joke. For example, the way Stripe delivered "instant" merchant accounts to its first users was that the founders manually signed them up for traditional merchant accounts behind the scenes.

Some startups could be entirely manual at first. If you can find someone with a problem that needs solving and you can solve it manually, go ahead and do that for as long as you can, and then gradually automate the bottlenecks. It would be a little frightening to be solving users' problems in a way that wasn't yet automatic, but less frightening than the far more common case of having something automatic that doesn't yet solve anyone's problems.

Big

I should mention one sort of initial tactic that usually doesn't work: the Big Launch. I occasionally meet founders who seem to believe startups are projectiles rather than powered aircraft, and that they'll make it big if and only if they're launched with sufficient initial velocity. They want to launch simultaneously in 8 different publications, with embargoes. And on a tuesday, of course, since they read somewhere that's the optimum day to launch something.

It's easy to see how little launches matter. Think of some successful startups. How many of their launches do you remember? All you need from a launch is some initial core of users. How well you're doing a few months later will depend more on how happy you made those users than how many there were of them. [10]

So why do founders think launches matter? A combination of solipsism and laziness. They think what they're building is so great that everyone who hears about it will immediately sign up. Plus it would be so much less work if you could get users merely by broadcasting your existence, rather than recruiting them one at a time. But even if what you're building really is great, getting users will always be a gradual process — partly because great things are usually also novel, but mainly because users have other things to think about.

Partnerships too usually don't work. They don't work for startups in general, but they especially don't work as a way to get growth started. It's a common mistake among inexperienced founders to believe that

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[10] There may even be an inverse correlation between launch magnitude and success. The only launches I remember are famous flops like the Segway and Google Wave. Wave is a particularly alarming example, because I think it was actually a great idea that was killed partly by its overdone launch.

a partnership with a big company will be their big break. Six months later they're all saying the same thing: that was way more work than we expected, and we ended up getting practically nothing out of it. [11]

It's not enough just to do something extraordinary initially. You have to make an extraordinary *effort* initially. Any strategy that omits the effort — whether it's expecting a big launch to get you users, or a big partner — is ipso facto suspect.

Vector

The need to do something unscalably laborious to get started is so nearly universal that it might be a good idea to stop thinking of startup ideas as scalars. Instead we should try thinking of them as pairs of what you're going to build, plus the unscalable thing(s) you're going to do initially to get the company going.

It could be interesting to start viewing startup ideas this way, because now that there are two components you can try to be imaginative about the second as well as the first. But in most cases the second component will be what it usually is — recruit users manually and give them an overwhelmingly good experience — and the main benefit of treating startups as vectors will be to remind founders they need to work hard in two dimensions. [12]

In the best case, both components of the vector contribute to your company's DNA: the unscalable things you have to do to get started are not merely a necessary evil, but change the company permanently for the better. If you have to be aggressive about user acquisition when you're small, you'll probably still be aggressive when you're big. If you have to manufacture your own hardware, or use your software on users's behalf, you'll learn things you couldn't have learned otherwise.

[11] Google grew big on the back of Yahoo, but that wasn't a partnership. Yahoo was their customer.

[12] It will also remind founders that an idea where the second component is empty — an idea where there is nothing you can do to get going, e.g. because you have no way to find users to recruit manually — is probably a bad idea, at least for those founders.

And most importantly, if you have to work hard to delight users when you only have a handful of them, you'll keep doing it when you have a lot.

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Chapter 17

The Hardest Lessons for Startups to Learn

April 2006

(This essay is derived from a talk at the 2006 Startup School.)

The startups we've funded so far are pretty quick, but they seem quicker to learn some lessons than others. I think it's because some things about startups are kind of counterintuitive.

We've now invested in enough companies that I've learned a trick for determining which points are the counterintuitive ones: they're the ones I have to keep repeating.

So I'm going to number these points, and maybe with future startups I'll be able to pull off a form of Huffman coding. I'll make them all read this, and then instead of nagging them in detail, I'll just be able to say: *number four!*

1. Release Early.

The thing I probably repeat most is this recipe for a startup: get a version 1 out fast, then improve it based on users' reactions.

By "release early" I don't mean you should release something full of bugs, but that you should release something minimal. Users hate bugs, but they don't seem to mind a minimal version 1, if there's more coming soon.

There are several reasons it pays to get version 1 done fast. One is that this is simply the right way to write software, whether for a startup or not. I've been repeating that since 1993, and I haven't seen much since to contradict it. I've seen a lot of startups die because they were too slow to release stuff, and none because they were too quick. [1]

One of the things that will surprise you if you build something popular is that you won't know your users. Reddit now has almost half a million unique visitors a month. Who are all those people? They have no idea. No web startup does. And since you don't know your users, it's dangerous to guess what they'll like. Better to release something and let them tell you.

Wufoo took this to heart and released their form-builder before the underlying database. You can't even drive the thing yet, but 83,000 people came to sit in the driver's seat and hold the steering wheel. And Wufoo got valuable feedback from it: Linux users complained they used too much Flash, so they rewrote their software not to. If they'd waited to release everything at once, they wouldn't have discovered this problem till it was more deeply wired in.

Even if you had no users, it would still be important to release quickly, because for a startup the initial release acts as a shakedown cruise. If anything major is broken– if the idea's no good, for example, or the founders hate one another– the stress of getting that first version out will expose it. And if you have such problems you want to find them early.

Perhaps the most important reason to release early, though, is that it makes you work harder. When you're working on something that isn't released, problems are intriguing. In something that's out there, problems are alarming. There is a lot more urgency once you release. And I think that's precisely why people put it off. They know they'll have to work a lot harder once they do. [2]

2. Keep Pumping Out Features.

[1] Startups can die from releasing something full of bugs, and not fixing them fast enough, but I don't know of any that died from releasing something stable but minimal very early, then promptly improving it.

[2] I know this is why I haven't released Arc. The moment I do, I'll have people nagging me for features.

Of course, "release early" has a second component, without which it would be bad advice. If you're going to start with something that doesn't do much, you better improve it fast.

What I find myself repeating is "pump out features." And this rule isn't just for the initial stages. This is something all startups should do for as long as they want to be considered startups.

I don't mean, of course, that you should make your application ever more complex. By "feature" I mean one unit of hacking– one quantum of making users' lives better.

As with exercise, improvements beget improvements. If you run every day, you'll probably feel like running tomorrow. But if you skip running for a couple weeks, it will be an effort to drag yourself out. So it is with hacking: the more ideas you implement, the more ideas you'll have. You should make your system better at least in some small way every day or two.

This is not just a good way to get development done; it is also a form of marketing. Users love a site that's constantly improving. In fact, users expect a site to improve. Imagine if you visited a site that seemed very good, and then returned two months later and not one thing had changed. Wouldn't it start to seem lame? [3]

They'll like you even better when you improve in response to their comments, because customers are used to companies ignoring them. If you're the rare exception– a company that actually listens– you'll generate fanatical loyalty. You won't need to advertise, because your users will do it for you.

This seems obvious too, so why do I have to keep repeating it? I think the problem here is that people get used to how things are. Once a product gets past the stage where it has glaring flaws, you start to get

[3]A web site is different from a book or movie or desktop application in this respect. Users judge a site not as a single snapshot, but as an animation with multiple frames. Of the two, I'd say the rate of improvement is more important to users than where you currently are.

used to it, and gradually whatever features it happens to have become its identity. For example, I doubt many people at Yahoo (or Google for that matter) realized how much better web mail could be till Paul Buchheit showed them.

I think the solution is to assume that anything you've made is far short of what it could be. Force yourself, as a sort of intellectual exercise, to keep thinking of improvements. Ok, sure, what you have is perfect. But if you had to change something, what would it be?

If your product seems finished, there are two possible explanations: (a) it is finished, or (b) you lack imagination. Experience suggests (b) is a thousand times more likely.

3. Make Users Happy.

Improving constantly is an instance of a more general rule: make users happy. One thing all startups have in common is that they can't force anyone to do anything. They can't force anyone to use their software, and they can't force anyone to do deals with them. A startup has to sing for its supper. That's why the successful ones make great things. They have to, or die.

When you're running a startup you feel like a little bit of debris blown about by powerful winds. The most powerful wind is users. They can either catch you and loft you up into the sky, as they did with Google, or leave you flat on the pavement, as they do with most startups. Users are a fickle wind, but more powerful than any other. If they take you up, no competitor can keep you down.

As a little piece of debris, the rational thing for you to do is not to lie flat, but to curl yourself into a shape the wind will catch.

I like the wind metaphor because it reminds you how impersonal the stream of traffic is. The vast majority of people who visit your site will be casual visitors. It's them you have to design your site for. The people who really care will find what they want by themselves.

The median visitor will arrive with their finger poised on the Back button. Think about your own experience: most links you follow lead to something lame. Anyone who has used the web for more than a couple weeks has been *trained* to click on Back after following a link. So your site has to say "Wait! Don't click on Back. This site isn't lame. Look at this, for example."

There are two things you have to do to make people pause. The most important is to explain, as concisely as possible, what the hell your site is about. How often have you visited a site that seemed to assume you already knew what they did? For example, the corporate site that says the company makes

> enterprise content management solutions for business that enable organizations to unify people, content and processes to minimize business risk, accelerate time-to-value and sustain lower total cost of ownership.

An established company may get away with such an opaque description, but no startup can. A startup should be able to explain in one or two sentences exactly what it does. [4] And not just to users. You need this for everyone: investors, acquirers, partners, reporters, potential employees, and even current employees. You probably shouldn't even start a company to do something that can't be described compellingly in one or two sentences.

The other thing I repeat is to give people everything you've got, right away. If you have something impressive, try to put it on the front page, because that's the only one most visitors will see. Though indeed there's a paradox here: the more you push the good stuff toward the front, the more likely visitors are to explore further. [5]

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[4] It should not always tell this to users, however. For example, MySpace is basically a replacement mall for mallrats. But it was wiser for them, initially, to pretend that the site was about bands.

[5] Similarly, don't make users register to try your site. Maybe what you have is so valuable that visitors should gladly register to get at it. But they've been trained to expect the opposite. Most of the things they've tried on the web have sucked– and probably especially those that made them register.

In the best case these two suggestions get combined: you tell visitors what your site is about by *showing* them. One of the standard pieces of advice in fiction writing is "show, don't tell." Don't say that a character's angry; have him grind his teeth, or break his pencil in half. Nothing will explain what your site does so well as using it.

The industry term here is "conversion." The job of your site is to convert casual visitors into users– whatever your definition of a user is. You can measure this in your growth rate. Either your site is catching on, or it isn't, and you must know which. If you have decent growth, you'll win in the end, no matter how obscure you are now. And if you don't, you need to fix something.

4. Fear the Right Things.

Another thing I find myself saying a lot is "don't worry." Actually, it's more often "don't worry about this; worry about that instead." Startups are right to be paranoid, but they sometimes fear the wrong things.

Most visible disasters are not so alarming as they seem. Disasters are normal in a startup: a founder quits, you discover a patent that covers what you're doing, your servers keep crashing, you run into an insoluble technical problem, you have to change your name, a deal falls through– these are all par for the course. They won't kill you unless you let them.

Nor will most competitors. A lot of startups worry "what if Google builds something like us?" Actually big companies are not the ones you have to worry about– not even Google. The people at Google are smart, but no smarter than you; they're not as motivated, because Google is not going to go out of business if this one product fails; and even at Google they have a lot of bureaucracy to slow them down.

What you should fear, as a startup, is not the established players, but other startups you don't know exist yet. They're way more dangerous than Google because, like you, they're cornered animals.

Looking just at existing competitors can give you a false sense of security. You should compete against what someone else *could* be doing, not just what you can see people doing. A corollary is that you shouldn't relax just because you have no visible competitors yet. No matter what your idea, there's someone else out there working on the same thing.

That's the downside of it being easier to start a startup: more people are doing it. But I disagree with Caterina Fake when she says that makes this a bad time to start a startup. More people are starting startups, but not as many more as could. Most college graduates still think they have to get a job. The average person can't ignore something that's been beaten into their head since they were three just because serving web pages recently got a lot cheaper.

And in any case, competitors are not the biggest threat. Way more startups hose themselves than get crushed by competitors. There are a lot of ways to do it, but the three main ones are internal disputes, inertia, and ignoring users. Each is, by itself, enough to kill you. But if I had to pick the worst, it would be ignoring users. If you want a recipe for a startup that's going to die, here it is: a couple of founders who have some great idea they know everyone is going to love, and that's what they're going to build, no matter what.

Almost everyone's initial plan is broken. If companies stuck to their initial plans, Microsoft would be selling programming languages, and Apple would be selling printed circuit boards. In both cases their customers told them what their business should be– and they were smart enough to listen.

As Richard Feynman said, the imagination of nature is greater than the imagination of man. You'll find more interesting things by looking at the world than you could ever produce just by thinking. This principle is very powerful. It's why the best abstract painting still falls short of Leonardo, for example. And it applies to startups too. No idea for a product could ever be so clever as the ones you can discover by smashing a beam of prototypes into a beam of users.

5. Commitment Is a Self-Fulfilling Prophecy.

I now have enough experience with startups to be able to say what the most important quality is in a startup founder, and it's not what you might think. The most important quality in a startup founder is determination. Not intelligence– determination.

This is a little depressing. I'd like to believe Viaweb succeeded because we were smart, not merely determined. A lot of people in the startup world want to believe that. Not just founders, but investors too. They like the idea of inhabiting a world ruled by intelligence. And you can tell they really believe this, because it affects their investment decisions.

Time after time VCs invest in startups founded by eminent professors. This may work in biotech, where a lot of startups simply commercialize existing research, but in software you want to invest in students, not professors. Microsoft, Yahoo, and Google were all founded by people who dropped out of school to do it. What students lack in experience they more than make up in dedication.

Of course, if you want to get rich, it's not enough merely to be determined. You have to be smart too, right? I'd like to think so, but I've had an experience that convinced me otherwise: I spent several years living in New York.

You can lose quite a lot in the brains department and it won't kill you. But lose even a little bit in the commitment department, and that will kill you very rapidly.

Running a startup is like walking on your hands: it's possible, but it requires extraordinary effort. If an ordinary employee were asked to do the things a startup founder has to, he'd be very indignant. Imagine if you were hired at some big company, and in addition to writing software ten times faster than you'd ever had to before, they expected you to answer support calls, administer the servers, design the web site, cold-call customers, find the company office space, and go out and get everyone lunch.

And to do all this not in the calm, womb-like atmosphere of a big company, but against a backdrop of constant disasters. That's the part that really demands determination. In a startup, there's always some disaster happening. So if you're the least bit inclined to find an excuse to quit, there's always one right there.

But if you lack commitment, chances are it will have been hurting you long before you actually quit. Everyone who deals with startups knows how important commitment is, so if they sense you're ambivalent, they won't give you much attention. If you lack commitment, you'll just find that for some mysterious reason good things happen to your competitors but not to you. If you lack commitment, it will seem to you that you're unlucky.

Whereas if you're determined to stick around, people will pay attention to you, because odds are they'll have to deal with you later. You're a local, not just a tourist, so everyone has to come to terms with you.

At Y Combinator we sometimes mistakenly fund teams who have the attitude that they're going to give this startup thing a shot for three months, and if something great happens, they'll stick with it– "something great" meaning either that someone wants to buy them or invest millions of dollars in them. But if this is your attitude, "something great" is very unlikely to happen to you, because both acquirers and investors judge you by your level of commitment.

If an acquirer thinks you're going to stick around no matter what, they'll be more likely to buy you, because if they don't and you stick around, you'll probably grow, your price will go up, and they'll be left wishing they'd bought you earlier. Ditto for investors. What really motivates investors, even big VCs, is not the hope of good returns, but the fear of missing out. [6] So if you make it clear you're going

[6] VCs have rational reasons for behaving this way. They don't make their money (if they make money) off their median investments. In a typical fund, half the companies fail, most of the rest generate mediocre returns, and one or two "make the fund" by succeeding spectacularly. So if they miss just a few of the most promising opportunities, it could hose the whole fund.

to succeed no matter what, and the only reason you need them is to make it happen a little faster, you're much more likely to get money.

You can't fake this. The only way to convince everyone that you're ready to fight to the death is actually to be ready to.

You have to be the right kind of determined, though. I carefully chose the word determined rather than stubborn, because stubbornness is a disastrous quality in a startup. You have to be determined, but flexible, like a running back. A successful running back doesn't just put his head down and try to run through people. He improvises: if someone appears in front of him, he runs around them; if someone tries to grab him, he spins out of their grip; he'll even run in the wrong direction briefly if that will help. The one thing he'll never do is stand still. [7]

6. There Is Always Room.

I was talking recently to a startup founder about whether it might be good to add a social component to their software. He said he didn't think so, because the whole social thing was tapped out. Really? So in a hundred years the only social networking sites will be the Facebook, MySpace, Flickr, and Del.icio.us? Not likely.

There is always room for new stuff. At every point in history, even the darkest bits of the dark ages, people were discovering things that made everyone say "why didn't anyone think of that before?" We know this continued to be true up till 2004, when the Facebook was founded– though strictly speaking someone else did think of that.

The reason we don't see the opportunities all around us is that we adjust to however things are, and assume that's how things have to be. For example, it would seem crazy to most people to try to make a better search engine than Google. Surely that field, at least, is tapped out. Really? In a hundred years– or even twenty– are people still going to search for information using something like the current Google? Even

[7] The attitude of a running back doesn't translate to soccer. Though it looks great when a forward dribbles past multiple defenders, a player who persists in trying such things will do worse in the long term than one who passes.

Google probably doesn't think that.

In particular, I don't think there's any limit to the number of startups. Sometimes you hear people saying "All these guys starting startups now are going to be disappointed. How many little startups are Google and Yahoo going to buy, after all?" That sounds cleverly skeptical, but I can prove it's mistaken. No one proposes that there's some limit to the number of people who can be employed in an economy consisting of big, slow-moving companies with a couple thousand people each. Why should there be any limit to the number who could be employed by small, fast-moving companies with ten each? It seems to me the only limit would be the number of people who want to work that hard.

The limit on the number of startups is not the number that can get acquired by Google and Yahoo– though it seems even that should be unlimited, if the startups were actually worth buying– but the amount of wealth that can be created. And I don't think there's any limit on that, except cosmological ones.

So for all practical purposes, there is no limit to the number of startups. Startups make wealth, which means they make things people want, and if there's a limit on the number of things people want, we are nowhere near it. I still don't even have a flying car.

7. Don't Get Your Hopes Up.

This is another one I've been repeating since long before Y Combinator. It was practically the corporate motto at Viaweb.

Startup founders are naturally optimistic. They wouldn't do it otherwise. But you should treat your optimism the way you'd treat the core of a nuclear reactor: as a source of power that's also very dangerous. You have to build a shield around it, or it will fry you.

The shielding of a reactor is not uniform; the reactor would be useless if it were. It's pierced in a few places to let pipes in. An optimism shield has to be pierced too. I think the place to draw the line is between what you expect of yourself, and what you expect of other people. It's ok to be optimistic about what you can do, but assume the worst about machines and other people.

This is particularly necessary in a startup, because you tend to be pushing the limits of whatever you're doing. So things don't happen in the smooth, predictable way they do in the rest of the world. Things change suddenly, and usually for the worse.

Shielding your optimism is nowhere more important than with deals. If your startup is doing a deal, just assume it's not going to happen. The VCs who say they're going to invest in you aren't. The company that says they're going to buy you isn't. The big customer who wants to use your system in their whole company won't. Then if things work out you can be pleasantly surprised.

The reason I warn startups not to get their hopes up is not to save them from being *disappointed* when things fall through. It's for a more practical reason: to prevent them from leaning their company against something that's going to fall over, taking them with it.

For example, if someone says they want to invest in you, there's a natural tendency to stop looking for other investors. That's why people proposing deals seem so positive: they *want* you to stop looking. And you want to stop too, because doing deals is a pain. Raising money, in particular, is a huge time sink. So you have to consciously force yourself to keep looking.

Even if you ultimately do the first deal, it will be to your advantage to have kept looking, because you'll get better terms. Deals are dynamic; unless you're negotiating with someone unusually honest, there's not a single point where you shake hands and the deal's done. There are usually a lot of subsidiary questions to be cleared up after the handshake, and if the other side senses weakness– if they sense you need this deal– they will be very tempted to screw you in the details.

VCs and corp dev guys are professional negotiators. They're trained to take advantage of weakness. [8] So while they're often nice guys, they just can't help it. And as pros they do this more than you. So don't even try to bluff them. The only way a startup can have any leverage in a deal is genuinely not to need it. And if you don't believe in a deal, you'll be less likely to depend on it.

So I want to plant a hypnotic suggestion in your heads: when you hear someone say the words "we want to invest in you" or "we want to acquire you," I want the following phrase to appear automatically in your head: *don't get your hopes up.* Just continue running your company as if this deal didn't exist. Nothing is more likely to make it close.

The way to succeed in a startup is to focus on the goal of getting lots of users, and keep walking swiftly toward it while investors and acquirers scurry alongside trying to wave money in your face.

Speed, not Money

The way I've described it, starting a startup sounds pretty stressful. It is. When I talk to the founders of the companies we've funded, they all say the same thing: I knew it would be hard, but I didn't realize it would be this hard.

So why do it? It would be worth enduring a lot of pain and stress to do something grand or heroic, but just to make money? Is making money really that important?

No, not really. It seems ridiculous to me when people take business too seriously. I regard making money as a boring errand to be got out of the way as soon as possible. There is nothing grand or heroic about starting a startup per se.

So why do I spend so much time thinking about startups? I'll tell you why. Economically, a startup is best seen not as a way to get rich, but as a way to work faster. You have to make a living, and a startup is a way to get that done quickly, instead of letting it drag on through your

[8]The reason Y Combinator never negotiates valuations is that we're not professional negotiators, and don't want to turn into them.

whole life. [9]

We take it for granted most of the time, but human life is fairly miraculous. It is also palpably short. You're given this marvellous thing, and then poof, it's taken away. You can see why people invent gods to explain it. But even to people who don't believe in gods, life commands respect. There are times in most of our lives when the days go by in a blur, and almost everyone has a sense, when this happens, of wasting something precious. As Ben Franklin said, if you love life, don't waste time, because time is what life is made of.

So no, there's nothing particularly grand about making money. That's not what makes startups worth the trouble. What's important about startups is the speed. By compressing the dull but necessary task of making a living into the smallest possible time, you show respect for life, and there is something grand about that.

Thanks to Sam Altman, Trevor Blackwell, Beau Hartshorne, Jessica Livingston, and Robert Morris for reading drafts of this.

[9] There are two ways to do work you love: (a) to make money, then work on what you love, or (b) to get a job where you get paid to work on stuff you love. In practice the first phases of both consist mostly of unedifying schleps, and in (b) the second phase is less secure.